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COUNTRY
GROWTH&JOBS
REPORT

North Macedonia: Escaping the Middle-Income Trap

Growth and Jobs Report

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GROWTH&JOBS
REPORT

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The Country Growth and Jobs Report is a core analytical report that helps client countries design and implement a strategy to accelerate economic growth and creation of new and better jobs.

Report No.

NORTH MACEDONIA

GROWTH AND JOBS REPORT

Escaping the Middle-Income Trap

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Weights and Measures:
Metric System

ABBREVIATIONS AND ACRONYMS

ALMP	Active Labor Market Policies		NEET	Not in Employment, Education, or Training
CBAM	Carbon Border Adjustment Mechanism		OECD	Organization for Economic Cooperation and Development
CESEE	Central East and South East Europe		PISA	Programme for International Student Assessment
CPC	Commission for the Protection of Competition		PIT	Personal income tax
CPI	Consumer Price Index		PPS	Purchasing Power Standard
EC	European Commission		PRO	Public Revenue Office
ECA	Europe and Central Asia		R&D	Research and Development
ESA	Employment Service Agency		SILC	Survey of Income Living Conditions
EU	European Union		SME	Small and Medium-sized Enterprise
EUR	Euro		SOE	State-Owned Enterprise
FDI	Foreign direct investments		SSO	State Statistical Office
GDP	Gross Domestic Product		TFP	Total Factor Productivity
GMI	Guaranteed Minimum Income		TFPQ	Quantity-based Total Factor Productivity
GNI	Gross National Income		TFPR	Revenue-based Total Factor Productivity
HBS	Household Budget Survey		TIDZ	Technological Industrial Development Zones
HHI	Herfindahl–Hirschman Index		TIMSS	Trends in International Mathematics and Science Study
ICT	Information and Communication Technology		UMC	Upper Middle-Income Countries
LFS	Labor Force Survey		WB6	Western Balkan countries
MKD	Macedonian Denar		WDI	World Development Indicators
MOF	Ministry of Finance			

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Key Messages

The Challenge: Stuck on a Plateau

North Macedonia has achieved a remarkable economic ascent since independence, rising from lower- to upper-middle-income status in under two decades. Yet for the past 17 years, the country has been stuck at that plateau. Convergence with EU income levels is slower than that of regional peers, and without reforms, GDP growth is projected to slow to just 2.4 percent by 2050 — far below the annual growth needed to fully converge with the EU within a generation.

The current growth model—driven by private consumption and low-value-added, import-intensive foreign direct investments (FDI), with high exports concentration on few markets and products—is reaching its limits. Three structural weaknesses lie at the heart of the problem:

- **Low-productivity low-investment trap.** The gap in private investments relative to peers is widening, while Total Factor Productivity (TFP) growth has been negative since the 2010s, recovering only marginally after the pandemic. Investments are overwhelmingly allocated to assets that marginally raise productive capacity or technological capability, yielding less competitive returns and discouraging further private accumulation.
- **Distorted markets and weak competition.** Market concentration is high: top firms in 10 out of 14 sectors control more than half of sector revenues, while half of the state portfolio is in competitive sectors. Entrenched incumbents, weak rule of law, and corruption suppress the dynamism needed for productivity gains.
- **Underutilized, shrinking and costly human capital.** The workforce is aging and shrinking due to emigration. Women participate in the labor market at rates 19 percentage points below men, while 24 percent of youth is not in employment, education, or training. Although educational attainment is increasing, 22 percent of workers are overqualified for their roles, and education quality is declining according to OECD PISA. Wages are rising faster than productivity, which erodes competitiveness.

The Path Forward: Three Reform Pillars

Despite these challenges, North Macedonia has significant opportunities to accelerate productivity growth, strengthen competitiveness, and raise living standards. Building on the ongoing reform plans and actions, the report identifies a reform agenda that can reverse current trends and unlock a more dynamic, resilient, and inclusive growth model, organized around three interconnected pillars:

- **Strengthen economic governance and competition.** Improve public spending efficiency, create a level-playing field and strengthen governance of SOEs, further improve the business environment, and enhance trade policy and invest in quality infrastructure—standards, metrology, and logistics—to position North Macedonia as a regional hub.
- **Foster technology infusion, energy efficiency, and innovation.** Boost firm innovation capacity and maximize the impact of FDI. Reduce persistently high energy intensity of firms and decarbonize the economy.
- **Develop skills and mobilize workers.** Enable women's equal participation and opportunities in the labor market, support an effective school-to-work transition for youth, and encourage lifelong learning of workers and longer stays in the labor market. Finally, reduce labor market frictions to support higher participation and formalization.

The Stakes: What Reform Can Deliver

While the cost of inaction is stagnation, equally the reward for policy action could be transformative in terms of faster growth, higher wages and more jobs. A combined reform package could raise average annual growth from **2.4 to 3.8 percent** by 2050, and generate additional **53,000 new jobs** and around **196,000 better-paid jobs** by 2050. The structural reforms needed are well understood. What is required now is the political commitment to pursue them—and the institutional capacity to implement them. EU accession process is a unique opportunity to make that leap.

Overview

A Remarkable Rise — and a Prolonged Plateau

Since gaining independence, North Macedonia has achieved one of the most notable economic transformations in the Western Balkans. In less than two decades, it ascended from lower-middle to upper-middle income status, reducing poverty to below 19 percent by 2024, and becoming one of the most open economies in the region with a trade-to-GDP ratio of 136 percent. These are real hard-won achievements, reflecting decades of policy discipline, institutional development, and integration with regional and global markets.

Yet the momentum has stalled. For the past 17 years, North Macedonia has remained in the upper-middle-income classification, slowing, rather than accelerating its transition to high-income status. The risk of a middle-income trap, where an economy loses the growth drivers of its earlier ascent before it has developed the capabilities needed to compete at the high end, is no longer a distant concern. It is a current reality.

The clearest measure of the challenge is convergence with EU living standards. Albania, Serbia, and Montenegro each narrowed the income gap with the EU27 average by 7 to 10 percentage points over the last decade. North Macedonia managed only 4 percentage points, reaching approximately 42 percent of the EU27 average GDP per capita in 2024. At this pace, full convergence is not a generational goal; it is an increasingly remote aspiration.

The human dimension makes the urgency sharper. The working-age population is shrinking due to aging and sustained emigration, and those who remain are not being fully utilized. Women participate in the labor market at rates 19 percentage points below men. Youth not in employment, education or training (NEET) stand at 24 percent. These gaps represent a significant untapped source of growth at a time when demographic trends are making labor increasingly scarce. By 2040, North Macedonia is projected to become 'super-aged', with at least 20 percent of population aged 65+, further eroding the labor contribution to growth.

The Stakes: Two Scenarios for 2050

The trajectory ahead is not predetermined, but is being shaped now by the choices made, or deferred. Two scenarios frame the decision. Under a business-as-usual scenario, annual GDP growth is projected to slow to 2.4 percent by 2050. By mid-century, North Macedonia would find itself at just 55 percent of the EU average income—still far short of the income convergence needed to anchor skilled workers and prevent an accelerating brain drain. In this scenario, the country falls further behind its peers.

Under the reform scenario, sustained structural change could generate average annual growth of 3.8 percent per year. This is the rate required for full EU income convergence within less than four decades. The simulation results are concrete: a combined reform package could increase real wages by 20 percent relative to the baseline and generate the equivalent of over 240,000 more and better jobs by 2050.

The difference between these futures is not luck or external condition. It is a policy.

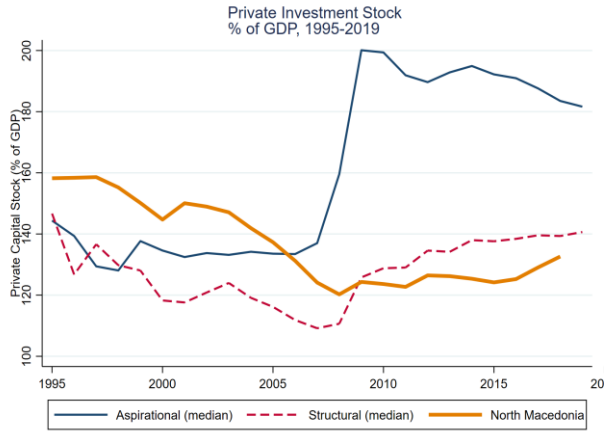
Three Structural Fault Lines

Understanding why the current growth model is running out of steam requires looking beneath the headline numbers. The three structural fault lines discussed below are distinct in their manifestations but together they point to a clear agenda: strengthening the governance foundations that underpin private-sector-led growth, from institutional quality and market regulation to the alignment of public policy with long-run productivity.

Productivity-Investment Stagnation. North Macedonia is investing, but not productively. Total Factor Productivity (TFP) growth has been negative since the 2010s, and only marginally improved post-pandemic. Investment, rather than flowing into high-tech manufacturing and skills-enhancing services such as ICT and professional services, is overwhelmingly concentrated in construction (almost two-thirds of the total). Returns on capital have steadily declined and now lag peer countries. The economy is accumulating assets without generating the efficiency gains that translate into sustained growth.

The steady decline in private investment since 1995 and the faster-than-peer deceleration in returns are by themselves rational response to an economy that does not offer a credible path to high returns on productive investment—because markets are not competitive enough to reward efficiency, institutions are not reliable enough to reduce uncertainty, and the industrial structure has not evolved to the point where domestic firms can capture the value that upgrading would generate. But these are not inevitable outcomes moving forward; rather, they represent untapped reservoir of growth waiting to be unlocked.

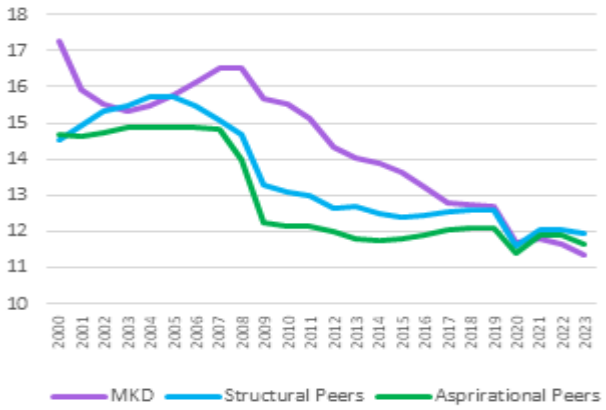
Private investment stock in North Macedonia had been declining since 1995



Source: World Bank calculations based on IMF's Investment and Capital Stock Dataset (ICSD). *Notes:* Structural and aspiration peers are as previously defined.

Rate of return on investment has been decelerating faster than with peers

(2000-2023, Percent)



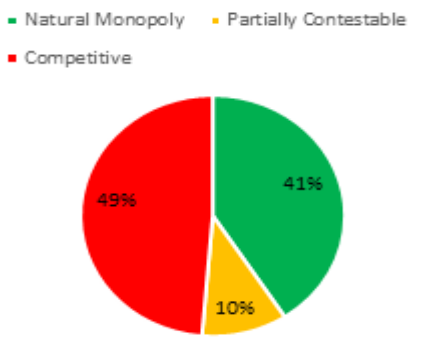
Source: World Bank calculations. *Notes*: The Rate of Return to Capital (RoRK) is the change in output brought about by a unit change in capital stock which is calculated using the perpetual inventory method and the amortization rate of 6 percent.

Limited Competition and Market Dynamism. Market structure in North Macedonia is heavily concentrated. In most sectors, the top four firms account for more than 50 percent of revenues, and these firms have held their dominant position for years. This level of concentration suppresses the dynamism that drives innovation, entry by new firms and overall productivity. Markets that are not contestable do not generate competitive pressure; and without competitive pressure, incumbents have little incentive to invest, upgrade, or improve.

The state's economic footprint compounds the problem. The government retains ownership stakes in over 190 entities, nearly half of which are in competitive sectors, potentially crowding out private investment and limiting the effectiveness of market signals. Corruption, weak rule of law, and a heavy regulatory burden make entry costly and compliance burdensome: 18.4 percent of firms cite giving gifts to public officials "to get things done" and 11 percent of firm cite corruption as the top obstacle to doing business. Together, these conditions represent a structural tax on the private sector, one that is not yet quantified in this report but unmistakable in its effects on investment and growth. This is, at its core, a governance weakness.

The state's economic footprint is extensive in competitive sectors…

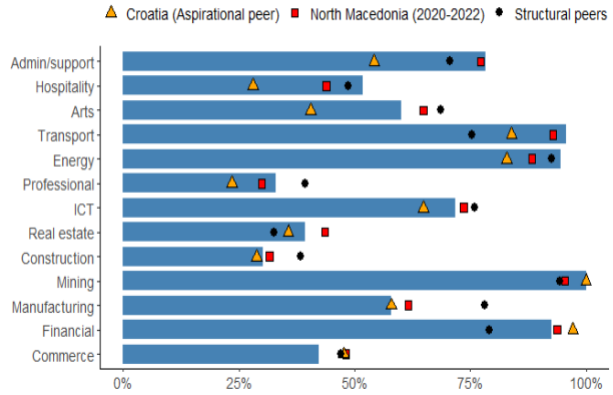
Revenue of firms in which the government has at least a 10% ownership stake as a share of GDP



Source: Based on the World Bank Global Business of the State database (2019).
Note: Data coverage on revenue for North Macedonia is 99%.

…while just four firms control more than half of the market most sectors.

Revenue share of top 4 firms by sector



Source: North Macedonia Central Business Registry.
Note: The blue bars denote North Macedonia (2017-2019). Data for structural and aspirational peers are for 2017-2019.

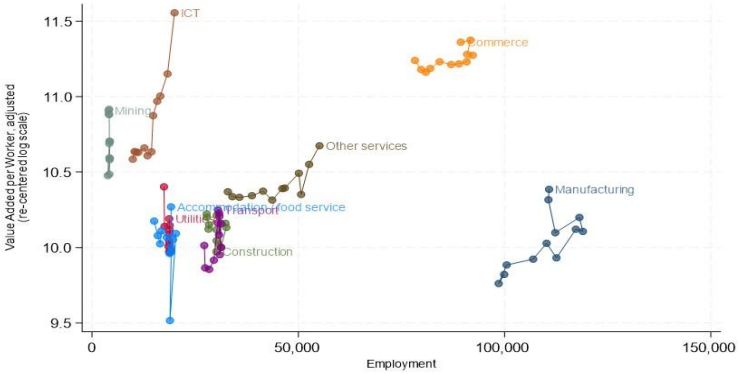
Underutilized, Shrinking and Costly Human Capital. North Macedonia faces a demographic challenge that is both immediate and long-term. The working-age population is projected to fall by more than half by 2070, driven by aging and sustaining emigration that shows no signs of reversals. This is a structural constraint on growth that is extremely difficult to fully offset, but it can be significantly mitigated through better use of those who remain.

The labor market gaps are large and addressable, and their persistence reflects governance weaknesses in education, social policy, and labor market regulation. Women participate in the labor market at rates 19 percentage points below men, a gap that reflects persistent barriers including inadequate childcare, inflexible work arrangements, and structural wage disparities. Achieving gender parity in employment could raise long-run GDP per capita by roughly 13.6 percent. Youth NEET rates at 24 percent point to a failed school-to-work transition. Twenty-two percent of workers are overqualified for their roles, a signal of misalignment between the education system and labor market demand.

Recent increases in labor costs, driven by rising minimum wage and public sector wages, have not been matched by corresponding productivity gains, increasing the real wage cost and eroding external competitiveness. And PISA scores reflect a declining quality of education that, if unreversed, will diminish the skills base on which future growth depends. Meanwhile, structural transformation is gravitating toward low-productivity or non-tradable services rather than high-value and tradable sectors that could anchor a more dynamic economy.

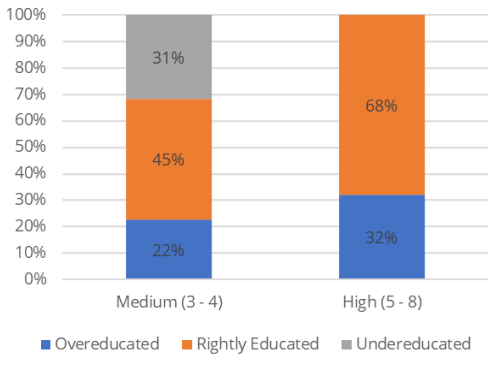
Employment expanded but in relatively low-productivity manufacturing and higher-productivity commerce, but not in ICT

Labor productivity and employment by sector, 2011-2022



…yielding modest returns to education and overqualified workers for their roles.

Over/underqualification by level of education



Source: World Bank staff calculation based on North Macedonia Central Business Registry data (panel a) and 2023 LFS collected by the State Statistical Office (panel b). *Note:* For panel a, agriculture is dropped for better visualization. Differences in capital-labor ratio is

accounted for by estimating the residual of log-value added per worker when regressed with capital labor ratio. For panel b, under- and overeducation are classified as per ILO skill levels for ISCO-08 major groups, though workers with professional experience may not be underqualified.

The Reform Agenda: Three Pillars

Despite these challenges, North Macedonia has significant opportunities to accelerate productivity growth, strengthen competitiveness, and raise living standards. By addressing governance constraints, fostering innovation and technology adoption, and better utilizing its human capital, already identified in existing reform agenda, the country can reverse current trends and unlock a more dynamic, resilient, and inclusive growth model.

The report identifies a comprehensive reform roadmap structured around three interlocking pillars. At the center of these three sits governance. More effective and consistent regulation, better aligned public policy, and stronger institutions are not one reform among many; they are the connective tissue that determines whether market competition generates dynamism, whether investment flows toward productive uses, and whether human capital translates into higher-value work.

These pillars are not independent; rather, they reinforce each other. Productivity gains require both competitive markets and skilled workers. Labor market reform is more effective when firms are investing and growing. And investment will not flow into higher-value activities without both market discipline and human capital to support it.

Pillar 1: Strengthening economic governance and competition

Productivity growth in North Macedonia will not be achieved through public investment alone that already amounts to 5.5 percent of GDP. Greater impact from public investment, combined with higher levels of private investment, standing 2 pp of GDP below the structural peers over the last decade, will be needed to support sustained growth. It requires open, competitive markets and an effective state that enables rather than distorts private activity. This pillar addresses the foundational conditions for private-sector dynamism:

- **Improving public spending efficiency** is ultimately about directing scarce fiscal resources toward the investments that most directly build the foundations of productivity growth. This means eliminating inefficiencies in public investment management and improving the efficiency of public education spending to halt the decline in education quality. Every fiscal denar misallocated to low-return or underperforming public programs is a denar not building the human capital and infrastructure on which long-run productivity depends. Meanwhile, firms may absorb infrastructure costs themselves while operating around unreliable public commitments; both of which suppress productivity and deter investments simultaneously.
- **Creating a level playing field and improving governance of state-owned enterprises (SOEs)** requires adopting a clear state ownership policy that articulates the economic rationale for retaining state stakes and applies good corporate governance principles to SOEs. Minimum wage policy should be aligned with productivity growth to preserve competitiveness. The competition regulatory framework should be strengthened to limit anticompetitive conduct and ensure effective merger review. State aid rules need to be reformed to reduce market distortions and foster genuine competitive neutrality.
- **Improving the business environment** through reducing the cost of doing business for firms of all sizes, with particular attention to SMEs and growing firms. This includes streamlining business permits, licensing, and customs procedures; accelerating digitalization of government-to-business services through online one-stop shops and business portals; supporting firms in their own digital transformation; and fighting corruption in public procurement, taxation, and access to government services, consistent with the National Strategy for Prevention of Corruption.
- **Enhancing trade policy and infrastructure** calls for addressing the regulatory and logistical barriers that limit North Macedonia's ability to deepen its trade integration. This includes reviewing and adopting transparent rules for cross-border e-commerce and digital trade, negotiating mutual recognition of professional qualifications with trading partners, improving the National Quality Infrastructure, and

reducing import tariffs on inputs and intermediate goods in sectors with high FDI potential—extending these benefits to domestic firms to ensure broad access to competitive inputs.

Table 1. Summary of reform packages to strengthen economic governance and competition

Improving public spending efficiency	Creating the level-playing field and improving governance of SOEs	Improving the business environment	Enhancing trade policy and infrastructure
Eliminate inefficiencies in public investment management (IMF-WB PIMA assessment)	Align minimum wage policy with productivity growth to retain competitiveness	Streamline business permits, licensing, and customs procedures to reduce regulatory burdens, especially for SMEs and growing firms	Reduce service trade restrictiveness and negotiate mutual recognition of professional qualifications to reduce technical barriers to trade
Improve public education efficiency to improve education quality (WB Public Finance Review)[1]	Enhance the competition and state aid regulatory framework to limit potential anticompetitive actions and ensure level-playing field	Accelerate digitalization of G2B services and one-stop-shops, and support firms in digitalization	Enhance the National Quality Infrastructure through targeted sector-level improvements and improve trade facilitation
	Adopt a clear state ownership policy to articulate the economic rationale for retaining state ownership and adopt good corporate governance principles for SOEs	Fight corruption in public procurement, taxation, and access to government services (per the National Strategy for Prevention of Corruption)[2]	Reduce import tariffs for input and intermediate goods to ensure access to high quality and cost competitive inputs and capital goods

Pillar 2: Fostering technology infusion, energy efficiency, and innovation

North Macedonia's export base remains heavily concentrated in low-value automotive supply chains, with domestic value-added amounting to only 18 percent of exports. This narrow, low-value export profile limits growth potential and vulnerability to external shocks. Reorienting trade and investment toward higher-value activities requires deliberate effort across three interconnected areas:

- **Boosting firm innovation** involves supporting continuous upgrading within firms, especially in manufacturing and services, through targeted programs for technology adoption and improved management practices, as articulated in the Smar Specialization Strategy. Instruments such as innovation vouchers and supply-chain financing platforms can lower the threshold for firm-level innovation. The impact of existing innovation funding programs can be enhanced by refining selection criteria, strengthening monitoring, and consolidating them into fewer, larger programs that build sustainable capacity in viable firms rather than spreading resources thinly. The ongoing consolidation of private-sector supporting institutions is welcome.
- **Maximizing the impact of foreign direct investment (FDI)** requires a strategic shift in how North Macedonia attracts and manages foreign investment. A targeted FDI promotion strategy should prioritize investment with high potential for job quality, technology transfer to local firms, and alignment with the green economy, which require moving decisively away from low-cost assembly toward high-value sectors, while actively retaining and deepening the footprint of investors already present. The real productivity gains, however, come from linkages: well-designed supplier development programs, technology matching, and co-location services that embed foreign investors into the domestic economy and lift the capabilities of local firms alongside them.
- **Reducing energy intensity and decarbonizing the economy** is not only an environmental imperative but a competitiveness issue. The impending Carbon Border Adjustment Mechanism (CBAM) from the EU poses a direct risk to North Macedonia's export competitiveness, making the transition to a low-carbon economy a matter of economic urgency. The country is also increasingly exposed to climate hazards, including flood events and heat extremes, that demand dedicated investment in resilience. In this

[1] https://www.worldbank.org/en/country/northmacedonia/publication/north-macedonia-public-finance-review
[2] https://dksk.mk/wp-content/uploads/2021/09/NACS-2021_25-and-Action-Plan-EN-final.pdf

context, carbon taxation serves a dual purpose: minimizing CBAM exposure while generating revenues that can be deliberately channeled into climate adaptation investments, such as flood protection infrastructure and heat-resilient public systems. The reform agenda calls for shifting from monitoring to deep investment in energy efficiency: incentivizing the adoption of energy-efficient technologies; developing energy service companies (ESCOs) and green credit lines that repay on the basis of verified energy savings; reallocating funding to build energy management skills among technical staff; and implementing carbon taxation to capture these revenue and resilience co-benefits.

Table 2. Summary of reform packages to foster technology infusion, energy efficiency, and innovation

Boosting firm innovation capacity	Maximizing the impact of FDI	Reducing persistently high energy intensity of firms and decarbonizing the economy
Support technology adoption and management upgrades, as articulated in the Smart Specialization Strategy, through innovation vouchers and supply chain platforms for financing	Develop targeted FDI promotion strategy focused on job creation, technology transfer to local firms via industrial and spatial linkages, and sustainability/green economy	Implement carbon taxation to minimize CBAM exposure[3] and fund resilience investments in energy-efficient technologies and capital upgrades
Consolidate innovation funding into fewer, larger programs with stronger selection and monitoring	Strengthen retention and expansion of existing FDI through grievance mechanism and IP rights	Develop/expand ESCOs and green credit lines repaid via verified energy savings.
	Link foreign and domestic firms through supplier development programs, joint innovation initiatives, and integration support (e.g., technology matching services, co-location services)	Reallocate funding programs to build energy management skills among technical staff and managers.

Pillar 3: Developing skills and mobilizing workers

Sustaining growth in the face of demographic decline requires a dual strategy: making better and fuller use of the workforce that exists today while investing in the workforce of tomorrow. This pillar addresses both dimensions, recognizing that labor market reform is inseparable from education, social policy, and demographic strategy:

- **Enabling women's equal participation and opportunity** is the single largest source of untapped labor supply. Expanding access to affordable childcare and eldercare services, especially in underserved areas, via mixed provision with quality standards and predictable financing addresses the most commonly cited barrier to women's employment. Labor code reforms should support flexible and inclusive work arrangements, including shared parental leave and flexible work modalities. Measures to reduce the gender wage gap, which simultaneously reflect and reinforce lower female participation, should be a core element of the reform package, including adopting and enforcing pay transparency and the principle of equal pay for work of equal value, to reduce the structural drivers of women's job sorting and wage gaps. Employment counseling and active labor market programs (ALMPs) should be explicitly designed to connect women to quality jobs and address skills gaps, including through scaled intensive case management, covering counselling, skills profiling, tailored training, and post-placement support, for long-term unemployed and social assistance beneficiaries, with Employment Service Agency (ESA) capacity and performance monitoring strengthened accordingly.
- **Supporting effective school-to-work transitions for youth** requires aligning vocational and tertiary education with labor market needs, scaling up workplace-based learning, and exploring the feasibility of tertiary vocational education. Youth employment can be directly stimulated through targeted hiring incentives (like the existing Youth Subsidy Scheme), subsidized apprenticeships, and employer participation in youth employment initiatives. Career guidance systems should be strengthened so that

[3] https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099060724174031856

young people make better-informed choices about education and career pathways, reducing the costly mismatch between qualifications and available jobs.

- **Encouraging lifelong learning and longer labor market participation** addresses the skills needs of the existing workforce and the fiscal pressures of an aging population. Tailored upskilling programs for unemployed individuals should focus on foundational, digital, and job-relevant skills. Financial and regulatory incentives should encourage employers, especially SMEs, to co-invest in workforce development. Labor market intelligence, regularly disseminated to employers, education providers, and policymakers, should underpin evidence-based decision-making. Increasing the statutory retirement age for both genders and introducing pension bonuses for continued work beyond retirement age would help sustain labor supply.
- **Reducing labor market frictions** involves reforming the labor code and labor tax system to reduce barriers to formal part-time/seasonal/gig hiring (including contribution rules and simplified contracts) while maintaining core worker protections, to formalize low-wage, seasonal and gig work. The social benefit system should be reformed to phase out assistance for recipients who take up employment gradually, rather than immediately, to eliminate the "benefit cliff" that discourages formal work, and reduce minimum contribution bases that make formal part-time and low-wage work disproportionately costly. Activation measures, including job search support and tailored training for social assistance recipients, backed by clear mutual obligations and data sharing between employment and social welfare service, should be strengthened to support a genuine transition into employment.

Table 3. Summary of reform packages to develop skills and mobilize workers

Enabling women's equal participation and opportunities in the labor market	Supporting effective School-To-Work Transition for youth	Encouraging lifelong learning of workers and longer stay at the labor market	Reducing labor market frictions to better support participation and formalization
Scale childcare and community-based long-term care in underserved municipalities, using a mixed delivery model with clear quality standards and predictable financing	Institutionalize employer-linked tertiary and dual/workplace-based learning pathways, and use graduate outcomes and vacancy data to continuously adjust program intake and curricula	Rebalance adult upskilling toward in-demand technical/green and digital skills, and introduce financial and regulatory incentives for employers to co-invest in the upskilling and reskilling of their workforce	Reduce barriers to formal part-time/seasonal/gig hiring (including contribution rules and simplified contracts) while maintaining core worker protections, to formalize low-wage, seasonal and gig work.
Adopt pay-transparency and enforcement tools and enable flexible work modalities and shared parental leave to reduce the structural drivers of women's job sorting and wage gaps	Strengthen career guidance & placement systems across secondary/tertiary education and ESA, with performance tracked through placement rates, retention, and study–job match metrics	Establish a Labor Market and Skills Observatory, and publish regular skills-need reports used to set training budgets and program intake	Make formal work pay by redesigning tax-benefit system and reducing minimum contribution bases that make formal part-time and low-wage work disproportionately costly
Scale intensive case management for long-term unemployed and social assistance beneficiaries, with ESA capacity and performance monitoring strengthened accordingly	Offer targeted wage subsidies and apprenticeships for NEETs and first-time jobseekers, conditional on training completion and minimum retention, with employer co-financing to reduce deadweight and improve placement quality.	Gradually raise retirement ages and reward delayed retirement, complemented by lifelong learning and healthy aging measures	

Implementation: From Agenda to Action

A reform agenda is only as valuable as its implementation. North Macedonia has produced sound policy strategies before. The challenge has rarely been diagnosis; it has been the execution. Three conditions determine whether this reform cycle will be different.

First, political commitment and policy continuity are essential for successful reforms implementation. Some of the measures outlined in this report will yield benefits quickly; others require years to mature. Both require a consistent political signal that the reform direction will not reverse with changes in government or shifting short-term priorities. This is not merely a governance aspiration, as empirical evidence confirm that political instability is among the most damaging constraints on firm productivity, and North Macedonia lags its region on this dimension. The EU enlargement process provides a powerful external anchor for this commitment, as demonstrated by the experience of EU peers in the years preceding and following accession, and should be used actively as a catalyst and accountability mechanism.

Second, reforms require strong institutions with the capacity to implement them effectively. Implementing reforms in public investment management, SOE governance, competition regulation, and labor market policy requires capable institutions with the mandate, resources, and independence to act. Capacity constraints have historically been as binding a barrier as political will. Investment in the institutions that will carry this agenda forward is not a peripheral activity; rather, it is central to success.

Finally, reform success depends on clear prioritization and sequencing. Not all reforms can proceed simultaneously, and some create enabling conditions for others. Foundational reforms in governance and competition should be prioritized early, as they shape the environment for private investment and innovation. Labor market reforms targeting female participation and youth employment can be implemented in parallel, given their high social returns and relatively contained fiscal cost. Energy and technology reforms may require longer lead times, regulatory changes, and private sector capital but should be initiated early to capture compounding returns.

A Crossroads Moment

North Macedonia stands at a genuine crossroads. The foundations are present: geographic position at the crossroads of Southeast Europe, a high degree of trade openness, good performance of hosting FDIs, and a track record of reform, however uneven. These are real assets.

But the window is narrowing. Demographic pressures are tightening with each passing year. Competitive erosion in key export sectors is advancing. The growth model that carried the country to upper-middle income status is running out of momentum. And the EU accession process, while a genuine opportunity, is not a substitute for domestic reform; it is an amplifier of it.

The analysis in this report is clear: North Macedonia can escape the middle-income trap. The reform agenda is detailed, evidence-based, and achievable. The simulated benefits, 3.8 percent annual growth, reaching 76 percent of the average EU income at purchasing power standards (PPS) by 2050, and additional 53,000 more and 197,000 better-paid jobs by 2050 —demonstrate the potential gains from implementing this reform agenda and reflect the modeled outcome of a defined set of policy choices.

Chapter 1. Macroeconomic features of growth and jobs

Over the past two decades, North Macedonia has made notable economic progress, transitioning from lower-middle to upper-middle-income status, yet it finds itself at a risk of being stuck in an upper-middle-income trap, with convergence toward EU income levels proceeding more slowly than regional peers. Growth has increasingly been driven by private consumption rather than productive investment, while total factor productivity growth has been largely muted since the 2010s — reflecting an economy where most investment flows into construction rather than higher value-added manufacturing or research and development. Structural transformation toward services is underway, but the majority of workers remain concentrated in low-value-added activities, and significant labor market challenges persist, including large gender participation gaps, low youth employment, and accelerating demographic decline due to emigration and aging. Under a business-as-usual scenario, growth is projected to slow to 2.4 percent by 2050, reaching only about 55 percent of the EU average — far short of the 5 percent annual growth rate that would be needed to achieve full convergence within a generation — pointing to an urgent need for comprehensive reforms in human capital, governance, private investment, and labor market policy.

1.1. Notable progress, but convergence slower than peers

North Macedonia is a small, landlocked country in the Western Balkans that gained independence in 1991 and subsequently began to transition from a centrally planned system to a market-based economy. With a population of 1.8 million and a strategic location along key regional trade corridors, the country's development path has been shaped by structural reforms, growing trade openness, and efforts to integrate with European markets. During the transition, the economy experienced substantial structural shifts, including privatization, industrial restructuring, and stabilization of the macroeconomic environment, which laid the foundation for subsequent economic development.

North Macedonia's economy has achieved substantial progress since independence, rising from lower-middle-income to upper-middle-income status in less than two decades. This led to a significant drop in poverty rate[4] to below 19 percent by 2024 lifting almost half a million people out of poverty. The country became one of the most open economies in the broader Europe region, with trade to GDP ratio reaching 136 percent in 2024. However, North Macedonia remains an upper-middle-income country for the past 17 years, as the advantages that have lift it to its current level, including the export-led FDI-driven growth (more elaborated in Chapter 2 and 4), have proven insufficient to sustain further rapid growth. In contrast, aspirational peers like Latvia and Croatia which joined the European Union (EU) took eight and 13 years, respectively, to reach high-income status.

Convergence with EU living standards is ongoing, but at a slower pace than in peer countries. Over the last decade, North Macedonia has narrowed the income gap with the EU27 average by just 4 percentage points (pp) to about 45 percent of GDP per capita in purchasing power standard (PPS) in 2025. During the same period, convergence of regional peers accelerated, ranging from 7 pp in Montenegro, to 11 pp in Albania and Serbia. North Macedonia's average growth rate has been below those of structural and aspirational peers[5] since 2000 (Figure 1), with GDP per capita above the upper-middle income average (Figure 2).

While maintaining resilience, North Macedonia's current growth model may be unable to secure long-term sustainability and achieve income convergence. Although consumption has long been the main driver of growth, since 2020 it has emerged as the main driver of growth, contributing 2.4 pp per year, largely supported by rising wages, pensions, and social transfers amid a tightening labor market. More importantly, while the economy has become more export-oriented over the last 15 years, the growth return has been limited by a narrow and import-

[4] Measured at PPP US$8.30 poverty rate poverty was 43.6 percent in 2009.

[5] Structural peers throughout the report include Albania, Bosnia and Herzegovina, Montenegro, Serbia, and Kosovo. Aspirational peers include Croatia, Estonia, Latvia, Lithuania, Slovenia, and Slovakia.

intensive export base, with nearly half linked to the automotive supply chain and German market. The high import content of both consumption and exports has also kept net exports consistently negative.

Figure 1. GDP growth has been slower than in peer countries…

(Average growth rates, North Macedonia and peer countries, percent)

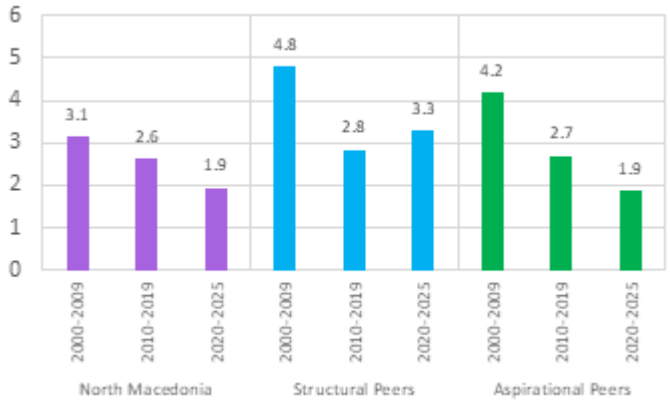
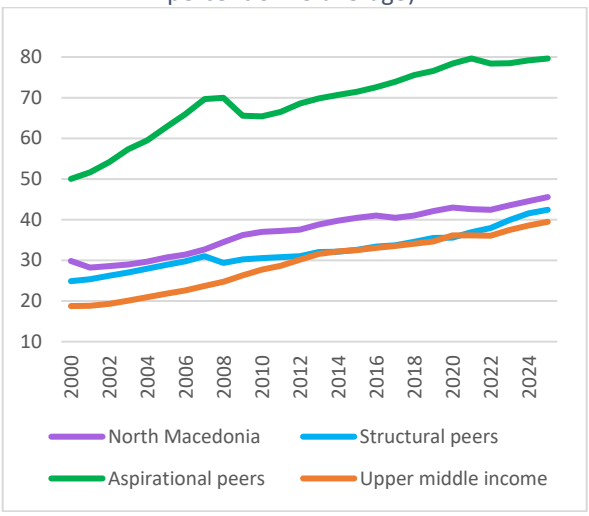


Figure 2. …but GDP per capita stood above the upper-middle income average.

(GDP per capita, PPP, constant 2021 international $, percent of EU average)

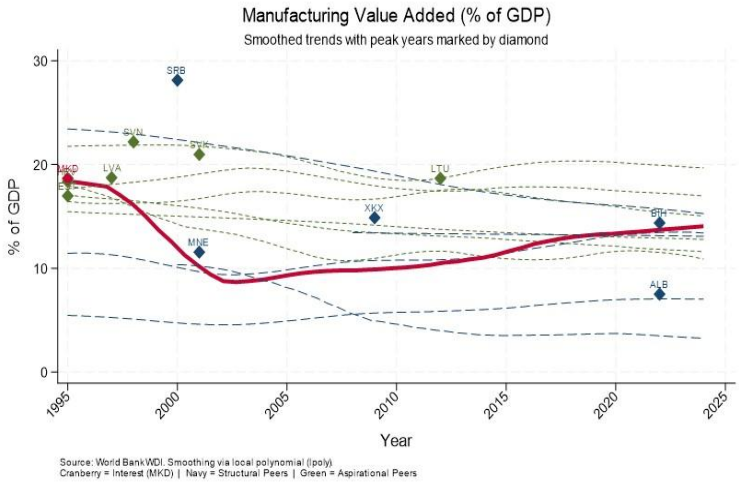


Source: WDI, national statistical offices, and World Bank staff calculations. *Notes:* Dots represent the global countries.

1.2. Structural transformation to service-based economy and suboptimal investments

North Macedonia experienced one of the sharpest deindustrialization episodes in the region. The country's manufacturing value added as a share of GDP started at roughly 19 percent in mid-1990 before bottoming out at around 9 percent in the early 2000s and heading off for a gradual albeit slow recovery (Figure 3). The decline followed as a result of the transition from a planned economy that shielded domestic industries, including state-owned enterprises (SOEs), to a market economy. The gradual recovery post 2010s was fueled by targeted FDI attraction into export-oriented manufacturing, particularly in the automotive components anchored around the country's technological industrial development zones (TIDZs). By 2024, North Macedonia's manufacturing share had climbed back to approximately 14 percent, but remained behind aspirational Central European and Baltic peers, whose shares hover closer to 18–22 percent of GDP.

Figure 3. North Macedonia is gearing towards slow reindustrialization, following a sharp decline in manufacturing share in the mid-1990s



North Macedonia's growth has historically been driven by capital investment, while productivity remained negative or muted. Throughout the 2010s, capital accumulation added around 2.5 percentage points to annual growth, making it the dominant engine of the economy (Figure 4). Labor contribution has been low and the quality of the workforce (skills and experience embodied in the workforce) weakened, and so did the quantity of labor, as an aging population and outward migration reduced the workforce while rising labor force participation compensated partly. Total factor productivity, a measure of how efficiently the economy uses its inputs, has

been subtracting from growth since the 2010s, although post-pandemic it turned slightly positive, adding an average of 0.2 percentage points per year below its structural peers in the Western Balkans.

Services have become the primary driver of growth, particularly in the aftermath of the pandemic. Since 2020, services contributed around 2.2 percentage points to annual growth (Figure 5). Low-productivity non-tradeable wholesale and retail trade have historically led this growth, but information technology and other knowledge-intensive sectors have emerged as increasingly important drivers in recent years (Box 1). Over time, agriculture and industry have become a drag on growth, especially in the post-pandemic period.

Figure 4. Growth has been mostly capital-intensive…

(Contributions to growth by factor, pp)

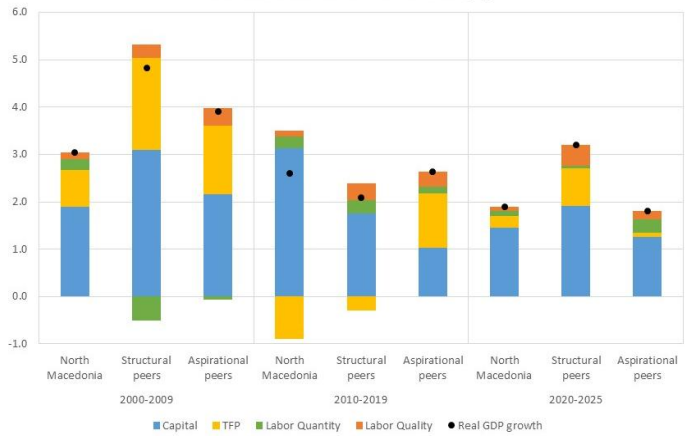
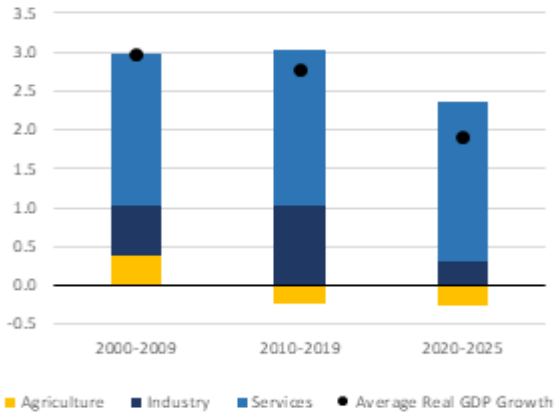


Figure 5.…and recently entirely led by services.

(Contributions to growth by sector, pp)



Source: Total Economy Database, State Statistical Office, and World Bank staff calculations.

Box 1. North Macedonia's ICT sector: a promising but constrained opportunity

North Macedonia's ICT sector has become one of the most dynamic parts of its economy. The sector reached an estimated revenue of 4 US$ billion in 2022, with employment growing by about 35 percent in that year to nearly 20,000 workers (Figure 6). While the sector recorded the largest employment growth of any sector in recent years, it still accounts for only around 5 percent of total employment. Its high-productivity profile stands out compared to the rest of the economy, and in 2022, ICT real wages were nearly 45 percent higher than the national average. ICT has also emerged as a meaningful driver of overall services growth, contributing alongside retail trade, finance, and real estate to output expansion in 2023,[6] and the leading sector in services exports (Figure 7). In the Western Balkans context, the employment multiplier from IT and software services is particularly striking: EUR 1 million in IT sector revenue generates nearly twice the employment of equivalent revenue in other industries, underscoring the sector's potential as a growth lever.[7]

However, the sector faces binding constraints that limit its scale: the most acute is a talent shortage. North Macedonia produced only 619 IT graduates in 2023, a number that is unlikely to meet employer demand and could suppress organic growth of digital entrepreneurship.[3] Brain drain compounds this: the gap between public and private sector salaries and well as the quality of local public services like health and education pushes skilled workers abroad, and the government itself struggles to attract and retain the technical talent needed to develop and operate digital public services. Gender underrepresentation further narrows the talent pool: a recent graduate tracer survey from public universities shows that ICT is male-dominated in the education pipeline (ICT accounts for 16.6 percent of male graduates vs. 6.5 percent of female graduates in the cohort), which limits the pool of potential entrants. Consistent with this, women represent only about 27 percent of the ICT workforce, with even lower representation in senior and managerial role.[8]

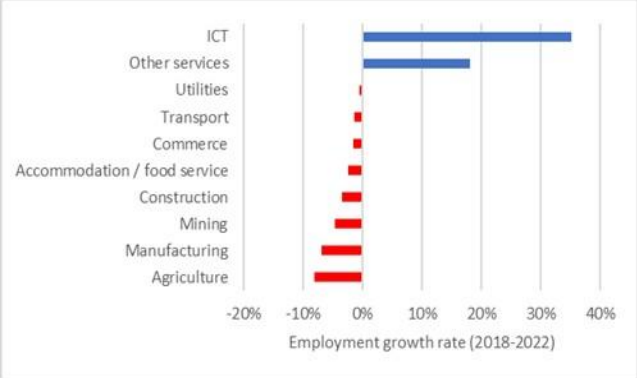
There is also the issue of limited access to technologies, including digitals, and know-hows due to underdeveloped technology services and quality infrastructure. There are currently some centers of excellence providing technology extension and innovation services, but they are limited in terms of scale and capacity. The demand from SMEs for such services is also low due to lack of awareness. This is reflected in North Macedonia's digital economy, where e-commerce

[6] Europe and Central Asia Economic Update, Spring 2024, p. 107.
[7] World Bank. Assessment of Startup Ecosystems in the Western Balkans. June 2023, p. 5.
[8] World Bank. North Macedonia — Policy Notes. September 2024, p. 85.

adoption lags behind all other Western Balkan economies and remains at an early stage of development[9], a missed opportunity for firms seeking to expand markets and modernize operations.

Figure 6. ICT recorded the highest employment growth since 2018

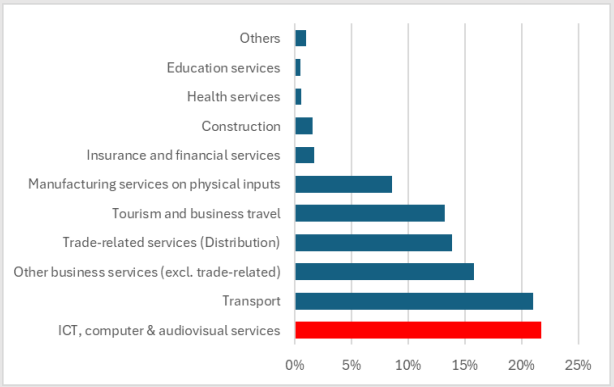
(Employment growth, 2018-2022, in percent)



Source: ICA 2.0 Toolkit based on Central Business Registry
Note: ICT covers NACE 2-digit classification 57-63.

Figure 7. ICT remains the major driver of services exports

(Share in total services exports, %, 2022)



Source: WTO's Trade in Services by Mode of Supply (TISMOS)
Note: Nearly 79% of telecommunications, computer, information and audiovisual services are attributed to ICT.

The upshot is that high-skilled programming and IT services can represent a genuine opportunity for North Macedonia to shift its growth model toward higher value-added, tradeable activities – but this opportunity is no longer automatic. Rapid advances in AI are expected to reshape task content and demand in ICT, with evidence suggesting that a large share of ICT jobs may be highly exposed with low complementarity, implying higher displacement risk for some roles. Realizing the potential at scale therefore requires not only expanding the talent pipeline, but also continuously upgrading skills toward AI-complementary work, strengthening university-industry links, and addressing brain drain and gender gaps that continue to constrain supply.

Private investment has been on a long declining trend in North Macedonia. The country's capital stock as a share of GDP has been eroding, losing an early advantage it held over peers in the mid-1990s (Figure 8). What is striking is that the country has now fallen behind not only its aspirational peers, such as more advanced economies like Croatia and the Baltic states, but also its structural peers with similar level of development. This suggests that the challenge is not simply one of catching up with more advanced countries, but of keeping pace with economies facing similar constraints. This does not only risk the country of being stuck in in a low-investment, low-productivity equilibrium that is difficult to escape, but it also signals that North Macedonia is not on a credible path toward the higher-income status it aspires to.

The declining returns to investment also suggest that the problem lies in how capital is used and that it goes into low-productivity sectors. While North Macedonia had a competitive advantage with higher returns on investment in the late 2000s, this advantage has diminished over time, faster than with peers, and now tends below the levels of its structural and aspirational peers (Figure 9). Low investment returns highlight several issues: the economy does not seem not to offer a credible path to high returns on productive investment because markets are not competitive enough to reward efficiency, institutions are not reliable enough to reduce uncertainty, and the industrial structure has not evolved to the point where domestic firms can capture the value that upgrading would generate. These factors contribute to structural inefficiencies.

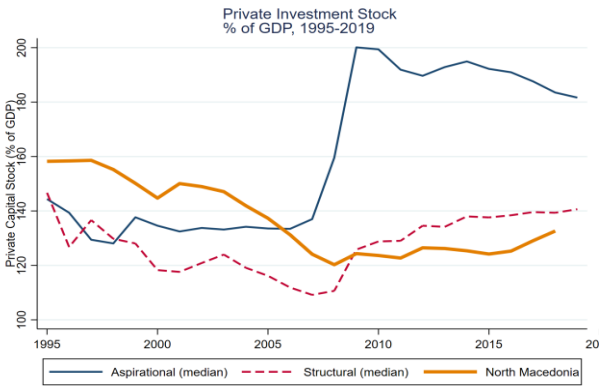
The composition of private investment reinforces the concern about declining returns. In 2013–25, nearly two-thirds of firm investment went to construction, much of it residential and driven by diaspora remittances, while only one-third was directed toward machinery and equipment. This suggests that firms are channeling resources into assets that do little to expand productive capacity or facilitate technological upgrading. Research and development spending has also stalled after modest early gains, pointing to a limited appetite for innovation and

[9] World Bank. Country Policy Note. North Macedonia Private Sector Development Roadmap: How to make firms more productive and competitive. October 2025.

upgrading. Indicators of innovation activity—including business R&D spending, technology adoption, and process upgrading corroborate that. This pattern can also explain why capital accumulation has not translated into stronger productivity growth (more elaborated in Chapter 2).

Figure 8. Private investment stock in North Macedonia had been declining since 1995

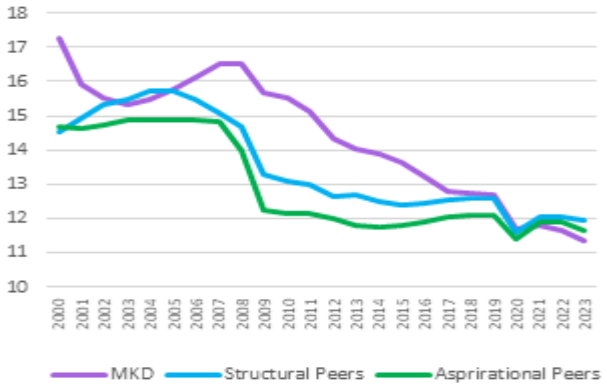
(2000-2023, Percent)



Source: World Bank calculations based on IMF's Investment and Capital Stock Dataset (ICSD). *Notes:* Structural and aspiration peers are as previously defined.

Figure 9. Rate of return on investment has been decelerating faster than with peers

(2000-2023, Percent)



Source: World Bank calculations. *Notes*: The Rate of Return to Capital (RoRK) is the change in output brought about by a unit change in capital stock which is calculated using the perpetual inventory method and the amortization rate of 6 percent.

Sectoral investment patterns reveal capital deepening in agriculture coupled with a labor-driven expansion in services. On average, agriculture has become more capital-intensive but also due to surplus labor moving out of agriculture, making production more capital intensive (Figure 10), whereas investment in industry and services has kept pace with rising employment but yielded only modest capital intensity gains. The estimated capital-labor elasticity of substitution varies sharply across sectors—highest in agriculture and lowest in services—as farmers are more able than service providers to substitute machinery for workers, as services have limited scope for capital–labor reallocation (e.g. software for clerks).

Figure 10. The agricultural sector has become significantly more capital-intensive

(Capital growth decomposition, in p.p., 2013-2025)



Source: State Statistical Office and World Bank staff calculations
Notes: Deepening means increasing the capital-to labor ratio, meaning each worker has access to more capital.

Historically, weak productivity in agriculture and industry dragged down overall total factor productivity (TFP) growth (within effects), **but improvements over time reveal an important structural transformation** (between effects).[10] Between-sector TFP growth has largely been driven by a rapid reallocation of labor into services in response to a growing wage premium (Figure 11).[11] After the pandemic, the gap between wages in the service sector and the average wage widened, while the industry wage premium narrowed, and wages in the agricultural sector fell further below the average, thus leading the structural transformation between sectors and through labor reallocation (Figure 12).

[10] The structural decomposition follows the methodology from Jooste and Tercioglu, 2025.
[11] Higher services TFP, falling rental-wage spreads, high labor mobility/transformation.

Figure 11. The reallocation between sectors has somewhat reduced the TFP decline

(TFP growth decomposition, in pp, 2012-2025)

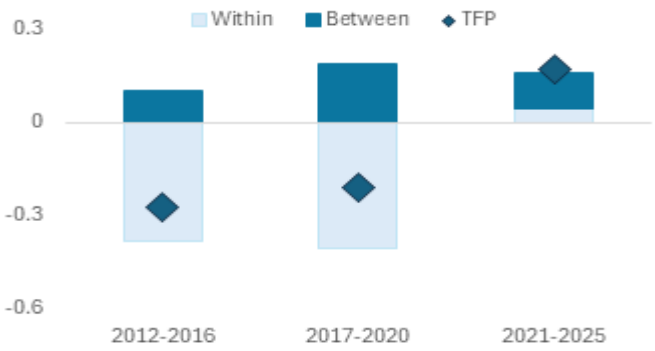
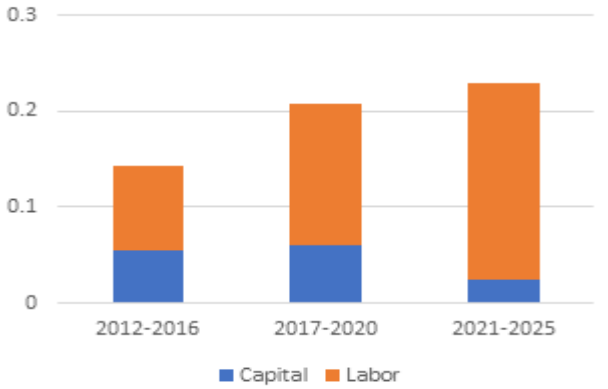


Figure 12. The labor reallocation, and to some extent capital, between sectors helped with containing the TFP decline

(TFP-between growth decomposition, in pp, 2012-2025)



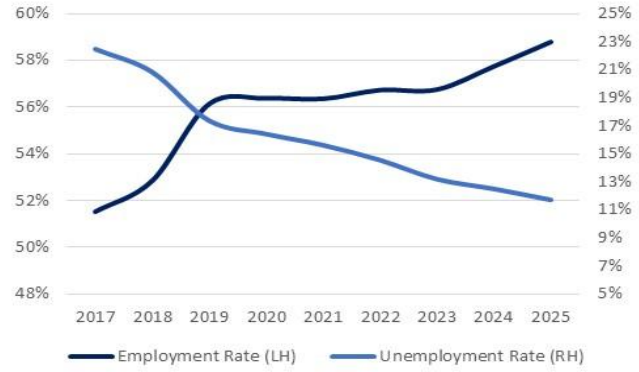
Source: State Statistical Office and World Bank staff calculations.

1.3. Unemployment declined, and labor contribution to growth increased amid demographic odds

North Macedonia's labor market has strengthened markedly over the past decade, with broad-based gains in employment, and halving unemployment. Between 2017 and 2025, the unemployment rate for people aged 15-64 fell from 21 to 11.7 percent, while the employment rate rose from 52 to 58.8 percent (**Error! Reference source not found.**). However, unlike in other Western Balkan countries, where employment has continued to improve since 2021, job creation in North Macedonia has slowed in the wake of the pandemic (**Error! Reference source not found.**).

Figure 13. The employment rate is rising

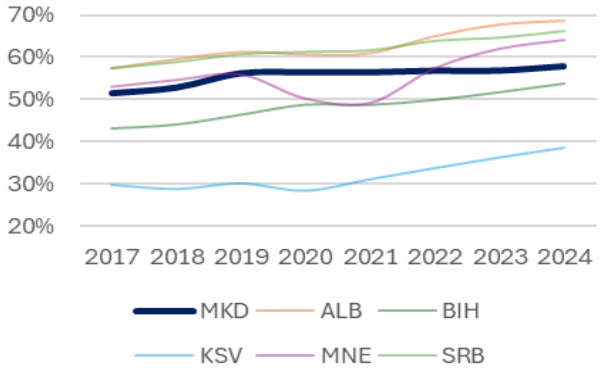
(Employment and unemployment rates (15-64), North Macedonia, 2017-24)



Source: State Statistical Office. *Note:* Data were revised by the State Statistical Office in accordance with the 2021 Census.

Figure 14. North Macedonia's labor market slowed post-pandemic unlike its peers

(Employment rates (15-64), Western Balkan countries, 2017-24)



Source: SEE Jobs Gateway for 2017–2023 and individual Country Statistical Offices for 2024.

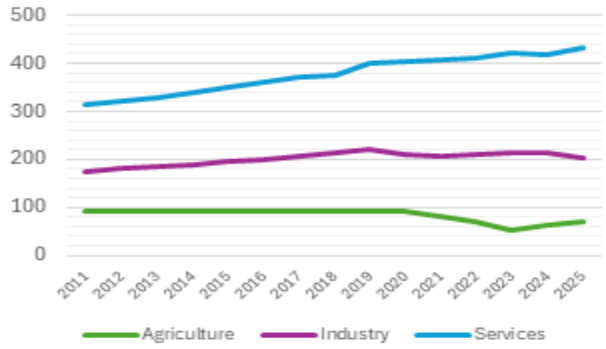
Labor-market dynamics have closely followed shifts in value added, with a notable employment reallocation to services (Figure 15). Since 2021, more than 30,000 formal workers and 25,000 informal workers have left the agricultural sector, either transitioning into services and industry, or exiting the labor force altogether. Total industrial employment remains below pre-pandemic levels, and workers have gradually moved away from traditional products like textiles and metals toward chemicals, pharmaceuticals, and automobile parts. Since

2017,[12] nearly 50,000 new jobs have been created in services, driven primarily by public administration, education, and health, followed by retail trade and information technology.

Employment remains largely concentrated in low and medium-productivity sectors, such as agriculture, manufacturing, and commerce (see Chapter 2). The structure of employment is skewed toward sectors with low value-added growth, slowing long-term income growth. In 2024, nearly two thirds of workers were employed in sectors with relatively low growth, while only about 10 percent were employed in high-value-added growth sectors. In recent years, however, a clear shift towards services (Figure 16) has been observed, with formal jobs declining in mining and stagnating in agriculture while expanding in ICT, accommodation and food, retail trade, and business services (including professional services).

Figure 15. Service sector has expanded employment most

(Employment, 000 workers, 2011-2025)



Source: State Statistical Office and World Bank staff calculations.

The largest expansion has been in the public sector, which also recorded one of the highest real wage increases despite being among the least productive. Rising public-sector wages have accounted for over a quarter of total wage growth since 2021. This growing wages and productivity gap is also observable in sectors facing rising labor shortages, including mining, construction, and manufacturing. Without further reallocation of labor toward more-productive and better-paid jobs, the economy risks becoming trapped in a lower-skill, lower-productivity, lower-income equilibrium.

Among the expanding sectors, only ICT and professional services offer wages above the national average. The commerce and manufacturing sectors, which include the automotive value chain, are North Macedonia's largest employers, and have also added jobs, but most firms in these sectors continue to pay below-average wages. High-productivity export-oriented firms are more dynamic but account for only a small share of total employment, limiting economywide spillovers. As a result, recent job creation has been concentrated where earnings potential and productivity growth remain minimal, which partly explains the growing disconnect between productivity and wages.

Figure 16. Workers are concentrated in sectors with lower value-added growth and lower wage growth

(Real value added, employment, and real wage growth, 2017-2024)

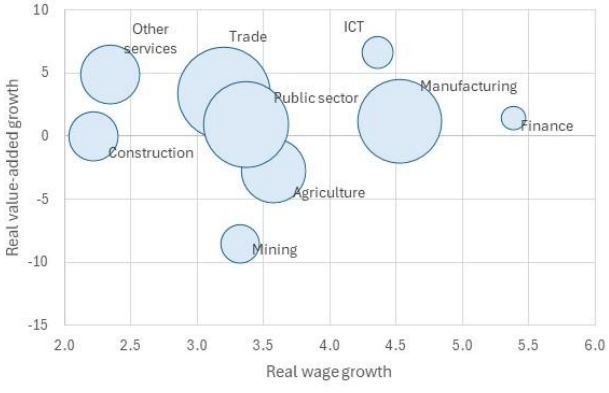
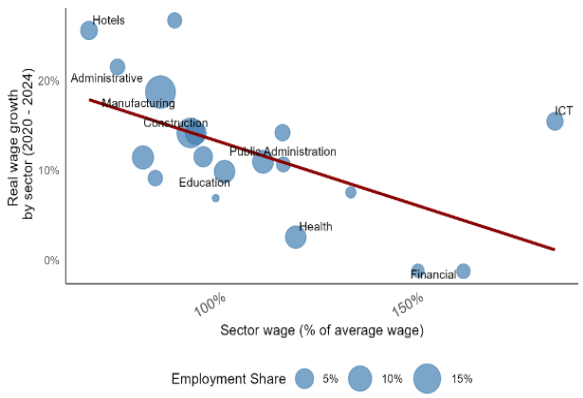


Figure 17. Real wage growth has been concentrated in sectors with relatively low levels of productivity

(Real wage growth of sectors relative to their wage, as percent of average wage)



Source: State Statistical Office and World Bank staff calculations, based on administrative wage data and LFS data. *Note:* Bubble represents the relative size of employment.

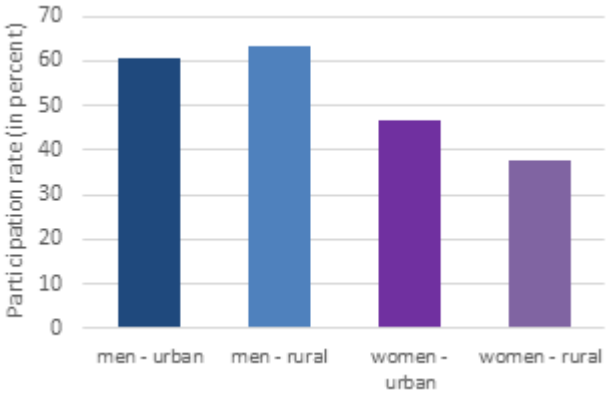
[12] The analysis uses 2017 as the last year of the historical revision of LFS data with the new population census from 2021.

Wage and productivity trends reveal growing imbalances that could erode future labor-market gains. Recent wage growth has been driven by repeated minimum-wage increases and ad-hoc public-sector wage increases, fueling the indexation of private sector wages, and as a result gains have been concentrated in lower-paying sectors (Figure 17). Meanwhile, wage growth has outpaced improvements in labor productivity, eroding competitiveness and discouraging investment in skills and innovation. State-owned enterprises and public institutions offer relatively high pay despite modest productivity, while frequent minimum-wage hikes and structural features of labor taxation have compressed the wage structure. In particular, minimum social security contribution (SSC) thresholds and minimum SSC bases make labor taxation regressive, implying higher effective tax rates for low-income earners and making formal part-time work expensive and scarce, thereby weakening work incentives at the bottom of the distribution and pushing activity into informality. At the same time, productive export-oriented firms remain constrained by global competition and weak bargaining power and continue to pay below-average wages. The resulting misalignment between productivity and wages weakens incentives both for firms and workers to upgrade skills.

The country is grappling with high rates of labor inactivity, including persistent gender gaps, especially in rural areas. In 2024, the gender participation gap (ages 15+) was 19 pp, with even wider gaps among women aged 30-34 (22 pp) and 55-64 (nearly 30 pp). After the pandemic, the female labor-force participation rate declined by over 2 pp from an all-time high of 47 percent in 2019, while men saw even a steeper decline, resulting in a modest narrowing of the gender gap. However, the disparity remained most acute in rural areas: among women aged 25-34, the participation gap with men reached 32.2 pp—three times higher than the urban gap (Figure 18). The large rural participation gap reflects greater barriers to labor-market entry and retention, including disproportionate childcare responsibilities, restrictive social norms, and limited access to good job opportunities (see Chapter 5).

Figure 18. The gender participation gap is wide

(Labor force participation rates, 2024, ages 15+)



Source: State Statistical Office and World Bank staff calculations.

Despite modest improvements, young people remain disconnected from the labor market. Labor-force participation among young people aged 15-24 stood at 26.9 percent in 2024, significantly below the EU average of 41.2 percent and slightly below the Western Balkan average of 30.9 percent. Young men in rural areas are 9.3 ppts more active than their urban peers, while the urban-rural gap for young women is much smaller at just 4 ppts. While the rate of young people not in employment, education, or training (NEET) declined from 24.5 percent (25.6 percent for women) in 2017 to 18.2 percent (17 percent for women) in 2024, it remains more than double the EU average of 9.2 percent. At the same time, education quality is deteriorating, as reflected in declining 2022 Program for International Student Assessment (PISA) scores, and nearly 30 percent of young adults perceive education as having little or no relevance to their jobs.[13]

Over the coming decades, a shrinking workforce is expected to further reduce the demographic dividend to growth. The dependent population's share of the total population is rising—the share of population 65+ grew from 13.3 percent in 2017 to 18.1 percent in 2024 (Figure 19). By 2070, the working-age population is projected to fall by more than half, approaching parity with the dependent population (Figure 20), which will further increase pressure on pension schemes, healthcare services, and social support programs. This demographic shift is compounded by outward migration, primarily for better job opportunities and particularly among individuals in their most-productive years. In the latest Balkan Barometer survey, over half of respondents expressed

[13] The figures are from the Labor Force Survey's ad hoc youth module for 2024.

interest in relocating to the EU, while more than a quarter indicated they would be willing to work elsewhere in the Western Balkans or in the United States.

Figure 19. The share of the dependent population is increasing…

(Demographic change, 2017-2024)

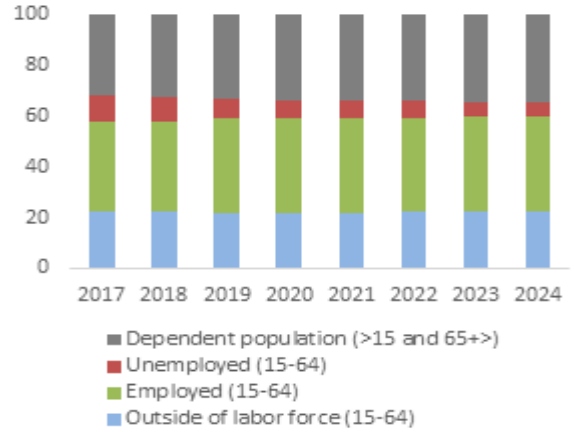
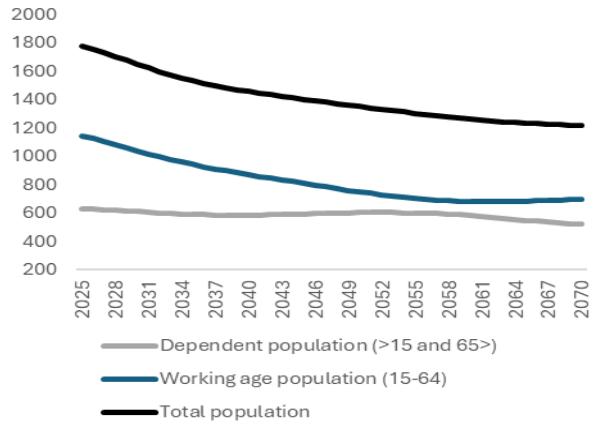


Figure 20. …and is projected to nearly match the working-age population by the middle of this century.

(Demographic projections, in 000s, 2025-2070)



Source: State Statistical Office and World Bank staff calculations.

The underutilization of the working-age population amid an accelerating demographic decline is becoming a critical drag on the country's growth prospects. Recent increases in labor costs, spurred by minimum-wage hikes and rising public-sector wages, have not been matched by productivity gains. This growing disconnect between productivity and wages threatens to erode the economy's competitiveness, fuel wage-price inflation, and further discourage private investment.

1.4. Achieving high-income status while creating better-paying jobs

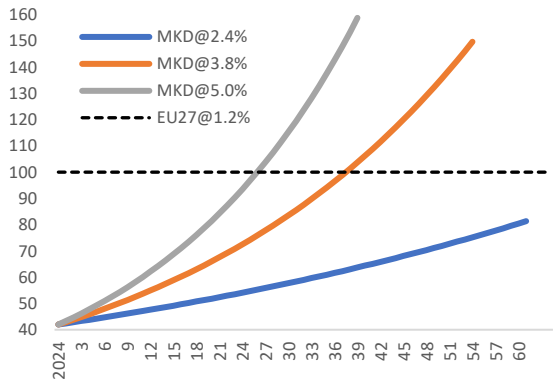
Can North Macedonia close the gap with high-income countries with the current pace of reforms? To answer this, the report simulates long-term growth prospects through 2050 using MFMOD Stand-Alone (MFMOD-SA), a macrostructural model calibrated for the country. Reflecting current reform efforts, the baseline (business-as-usual) scenario takes into account the historical average total factor productivity (TFP) growth rate of 0.3 percent, an investment-to-GDP ratio of approximately 26 percent, and a gradual deceleration of the working-age population of -0.2 percent on average per year. Under these assumptions, growth is projected to settle at the average recorded over the last decade by 2050.

With baseline growth, North Macedonia's GDP per capita (at PPS) is projected to reach only 55 percent of the EU average by 2050. Capacity constraints, diminishing rates of human and physical capital accumulation, a low TFP growth rate, and an aging population are expected to drive this trend. Without stronger reforms, slowing growth would constrain the country's potential to achieve structural transformation and converge with EU living standards. A 3.8 percent growth, reflecting the simulated reforms proposed in the report, would enable North Macedonia's economy to converge with the EU average by 2060, while even more ambitious 5-percent average growth rate could enable full convergence by 2050 (Figure 21).

Reaching 3.8 percent compound annual growth rate for upper-middle-income country (UMIC) like North Macedonia is not implausible. Yet, it will require consistent reform efforts to reach the top third of historically best performing UMIC instead of being at the bottom one third with the current growth rate. The largest gains should come from boosting TFP by 2.4 pp on average over the period of next 25 years (Figure 22), with private capital contribution of 1.1 pp over the period. Equally important, although with smaller contributions, are improvements in the quality of human capital and a rise in labor force participation.

Figure 21. Convergence scenarios

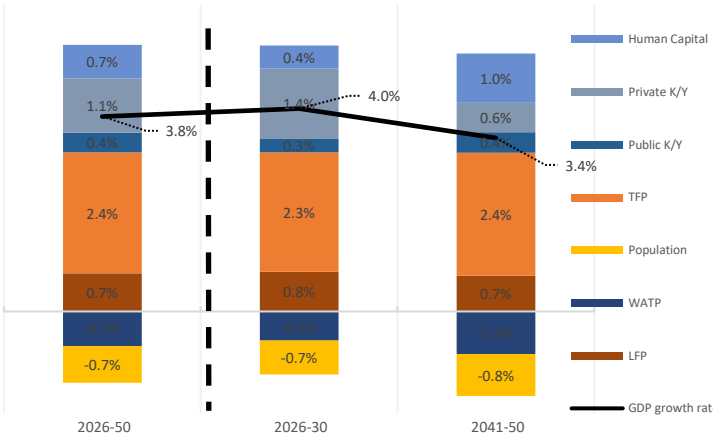
(North Macedonia and EU, GDP per capita in PPS prices)



Source: State Statistical Office and World Bank staff calculations.
Notes: GDP per capita in PPS current prices.

Figure 22. Projections of ambitious convergence scenario

(Contributions of drivers to GDP growth, pp)



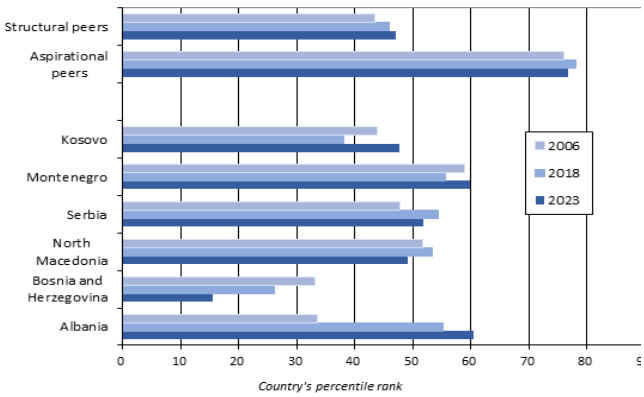
Source: State Statistical Office and World Bank staff calculations. *Notes:* Simulations were made with the Long-Term Growth Model with human capital extension, based on Steven Pennings, Arthur Mendes and Jorge Luis Guzman, http://www.worldbank.org/LTGM

Such a reform package would have important impact on labor markets. It would lead to a rise in real wages by 24.6 percent above the baseline, with employment being 7.2 percent above the baseline projection (Chapter 6 provides more details). These percentages would translate into over 53,300 new additional jobs created above the baseline projection, with over 196,700 jobs being better paid.

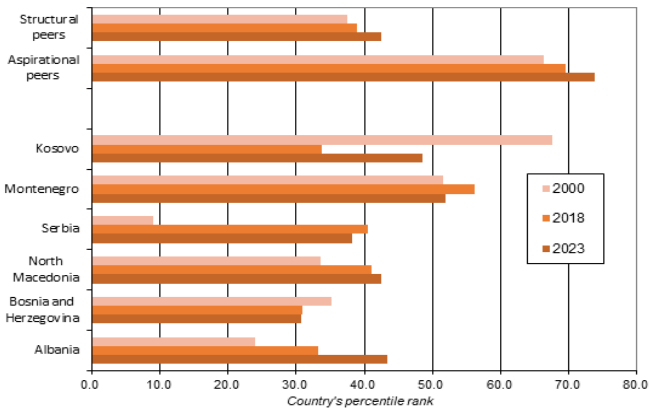
North Macedonia's most binding constraint for accelerating convergence will be governance, the rule of law and fight against corruption. The gap to the aspirational peers on these metrics is wide—over 25-30 percentage points (Figure 23). In the five years preceding EU accession, aspirational peers recorded average growth rates of around 5 percent, lifted by robust structural reform efforts aimed at aligning national institutions, rule of law, control of corruption, and regulations with EU standards. Those on the other hand were critical to boost productivity, attract investment, and narrow the income gap with peer EU member states. Using a stochastic frontier approach, improving government effectiveness (WGI) to the average level of aspirational peers (Croatia, Estonia, Latvia, Lithuania, Slovakia, and Slovenia) within 15 years would correspond to an average annual increase in TFP growth of around 0.56 percentage points.

Figure 23. Governance challenges need to be addressed to unlock investment and growth

a. Government Effectiveness



b. Control of Corruption



Source: World Bank World Governance Indicators.

Institutions are critical for transitioning the economy to high-income status: for (i) completing the transition to a functioning market economy, (ii) revamping energy and climate resilience and improving energy efficiency, (iii)

improving the education system and labor market outcomes, and (iv) improving the innovation and technology adoption. Corruption directly diminishes the growth dividend by diverting long-term productive investments. Addressing these constraints is not merely a matter of closing structural gaps; it is the precondition for unlocking the private investment that would put North Macedonia on a materially faster convergence path.

Chapter 2. Strengthening firm productivity

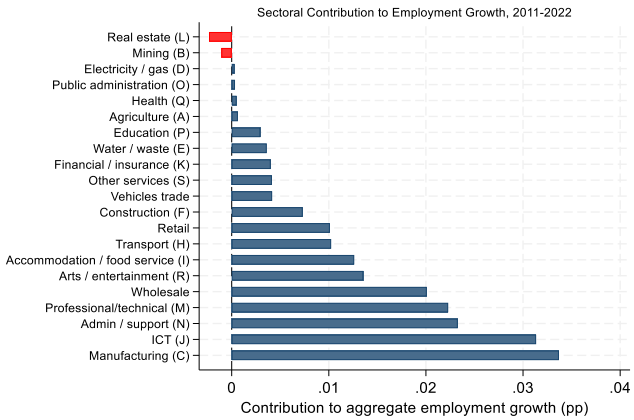
North Macedonia's economy has undergone a structural shift toward higher-value services — notably ICT — and technology-intensive manufacturing, yet it continues to lag behind aspirational peers in firm productivity. The productivity gap is driven less by a lack of capital than by deep inefficiencies: resources are misallocated away from the most productive firms, new entrants replace rather than upgrade existing businesses ("movement without transformation"), and foreign-owned firms — the clear productivity leaders — generate limited broader spillovers due to the weak absorptive capacity of domestic firms. The business environment compounds these challenges through widespread informality, corruption, regulatory uncertainty, and operational barriers that constrain both domestic entrepreneurship and FDI. Simulation results suggest that the greatest productivity gains would come from tackling informal sector competition, political instability, and infrastructure deficiencies — pointing to the conclusion that closing the gap with the EU requires not more capital, but stronger market efficiency, institutional reform, and deliberate investment in the management and workforce capabilities of domestic firms. Such persistent distortions directly result in weaker employment outcomes as discussed above.

2.1. Patterns of productivity growth and employment dynamics

North Macedonia entered the reform era with the right building blocks. Apart from attracting investment and integrating into global markets, the economy was also restructuring its workforce toward higher-value services and technology-intensive activities and away from primary sectors.[14] ICT emerged as the second largest contributor to aggregate employment growth between 2011 and 2022[15] (Figure 24), a notable development given its small base (about 3 percent in 2011), signaling that the sector is not only growing rapidly but beginning to punch above its weight in the economy. Professional and technical services and administrative and support services also featured among the top contributors, suggesting a broadening of the economy's service base toward activities with higher skill content.

Figure 24. Employment has shifted toward services and technology…

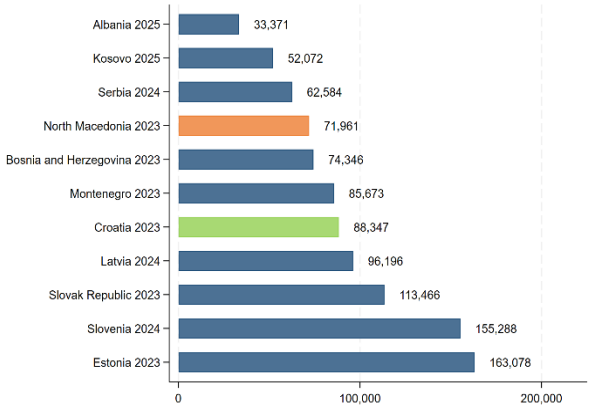
(Contribution to employment growth by sector, 2011-2022)



Source: ICA 2.0 Toolkit based on national business registry data.
Note: Overall employment growth between 2011 and 2022 is 0.20 or 20.2%. Contribution = base year sectoral employment share x sectoral employment growth.

Figure 25. …but structural change has had a very limited effect on productivity.

(Sales per worker by select country)



Source: ICA 2.0 Toolkit based on WB Enterprise Survey.
Notes: Most recent available survey was shown for each country. Average sales per worker is weighted by sampling weights. Sales per worker is in 2009 USD.

Yet, the picture is not one of unambiguous structural upgrading. Manufacturing remained the single largest contributor to employment growth, much of it in relatively labor-intensive automotive components, rather than

[14] Since 2018, however, all other sectors contracted, except ICT and other services (see Figure 88 in Annex 3).
[15] The report relies on the firm-level data from the national business registry obtained for 2011-2022 period only that covers all formal firms.

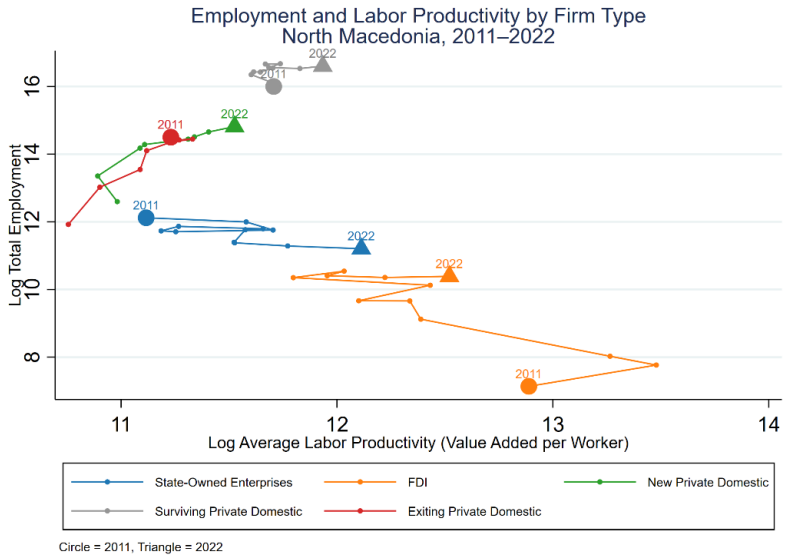
a decisive move up the value chain. Meanwhile, sectors such as real estate and mining contracted their employment shares, contributing negatively to aggregate growth. Overall, the pattern points to an economy in transition where the seeds of a more knowledge-intensive services sector are visible in the strong contributions of ICT and professional services, but the center of gravity in employment creation remains in manufacturing and lower-value activities.

Firm productivity remains lower than with aspirational peers.[16] Despite the structural change towards higher-value and technology-driven sectors, it takes 1.2 Macedonian workers to produce the same output as one Croatian[17] worker (Figure 25). Moreover, the correlation between changes in labor shares across sectors[18] and initial value added per worker over 2011–2022 is positive but close to zero, suggesting that labor reallocation across these sectors has contributed little to productivity growth.

The reallocation of resources toward the most productive activities remains incomplete. Figure 26 traces employment and labor productivity trajectories across firm types between 2011 and 2022. Foreign-owned firms were already the productivity leaders at the start of the period and expanded employment significantly by 2022, though their average productivity edged down as more labor-intensive operations scaled up. A striking pattern emerges among domestic firms: survivors exhibit systematically higher labor productivity than entrants and exiting firms, suggesting that a productivity threshold governs which firms endure. Yet, despite this selection, surviving private domestic firms, the economy's largest employers, made only modest gains on both employment and productivity dimensions, indicating that survival alone does not drive meaningful upgrading.

State-owned enterprises have been the least productive firms in the economy. While they shed employment over the period, this contraction may have mechanically lifted their average labor productivity by concentrating activity among fewer, relatively better-performing workers, a gain driven by shrinkage rather than technical upgrading. Exiting firms, which carried low productivity and a sizeable employment share in 2011, were replaced by new entrants, but ones that are not substantially more productive than their predecessors and not entering at sufficient scale to shift aggregate outcomes. This gap between structural change and productive reallocation explains why employment shifts have contributed so little to closing the productivity gap with aspirational peers.

Figure 26. Productivity gains are driven by foreign-owned firms, while domestic firm dynamics remain limited



Source: ICA 2.0 Toolkit based on national business registry data.

[16] The estimates rely on WBES data, and the same pattern emerges when productivity is measured using GDP per worker from the Penn World Tables. This consistency indicates that the findings are not driven by data source or measurement choice but reflect broader productivity dynamics.

[17] An aspirational peer and the EU member state for which firm-level data were obtained.

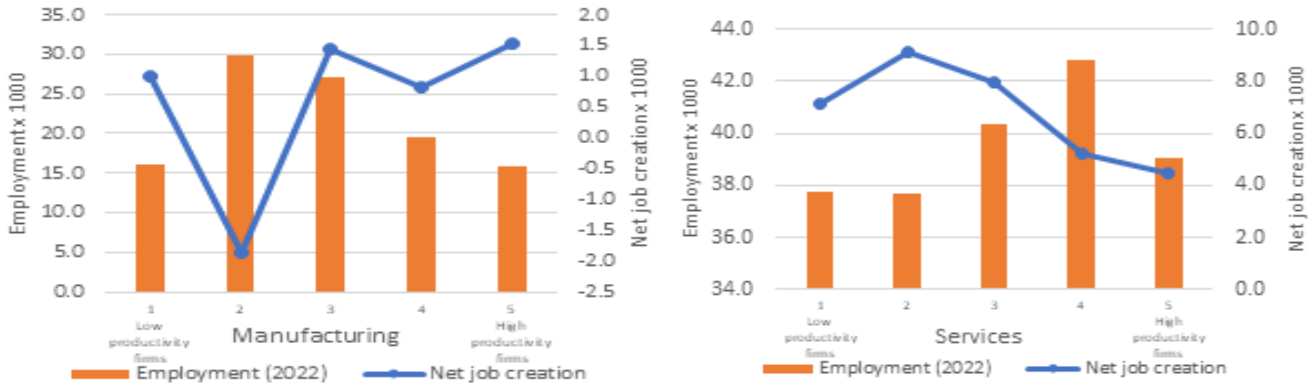
[18] NACE 2-digit.

Notes: In the absence of observed entry (exit) year, firms are assumed to have entered (exited) the year they were first (last) observed in 2014 (2019), 3 years after (before) the start (end) of the dataset.

The disconnect between where jobs are being created and where productivity is highest carries through to the sectoral level. In 2022, high productivity manufacturing firms in each narrowly defined industry accounted for only about 16 percent of the sector's total employment, despite posting positive employment growth (Figure 27a). In services, jobs are more concentrated in high productivity firms. High productivity service firms account for nearly 40 percent of the sector's total employment but show modest employment growth (Figure 27b). The pattern is consistent: neither employment nor job creation is systematically flowing to the most productive firms.

Figure 27. Employment and net job creation within industry tend to be low with the most productive firms

(Employment and net job creation by within-industry TFPR quantile, 2018-2022)

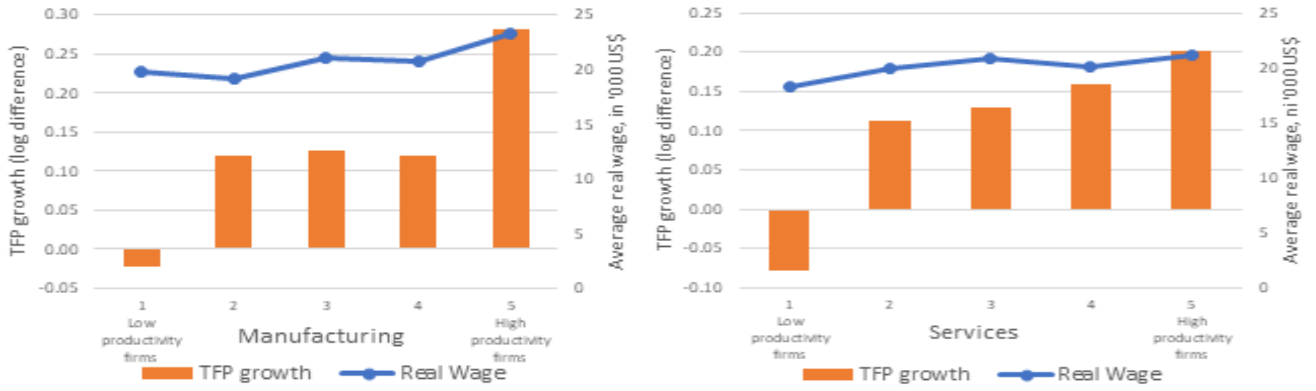


Source: ICA 2.0 Toolkit based on national business registry data. *Notes*: TFPR quantile is calculated within 2-digit NACE. Net job creation follows Davis and Haltiwanger (1992) methodology. Employment level is in 2022, while net job creation is calculated between 2018 and 2022. For entrants (first observed year), previous employment is treated as 0.

High-productivity firms offer better paid jobs. Although more-productive firms account for a smaller share of employment than less-productive counterparts, they tend to pay higher wages. From 2018 to 2022, the most-productive firms in manufacturing and services paid wages that were 17 and 15 percent higher than those paid by the least-productive services firms, respectively (Figure 28). This means that when workers do reach productive firms, they benefit; but because those firms are not expanding their employment share, the wage premium is not being broadly transmitted across the economy.

Figure 28. More-productive firms tend to offer higher wages

(TFP growth and the average level of wage within-industry TFPR quintile, 2018-2022)



Source: ICA 2.0 Toolkit based on national business registry data. *Notes*: TFPR quantile is calculated within 2-digit NACE. Wage rate is in 2022, while TFP is calculated between 2018 and 2022. Both are in terms of average, weighted by employment.

Despite comparable firm dynamism relative to an aspirational benchmark, the entrepreneurial base appears limited, and what dynamism exists is not generating productive renewal. In 2024, both total and new business densities were substantially lower relative to Croatia (not the best performing EU peer), suggesting fewer new

and existing businesses to accommodate its working population (Figure 29). Entry and exit rates have been declining since 2021, indicating decelerating firm dynamism. This matters because, as Figure 26 shows, new entrants are the most employment-dynamic group, yet they enter at productivity levels no higher than the firms they replace. The nearly identical productivity distribution of entering and exiting firms confirms that more-productive firms are not systematically displacing less-productive ones (Figure 30). Survivors do exhibit higher productivity, suggesting a selection threshold governs which firms endure; however, when the pool of entrants is shallow and their scale is too small to shift aggregate outcomes, this selection process alone cannot drive meaningful upgrading. The result is an entrepreneurial ecosystem that generates movement without transformation.

Figure 29. Despite comparable entry rates with the aspirational peers…

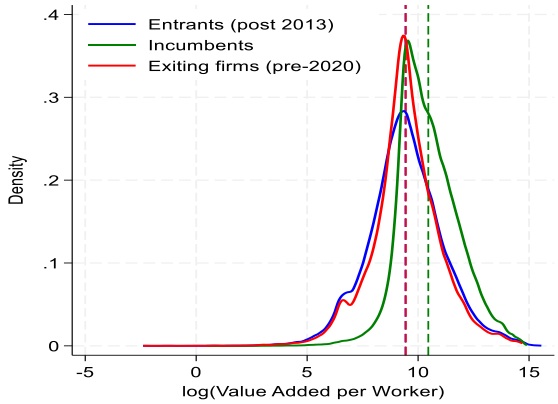
(Firm entry and exit rates, 2024)



Source: ICA 2.0 Toolkit based on Entrepreneurship database[19].
Notes: Entry rate is calculated as percentage share new firms to total number of firms. The total business density is defined as the total number of registered firms per 10,000 working-age people (those ages 15–64); new business density is the number of new businesses per 1,000 working-age people. The units of measurement are private companies with limited liability.

Figure 30. …entrepreneurial depth and productivity reallocation remain limited.

(Distribution of value added per worker across entering, exiting, and incumbent firms, 2011-2022)



Source: ICA 2.0 Toolkit based on national business registry data.
Note: In the absence of observed entry (exit) year, firms are assumed to have entered (exited) the year they were first (last) observed in the data. Firms observed for the first time in 2013 are classified as entrants, while firms last observed in 2020 are classified as exiters.

Capital accumulation is necessary but not sufficient to close the gap; it should be complemented by improvements in efficiency. North Macedonia's capital-to-labor ratio stands at roughly seven times lower than Croatia's (Figure 31), a gap that is mirrored almost exactly in terms of sales per worker, both measured in PPP terms. North Macedonia's capital-labor ratio shortfall relative to EU peer is real and substantial, but they are unlikely to deliver sustained productivity convergence unless accompanied by complementary reforms that improve how capital is used. Within North Macedonia, approximately 53 percent of the labor productivity gap between the typical Macedonian firm and top-performing ones (above 90th percentile) is explained by differences in capital per worker (Figure 32).[20] The remaining 47 percent of the gap reflects factors other than capital deepening, such as TFP, managerial quality, and workforce skills that enable technology adoption and learning. These capabilities can influence whether firms benefit from spillovers and upgrading. Together, the

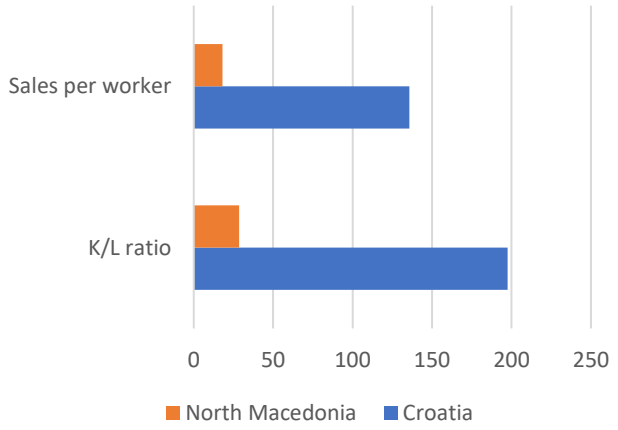
[19] The data from State Statistical Office on the structure of enterprise births by legal form show that the majority were enterprises categorized according to their legal form as limited liability companies and joint stock companies (84.8 percent).

[20] Using firm-level data from North Macedonia, we assess how much of the dispersion in labor productivity across firms is explained by differences in capital per worker. We first estimate a log–log regression of labor productivity on capital per worker, controlling for 2-digit NACE industry, year, size classification, and ownership effects, and obtain an elasticity of 0.467, implying that a 10-percent increase in capital per worker raises labor productivity by about 4.67 percent. We then compare each firm's productivity to that of top-performing firms within the country (top decile) and compute the share of the productivity gap that is mechanically explained by lower capital per worker.

evidence suggests that productivity convergence will require improvements in efficiency and firm capabilities, not just further capital accumulation.

Figure 31. Productivity convergence may require policies that boost firm efficiency and capabilities…

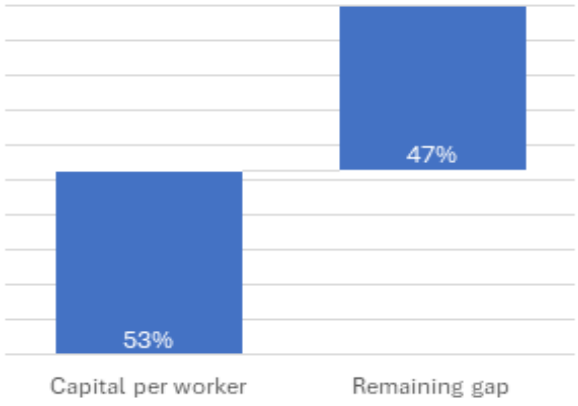
(Sales per worker and capital-labor (K/L) ratio, North Macedonia vs Croatia, 2023)



Source: ICA 2.0 Toolkit based on WB Enterprise Survey
Note: sales per worker is in 10,000 and K/L is in 1,000 PPP terms and in terms of number of permanent, full-time employees at the end of the fiscal year. Capital is calculated as the sum of annual expenditure on equipment and on land and buildings, converted in PPP terms in 2023 using data from WDI. Sales per worker and K/L ratio are country averages, weighted by sampling weights.

Figure 32.…not just more capital investment.

(Distribution of labor productivity, 2011-2022)



Source: ICA 2.0 Tookit based on North Macedonia's Central Business Registry data. *Notes:* labor productivity is in log-transformed value added per worker.

Markets in North Macedonia are not rewarding the most productive firms, and this has been the case for nearly a decade. The earlier evidence on firm entry, survival, and employment dynamics pointed to a market that was not effectively channeling resources toward the most productive firms. The productivity decomposition (Figure 33) shows that in the years leading up to 2016, aggregate productivity, measured by value-added weighted TFP, grew modestly and markets show early signs of working as intended: more productive firms were winning market share. Between 2017 and 2020, aggregate productivity fell sharply, with deterioration visible across every dimension: firms became less efficient internally, market shares shifted toward less productive firms, and firm entry and exit dynamics offered no offsetting benefit. The COVID-19 period likely amplified these pressures, but the underlying trends predate the pandemic. The most recent period (2021–2022) shows that firms made meaningful internal productivity improvements through upgrades, but these gains were effectively cancelled out by continued misallocation of market shares. In other words, productive firms got better but they just did not grow. The market mechanism that should reward efficiency with expansion was not functioning, and the result was aggregate productivity growth decline despite firm-level progress.[21]

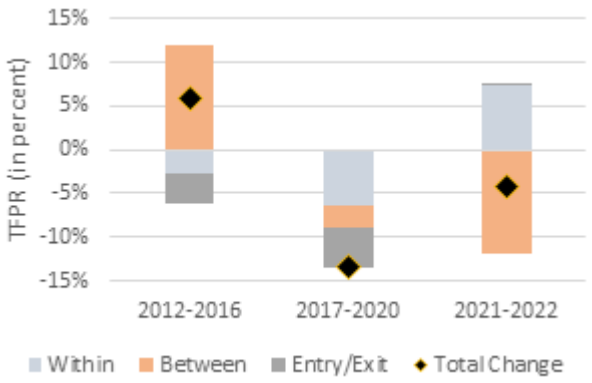
Misallocation, then, is the symptom; and it is persistent, measurable, and worsening. The standard deviation of log revenue-based TFP (TFPR) remains consistently high over the sample period, fluctuating between 1.36 and 1.43 (Figure 34).[22] To put this in perspective, in a well-functioning market, competition should drive revenue productivity toward equalization across firms: more efficient firms expand, inefficient ones shrink or exit, and the dispersion of TFPR narrows over time. A persistently high standard deviation signals the opposite, that firms operating in the same sector face very different returns to their inputs, not because of genuine differences in capability, but because distortions may be present in the business environment that prevent the competitive

[21] More recent trends, consistent with the aggregate picture presented in Chapter 1, suggest some tentative signs of improvement. Unfortunately, the micro-level data used in this report only runs through 2022, making it impossible to validate whether these more recent developments have translated into actual firm-level gains.
[22] For context, comparable studies find TFPR standard deviations of around 0.49 for the United States, 0.63 for China, and 0.67 for India (Hsieh and Klenow, 2009) and between 0.97 and 0.75 for Turkey (Nguyen, Taskin and Yilmaz, 2016).

process from running its course. In North Macedonia, this dispersion has not narrowed; if anything, it has edged upward since 2016, suggesting that the distortions are deepening rather than dissipating. The drivers driving this pattern, such as regulatory burdens that protect the incumbents, insufficient infrastructure and facilities to nurture young firms to grow, and institutional inefficiencies, are areas that will be revisited later in this subsection.

Figure 33. More recent productivity gains arise from within-firm improvements rather than from the expanding market share of more-productive firms…

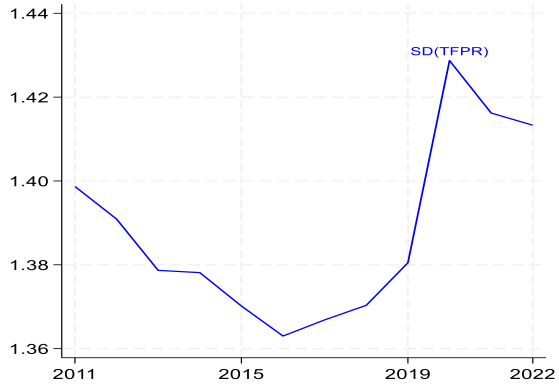
(Decomposition of productivity growth)



Source: ICA 2.0 toolkit based on national business registry data.
Notes: In the absence of observed entry and exit dates, the productivity decomposition based on Melitz and Polanec (2015) assumes all firms observed in the beginning and end year are the surviving firms, and entering and exiting firms are those only observed in the end and beginning year, respectively. TFPR = revenue based TFP growth in percent. Real value added is used as weights in calculating aggregate productivity.

Figure 34.…likely indicating deterioration of allocation of resources among firms.

(Standard deviation of TFPR)



Source: World Bank Staff calculation based on national business registry data.
Note: The standard deviation of revenue-based total factor productivity (TFPR) is derived from all firms in the sample.

The consequences of misallocation extend beyond productivity: it suppresses entrepreneurial dynamism and job creation. A simple binned scatter analysis shows a clear negative relationship between misallocation, and rates of firm entry and job creation (Figure 35 and Figure 36). This pattern implies that inefficient resource allocation undermines the ability of new, productive firms to enter the market and generate employment.

Figure 35. High levels of resource misallocation are associated with low rates of firm entry…

(Firm entry)

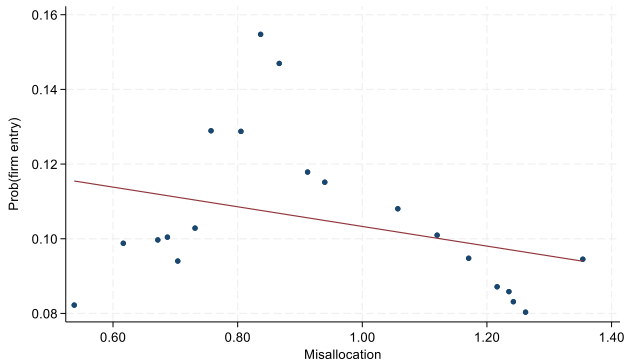
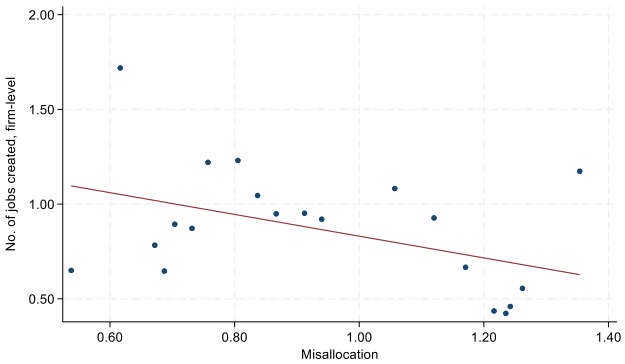


Figure 36.…and low level of job creation.

(Job creation)



Source: ICA 2.0 toolkit based on national business registry data. *Note:* Misallocation is calculated as the standard deviation of log-transformed revenue-based total factor productivity (TFPR) at the 2-digit NACE level. Binned scatter plots include year-fixed effects.

2.2. FDI can boost productivity, but only when local firms are ready to absorb it

North Macedonia's FDI trajectory reflects both its successes and emerging challenges in leveraging foreign investment for growth. After initially lagging upper-middle income peers, inflows surged in the early 2000s, peaking at 12 percent of GDP (Figure 37). Driven by major investment deals in energy and utilities and export-oriented manufacturing, this gave the country a clear FDI advantage over upper-middle-income peers. FDI inflows, however, have moderated since, and the FDI stock, which captures the accumulated presence of foreign investors, has gradually converged with peer levels (Figure 38), eroding some of the early 2000s edge the country had secured.

Figure 37. North Macedonia's FDI stock remains strong…

(FDI inflows as share of GDP, percent)

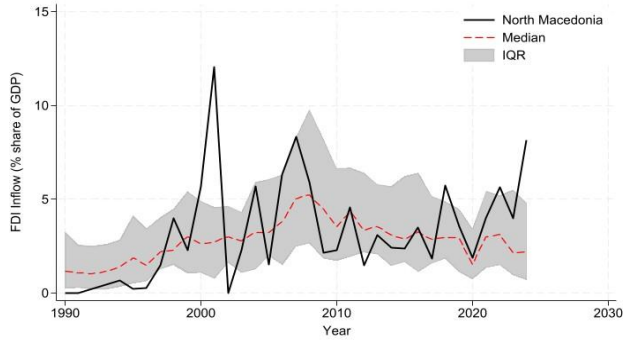
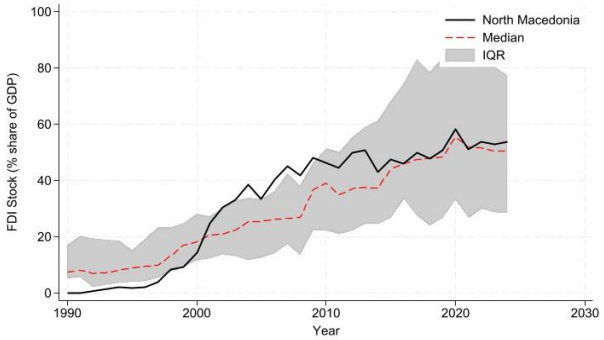


Figure 38. …but inflows are leveling, signaling a need to sustain its edge.

(FDI inward stock as share of GDP, percent)



Source: Invest_Data online tool based on UNCTAD data. *Notes:* The median and interquartile range are based on the World Bank's FY2025 definition of UMICs.

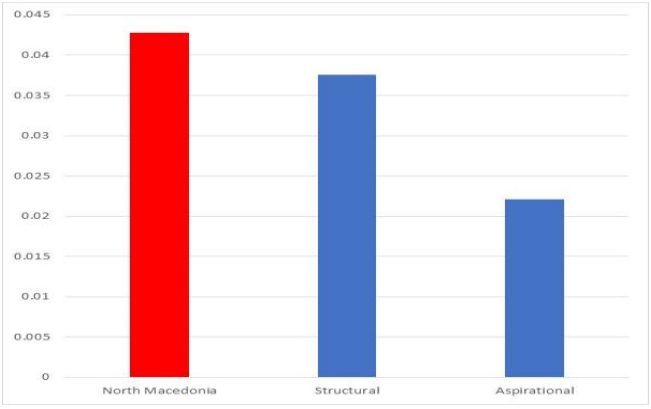
Yet, beneath this broadly open headline investment climate, the regulatory environment for foreign investors is more restrictive than it appears. North Macedonia's FDI Restrictiveness Index (FDIRI) stood at almost twice as much as the average of the aspirational peers (Figure 39). The restrictions driving this gap are not the most visible ones; such that screening and approval mechanisms and restrictions on key foreign personnel are at or near zero, suggesting that the formal gatekeeping of foreign investment is limited. Instead, the burden falls on operational constraints that are less monitored (i.e., "other restrictions") but no less consequential, such as discriminatory government procurement rules, along with land and real estate ownership, limits on profit repatriation, and barriers to establishing branches (Figure 40). These are the kinds of behind-the-scenes frictions that do not show up in headline rankings but accumulate into a meaningfully more difficult environment for foreign investors seeking to establish and expand operations. Combined with the moderating trend in FDI inflows documented earlier, this suggests that sustaining foreign investment will require addressing not just formal entry barriers but the operational environment that foreign firms encounter once they are in the country.

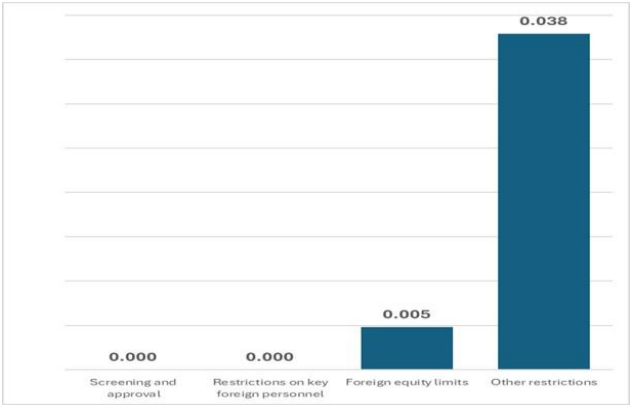
Figure 39. North Macedonia's FDI regulation is relatively more restricted…

(FDI Restrictiveness Index)

Figure 40. …driven by restrictions on land ownership, profit repatriation, and discriminatory government procurement rules.

(FDI inward stock as share of GDP, percent)



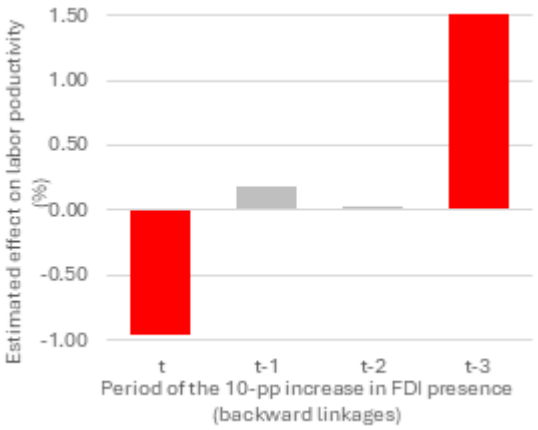


Source: ICA 2.0 toolkit based on OECD's FDI Restrictiveness Index. *Notes:* FDIRI for structural and aspirational peers are unweighted averages. The FDIRI measures how restrictive a country's laws and regulations across 22 economic sectors are toward FDI. Each measure is scored from 0 (fully open) to 1 (fully closed), with sectoral scores aggregated into a composite index. A lower score indicates a more investment-friendly regulatory environment. The FDIRRI captures four main types of FDI restrictions: foreign equity limits, screening and approval mechanisms, restrictions on key foreign personnel, and 'other restrictions' faced by foreign investors, such as restrictions on the acquisition of land and real estate for business purposes. See OECD (2024) FDIRI Methodological Framework for details.

FDI in North Macedonia appears to generate positive productivity spillovers. The estimated spillover effect on domestic firms in the supplying sectors via backward linkages is around 1.5 percent rise in labor productivity. This pattern contrasts those of regional comparator countries in the region, where FDI spillovers are often limited.[23] However, these effects tend to emerge with a lag of about three years following the entry or expansion of multinational firms that are backward-linked to local suppliers (Figure 41). These delayed but tangible gains suggest that local firms require time to absorb new technologies, adapt business practices, and build stronger supplier relationships with multinationals before productivity benefits fully materialize.

Figure 41. In North Macedonia, FDI boosts productivity but with a lag …

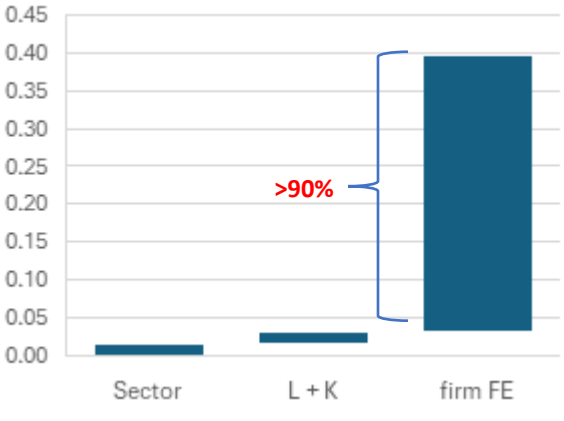
(Estimated spillovers effect on domestic firms in the "supplying" sectors, 2011-22)



Source: ICA 2.0 Toolkit based on Central Registry data.
Notes: The plot shows the estimated effect of a 10-percentage point increase in the average share of FDI's output in downstream "buying" sectors (currently 25.85% in the sample) on labor productivity of domestic firms in the "supplying" sectors, following the methodology developed by Javorcik

Figure 42.…as local firms take time to build capacity, absorb technologies and build stronger links with multinationals.

(Contribution of each factor to explained variation behind higher-than-predicted labor productivity)



Source: ICA 2.0 Toolkit based on Central Registry data.
Notes: The figure shows how much of the variation in firms' higher-than-expected productivity growth over the 3-year horizon can be explained by progressively richer sets of factors. The adjusted R-squared, which measures how much of the variation each model accounts for, remains extremely low across all three specifications,

[23] Iacovone et al. (2025). "TIDES of Change: Igniting Productivity Growth in Europe and Central Asia". *Europe and Central Asia Studies*. Washington, DC: World Bank.

(2004). Red bars denote statistical significance at the 5% level. Foreign ownership is proxied by locators in economic zones.

even after controlling for sector conditions (Sector), capital, and labor inputs (L + K). It is only when firm fixed effects (firm FE) are added in the final step that the R-squared jumps substantially, meaning that the overwhelming share of what drives above-expected productivity growth is captured by stable, unobserved firm-specific characteristics.

Not all domestic firms benefit equally from foreign investment, and the difference comes down to what is inside the firm. Figure 42 shows that more than 90 percent of the variation in why some firms achieve higher-than-expected labor productivity growth cannot be explained by observable factors like sector conditions, capital, or workforce size. What explains it instead are firm-specific strengths that do not show up in standard data, such as the quality of management, the skills and experience of the workforce, and the organizational capacity to absorb and apply new knowledge. In practical terms, this means that FDI spillovers do not spread evenly across the economy. They accrue disproportionately to firms that are already well-positioned to learn from their multinational partners. Firms that lack these internal capabilities are likely to remain on the sidelines, even as foreign investors expand around them. This has a clear policy implication. Broadening the benefits of FDI requires investing in the absorptive capacity of domestic firms through management support, skills upgrading, quality certification, and supplier development programs that build the internal capabilities needed to turn foreign presence into genuine productivity gains.

2.3. Misallocation is the symptom; weak governance is the underlying disease

North Macedonia's business environment is relatively more constrained than of its peers across several business environment indicators. In 8 out of 15 obstacles measured under the Enterprise Survey, North Macedonia reports higher shares of firms identifying constraints than both structural and aspirational peers (Figure 43). The gaps are particularly stark in corruption, bribery incidence and informal payments (11 percent compared to 4.3 percent for aspirational groups), business licensing, and crime, theft and disorder. These are not mere inconveniences; rather, they are the distortions that prevent markets from allocating resources to their most productive uses. When regulations are applied unevenly, informality goes largely unchecked, and political uncertainty clouds the investment horizon, the competitive signals that should reward productive firms and discipline underperforming ones are blunted. Firms that should expand cannot; firms that should exit persist. The result is precisely the pattern documented earlier in this chapter: productive firms are constrained below their efficient scale, market shares misaligned with efficiency, and aggregate productivity gains that fall short of what the economy's structural transformation would otherwise promise.

Figure 43. North Macedonia's business environment is broadly more constrained than of its peers…

(Percent of firms reporting the biggest obstacle they face)

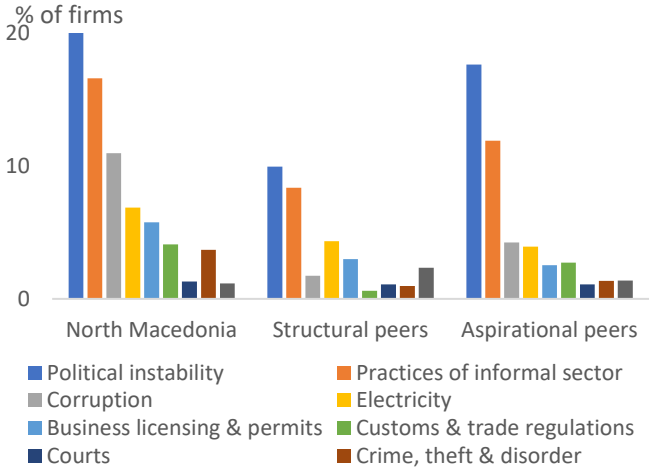
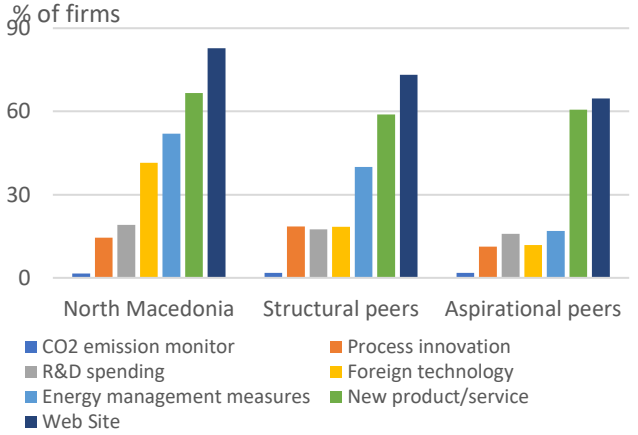


Figure 44.…yet, firms have appetite for modernization and innovation.

(Percent of firms reporting investment measures)



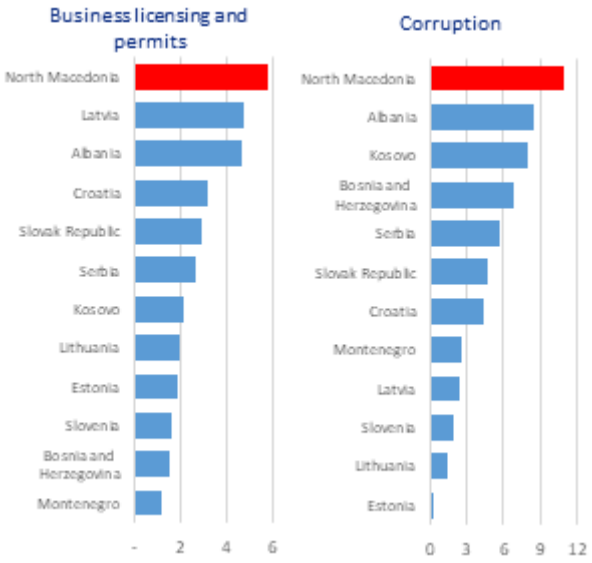
Source: World Bank Enterprise Survey, 2023. *Notes*: Averages for structural and aspirational peers are computed as the unweighted average of country-level weighted averages. Structural peers (survey year): Albania (2025), Bosnia & Herzegovina (2023), Georgia (2023), Armenia (2024), Moldova (2024) and Serbia (2024). Aspirational peers: Croatia (2023), Estonia (2023), Latvia (2024), Lithuania (2019), Slovak Republic (2023), and Slovenia (2024).

The good news is that its firms show a genuine appetite for productivity upgrading, even against the backdrop of a constrained business environment. More than four in five firms maintain a website, and over half have adopted energy management measures, rates that surpass both peer groups and signal a private sector willing to invest in its own competitiveness (Figure 44). Similarly, the high uptake of foreign technology and investment in research and development suggests that firms are actively seeking to upgrade and catch up. Yet this dynamism has its limits. Process innovation and environmental monitoring, the kinds of activities that drive sustained productivity growth and long-term competitiveness, remains weak and lag structural peers.

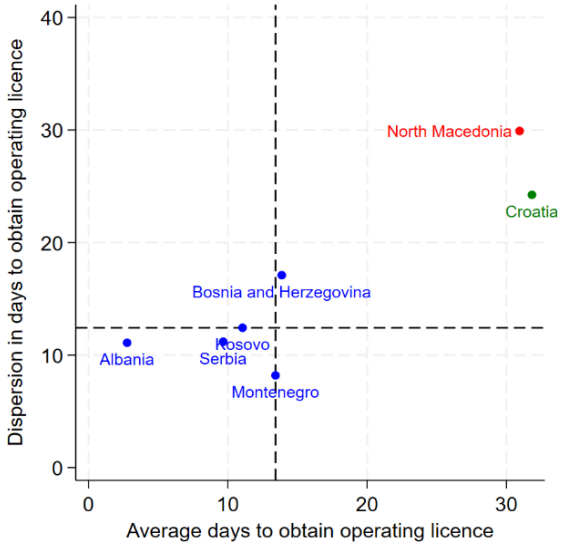
Regulatory and administrative barriers increase costs and uncertainty. A growing number of firms report that delays and administrative burdens hinder investment and expansion (Figure 45a), as evidenced by both significantly long approval times and high dispersion in obtaining operating licenses in the country relative to others in the region (Figure 45**Error! Reference source not found.**b), indicating not only delays but also significant regulatory uncertainty across firms. Restrictive sectoral regulations (notably rail transport), complex licensing procedures, and the absence of "silence-is-consent" principle[24] increase costs and uncertainty for firms.

Figure 45. Many firms face investment and expansion constraints due to lengthy, inconsistent licensing and high regulatory uncertainty

a. Shares of firms citing business regulations and corruption as a major obstacle to business operations



b. Standard deviation and mean for time to obtain business licensing and permits, Western Balkans and Croatia



Source: ICA 2.0 Toolkit based on WB Enterprise survey. *Notes*: For WBES, the latest WBES survey year was used. Dispersion is measured by the standard deviation.

Among the business climate dimensions where North Macedonia lags Croatia[25], power outages and political instability show the strongest negative association with firm-level labor productivity. Regression estimates confirm that power outages and political instability are among the most damaging constraints on firm productivity (Figure 48), and North Macedonia lags Croatia on both. If these gaps were closed, a simulation based on estimated elasticities show productivity gains of 3.2 and 3.3 percent, respectively (Figure 49). Competition from the informal sector compounds this, with a simulated gain of 4.3 percent, the largest single lever. Notably,

[24] A "silence is consent" rule is a form of tacit approval under which a business license or permit application is considered automatically granted if the competent licensing authority fails to react or issue a decision within the prescribed time limit.
[25] Croatia is used as a comparator given its proximity to North Macedonia, similar heritage, and being an aspirational high-income peer.

North Macedonia outperforms Croatia in credit access and formal training, meaning the large regression coefficients on these variables (-0.63 and -0.65) represent strengths to be preserved rather than gaps to be closed. The priority, therefore, is not broad-based business environment reform but targeted action on the specific fault lines where North Macedonia demonstrably underperforms (see Box 2).

Box 2. Energy inefficiency at the firm level: market failure left uncorrected

North Macedonian firms are measuring their energy and emission problem but not solving it. The WB Enterprise Survey data reveal a striking paradox in North Macedonia's private sector. Firms rank third among twelve European comparators in CO2 monitoring, suggesting meaningful awareness of their energy footprint (**Figure 52**Figure 46). Yet, power intensity, measured by the share of electricity costs to annual sales, sits among the highest in the group at approximately 4.37 percent (Figure 47). Adoption of energy management measures is only moderate**Figure 52**. This combination of high-power intensity, moderate efficiency action, and relatively strong emission monitoring points to a private sector that has the awareness but not the enabling condition to act on it.

This gap between awareness and action reflects a classic market failure. Energy use carries unpriced negative externalities, such that firms do not bear the full social cost of their carbon emissions and therefore have insufficient incentive to reduce them. At the same time, upfront investment in energy-efficient equipment is often beyond the reach of firms that face capital market imperfections, particularly smaller enterprises for whom financing costs are high and collateral constraints binding. Where energy prices are additionally distorted by subsidies or regulated tariffs, the price signal that would otherwise drive efficiency investment is further weakened. The result is a private sector that is aware of its energy footprint but rationally under-invests in reducing it.

Figure 46. Climate adaptation incidence is high, and firms monitor emissions

(Percent of firms reporting action)

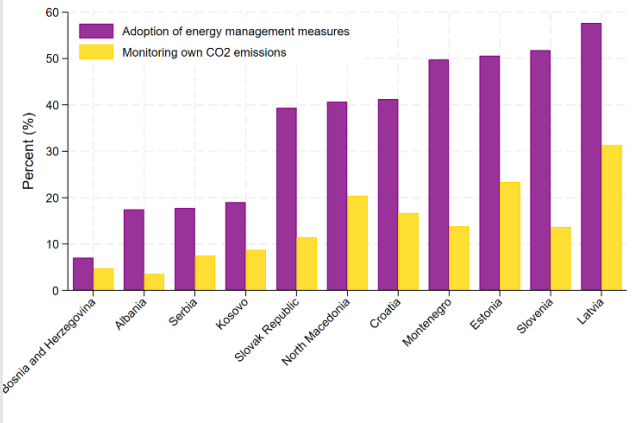
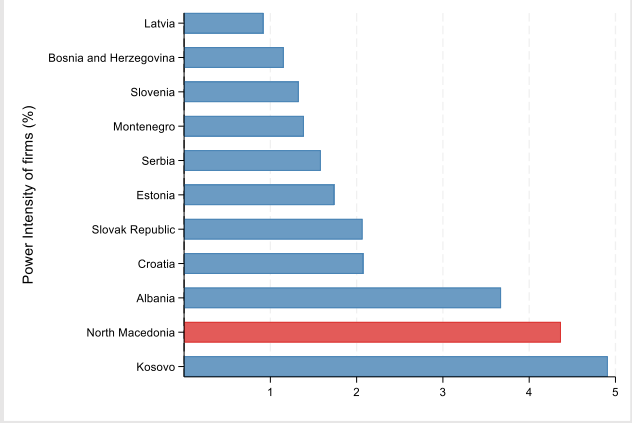


Figure 47. Yet, electricity intensity of firms is high

(Power intensity, % of electricity cost in annual sale revenues)



Source: ICA 2.0 Toolkit based on WB Enterprise Survey, based on April 1, 2026 data.
Note: Country-year coverage: Albania (2025), Bosnia & Herzegovina (2023), Croatia (2023), Estonia (2023), Latvia (2024), North Macedonia (2023), Montenegro (2023), Serbia (2024), Slovak Republic (2023), Slovenia (2024), Kosovo (2025). For countries with 2025 survey, power cost is calculated as average monthly costs multiplied by 12 months. Power intensity is annual cost divided by annual sales, both are in real local currency units. Averages are weighted by sampling weghts. The data capture electricity costs only. Fuel costs, which may cover heating and industrial processes, are unobserved, and given North Macedonia's historical dependence on coal and solid fuels, the true energy burden on firms is likely considerably higher than electricity intensity alone suggests.

The challenge is particularly acute in high-emitting sectors, where the capacity to finance mitigation investments is severely constrained. The CCDR for North Macedonia (2024) finds that nearly half of the businesses of the state operating in these sectors are not generating profits, undermining their ability to fund the capital investments that decarbonization requires. The problem is compounded by a partly decentralized ownership structure (i.e., around 30 percent of these firms are subordinated to municipal governments), making it difficult to mandate and coordinate climate mitigation objectives from the center. Where cash flows are thin and ownership fragmented, the fiscal space for efficiency upgrades and clean technology adoption is limited regardless of policy intent.

The presence of market failures creates a clear rational for public intervention, such as concessional financing, efficiency standards and targeted incentives, among others. The problem in North Macedonia is that the governance architecture to deliver this intervention is not fully developed. Policy framework for industrial efficiency lacks the coherence needed to shift firm behavioral scale. Firms are thus caught twice: first by the market failure that makes private efficiency

investment unattractive without support, and then by the governance failure that prevents public support from materializing.

The appropriate response is not simply to expand public spending on climate action in sectors where private operators are better placed to deliver. Rather, the government will need to right-size its presence in competitive markets and create a level playing field between state-owned and private actors — particularly in energy markets. This requires careful market liberalization, streamlined energy audit requirements, and energy pricing closer to market values. Critically, enabling a shift toward renewable energy and greater efficiency implies allowing third-party access to transmission and distribution networks, so that private investors can sell directly to eligible customers without discrimination.

The weaker integration of North Macedonia's firms into international markets represents a second, distinct policy challenge as compared to aspirational peers. Despite having superior customs procedures on average, the country lags Croatia in the share of exporters, foreign-owned firms, and firms with foreign-licensed technology, suggesting the bottleneck in the broader investment climate, where there are factors, including political uncertainty and inconsistent implementation, which may deter foreign investors and discourage firms from pursuing international certification or technology partnerships. Regression estimates show these international linkages, foreign ownership, licensed technology, certification, and export participation, carry some of the largest productivity premiums. Closing these gaps, as simulated, could deliver an additional 0.4 to 3.1 percent in productivity gains. Together with the previous results, the implication is that improving political stability and reducing informality are not only direct productivity interventions, but they can also be preconditions for unlocking the international integration that North Macedonia's customs infrastructure is already positioned to support.

Figure 48. Because the quality of the business environment supports firm-level productivity…

(Link between business climate on firm productivity, regression coefficients to value added per worker)

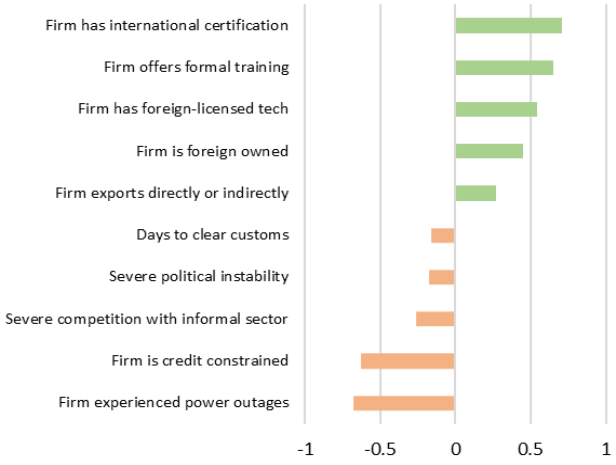
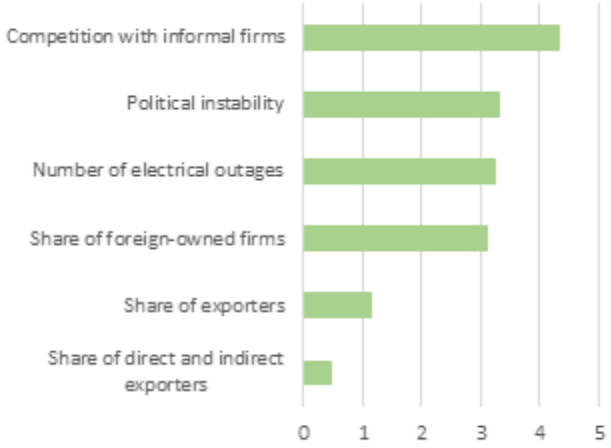


Figure 49. …improving the business environment can yield significant productivity gains.

(Labor productivity rise from closing the gap between North Macedonia and Croatia, in %)



Source: ICA 2.0 Toolkit based on data from the World Bank Enterprise Surveys 2023 for North Macedonia. *Notes*: Panel a: Size and sector fixed effects have been used. Each regression used sampling weights in calculating coefficients. Regressions involving power outages, foreign ownership, formal training, and international certification are all statistically significant. Panel b illustrates the simulated change in labor productivity if North Macedonia closes the gap in select indicators.

Clearly, the reach of governance reform extends well beyond the investment climate. The evidence presented in this chapter makes this case on multiple fronts: misallocation persists because weak governance fails to reward productive firms and penalize inefficient ones; energy market failures go uncorrected because the institutional architecture to allow more private participation and engagement is underdeveloped; and weaker international integration reflects an investment climate where political uncertainty and regulatory inconsistency deter foreign investors and limit firms' engagement with global markets. Addressing these failures through governance reform would therefore not only correct the domestic misallocation of resources but also unlock the international

integration that North Macedonia's trade infrastructure is already positioned to support. More fundamentally, better governance may further open the channels through which the most durable productivity gains materialize: (i) the infusion of FDI spillovers and foreign knowledge into domestic firms, where linkages are present but remain slow and uneven, and (ii) stronger incentives for incumbent firms to innovate. The last dimension is examined more deeply in Chapter 3.

Chapter 3. Reinforcing competition and improving business environment

Competition is essential for sustainable and inclusive economic growth. Robust competition creates incentives for firms to operate efficiently, innovate, and increase their productivity. In the near term, competition boosts growth through its effect on prices, investment, consumption, and exports. In the long term, competition accelerates growth and thus job creation by promoting an efficient allocation of resources and by encouraging the entry of more-productive firms and the exit of less-productive ones, which produces better economic outcomes. Evidence from perception indices and firm-level data indicates that North Macedonia's competitive environment remains weaker than of aspirational EU peers, with comparatively ineffective anti-monopoly policy, elevated business risks linked to vested interests and cronyism, and increasingly concentrated non tradable markets dominated by entrenched incumbents. A relatively sizable state footprint in competitive sectors and fragmented SOE oversight risk crowding out private investment and constraining labor reallocation. In parallel, the state aid framework allows potentially distortive measures and suffers from incomplete information, unclear allocation criteria, and weak monitoring,

3.1. Constrained competition limits productivity-enhancing reallocation

Perception indicators suggest room to enhance the competition environment in North Macedonia exists. According to the Bertelsmann Transformation Index (BTI), North Macedonia domestic markets are perceived as relatively competitive overall, ranking 14th out of 136 comparator countries in terms of market organization and competition, outperforming structural peers and placing it broadly in line with some aspirational peers such as Latvia and Croatia but lagging other aspirational peers such as Estonia, Lithuania, and Slovenia (Figure 50). By contrast, the country's anti-monopoly policies compared well to its structural peers, they are perceived to be less developed than those of its aspirational peers (Figure 51), pointing to potential gaps in the legal and institutional frameworks supporting competition. Combined with relatively high market concentration in some sectors, this may indicate weaker competitive pressures and barriers to market entry.

Figure 50. Perception of market competition is similar to aspirational peers

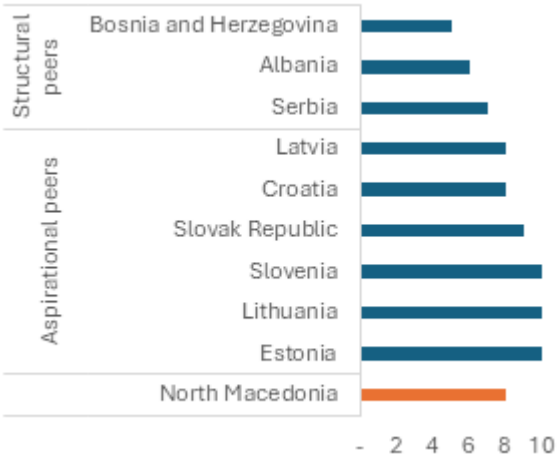
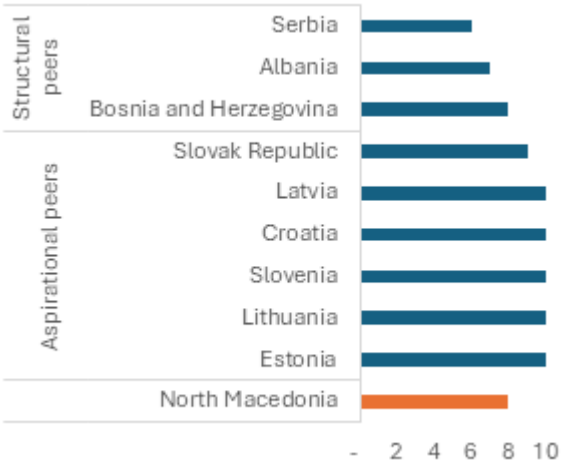


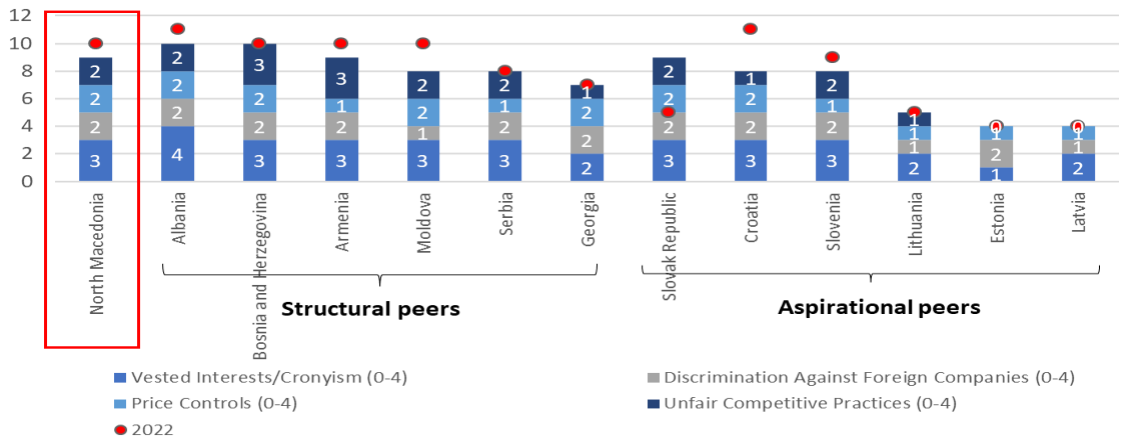
Figure 51. Perception of anti-monopoly policy points to less effectiveness



Source: Bertelsmann Transformation Index (BTI), 2024. *Notes:* The BTI evaluates how developing countries and countries in transition are working towards a market economy and democracy based on three dimensions: (i) political transformation, (ii) economic transformation, (iii) governance. Score ranges from 1 (worst) to 10 (best). A higher value on market-based competition means a better competition enabling environment. A higher value on anti-monopoly policy means a stronger policy in place.

In addition, while the private sector perceives the risk of doing business in North Macedonia favorably compared to some of structural peers, there remains scope for further improvement to reach the levels observed in aspirational peers. According to the Economic Intelligence Unit, these risks stem primarily from the power of vested interests and the prevalence of cronyism (Figure 52), which can create an uneven competitive playing field. Additional factors such as price controls, discrimination against foreign companies, and unfair competitive practices may further elevate business risks and deter investment. According to the US State Department's 2024 Investment Climate Statement, corruption frequently influences government tenders and licensing processes.[26]

Figure 52. Perceived operational business risks faced by the private sector, 2024

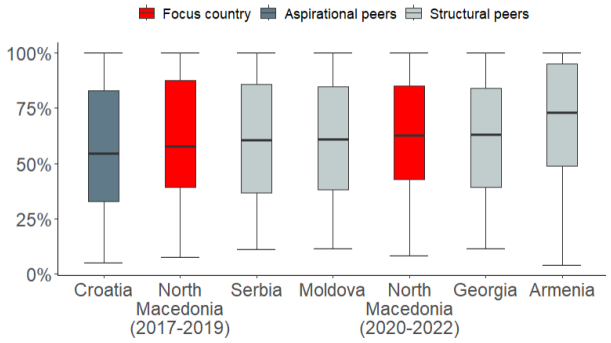


Source: Economist Intelligence Unit, Q4 2024 and average 2022. *Notes:* The score for each component varies from 0 (very little risk) to 4 (very high risk). The maximum risk level is 16.

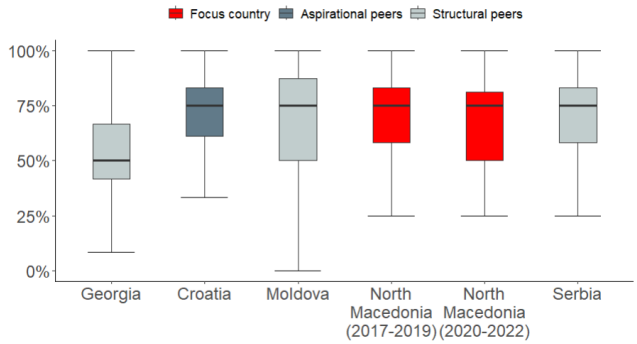
Hard data supports some of the perception-based findings: an analysis of firm-level data indicates that North Macedonia's markets tend to be relatively concentrated and characterized by entrenched incumbents. During 2017-19, the average revenue share of the top four firms in non-tradable sectors was higher in North Macedonia than in Croatia, an aspirational peer, but lower than in structural peers such as Armenia, Georgia, Moldova, and Serbia (Figure 53a).[27] However, North Macedonia's average market concentration has increased since 2019. Moreover, the top four firms have held their positions year after year,[28] demonstrating more persistent entrenchment than in nearly all peers (Figure 53b). These findings suggest limited market dynamism, at least among the top firms.

Figure 53. Concentration, entrenchment, and price-cost margins in North Macedonia and selected peers

a. Revenue share of top four firms in non-tradable sectors



b. Entrenchment among top 4 firms

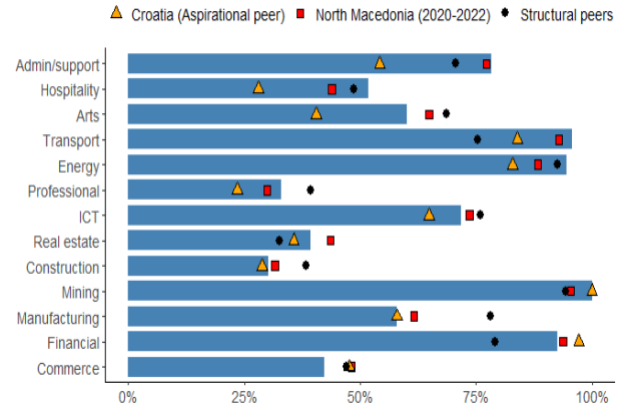


[26] https://www.state.gov/reports/2024-investment-climate-statements/north-macedonia/

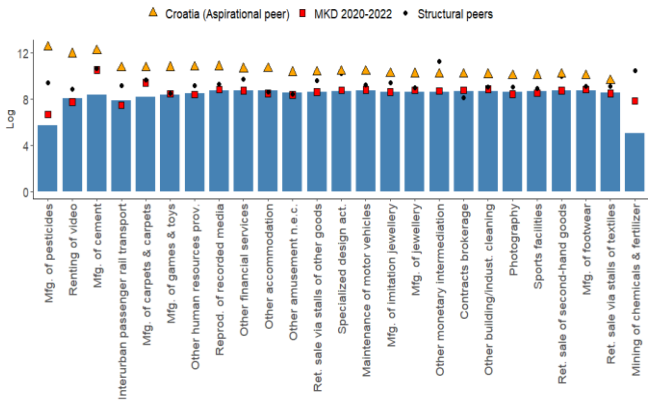
[27] Concentration is only calculated for the non-tradable sectors, as calculating concentration for tradables would require granular information on imports and exports that is not currently available for all countries in the comparison growth.

[28] Entrenchment measures the probability that one or more of the top four firms in an industry in year *t-3* will remain in the top 4 in year *t*. Higher entrenchment levels imply less business dynamism at the top.

c. Revenue share of top 4 firms by sector 2017-19 (blue bar indicates North Macedonia)



d. Bottom 25 industries with the lowest levels of labor productivity, 2017-19 (blue bar indicates North Macedonia)



Source: WB Staff calculations based on firm-level dataset from the Republic of North Macedonia and peer countries. *Notes:* The source of the data for North Macedonia is the Central Registry of North Macedonia, which includes all registered (formal) firms, except self-employed (one person) firms. Comparator countries are those with appropriate firm-level datasets for this analysis. The datasets used are averages for 2017-19, except for Armenia's, which is for 2018-19. As a result, Armenia's entrenchment rate cannot be calculated for 2017-19, and labor productivity estimate is for 2018-19. North Macedonia's dataset is for two periods, 2017-19 and 2020-22.

High market concentration is strongly correlated with, and likely contributes to, suboptimal productivity across industries. In North Macedonia, in 10 out of 14 sectors analyzed, the top four firms account for 50 percent or more of total revenues—rising to at least 90 percent in enabling sectors such as finance, energy, transport, and mining, which are also more concentrated than in Croatia (Figure 53c). Across 25 subsectors within the most concentrated sectors (including financial services, mining, transportation, and parts of manufacturing), many are among the weakest performers (Figure 53d). This pattern suggests a broad-based productivity challenge, with limited competitive pressure potentially weakening incentives to innovate, improve efficiency, adopt new technologies, and create jobs.

While North Macedonia's regulatory and institutional framework for market competition aligns with the EU, its design and implementation continue to pose challenges. The country's main competition law[29] contains key provisions regarding cartelization, abuse of dominance,[30] and mergers. State aid is regulated by a separate law.[31] Both laws allocate enforcement powers[32] to Competition Protection Commission (CPC), an independent agency. However, CPC has limited budgetary resources; enforcement of anticompetitive provisions is weak; and significant resources are devoted to reviewing mergers that have little potential to affect markets.

Legal ambiguities and high evidentiary burdens can enable cartels and unchecked dominance. The law prohibits cartels and abuse of dominance on efficiency grounds[33], but secondary rules do not define efficiency criteria,[34] and it allows cartels deemed "necessary to protect the public interest" without clarifying how the exemption applies. At the same time, proving abuse of dominance is comparatively rigid: the CPC must establish specific abuses (e.g., refusal to deal or denial of access to essential facilities), including an "intention to harm […] in a dishonest manner" for refusal to deal, and refusals may be lawful if objectively justifiable.[35] The law also sets a 40-percent market-share "presumption of dominance," which could deter investment and merit-based competition. Although the law follows the EU model in prohibiting exclusionary and exploitative abuses of

[29] Law on the Protection of Competition, No. 145/2010 (Unofficial Consolidated text).
[30] Article 11(2)(i)-(iv) of the law mirrors Article 102 TFEU.
[31] Law on State Aid Control, Official Gazette of RM 145/10.
[32] European Commission, 2024. Report on Northern Macedonia, Communication on EU Enlargement Policy, page 65.
[33] Article 7 of the Law mirrors Article 101 TFEU.
[34] As permitted under Article 7(3) of the Law. No further explanation is provided in the secondary rules (see Guidelines on the application of Article 7(3) of the Law on Protection of Competition) which copy the EC's Guidelines on the application of Article 101(3) TFEU, 2004/C 101/08.
[35] Article 11(3) states that a refusal may be permitted if access is "not possible due to certain operational or other reasons, or that it should not be allowed due to certain justified reasons".

dominance, it may unduly restrict pro-competitive conduct (e.g., price undercutting) that would benefit consumers. By contrast, EU practice treats shares above 40 percent only as indicative and weighs in factors such as entry barriers, financial independence, durability of market power, and buyer power.

The merger-control regime is onerous for firms and the CPC due to low notification thresholds and the lack of a simplified process for mergers with little effect on competition. Notifications are required for mergers with combined worldwide turnover of €10 million, combined national turnover of €2.5 million, combined market share of 60 percent, or an individual market share of 40 percent. The national turnover threshold is low—almost half the benchmark estimate (Box 3)—and is not cumulative, so deals without a clear local nexus may be notifiable. The lack of individual turnover thresholds[36] can also capture transactions with limited impact. Market-share thresholds can be subjective and difficult to apply.[37] As a result, over 100 mergers are reviewed each year, yet on average only two require in-depth investigation.[38]

The merger-notification procedure could be improved to enhance legal certainty. Extensive disclosure requirements in the merger-notification regulations[39] applied by the CPC on a case-by-case basis, increase compliance burdens. The law states the review period starts when the CPC receives all required data[40], whereas EU practice begins only once a notification is deemed complete. Reviews are capped at 90 working days, with possible extensions in blocks of 20 working days. The law does not specify Phase I or II procedures and secondary rules remain incomplete, undermining due process. Best practices include state-of-play meetings, market testing, and a right to be heard before Phase II.

The CPC's effectiveness, transparency and independence are constrained by limited budget and gaps in the institutional framework. While the legal framework includes safeguards (e.g., five-year renewable terms, qualification requirements, and conflict-of-interest checks), it lacks detail on staggered terms and cooling-off periods and provides limited grounds for dismissal beyond criminal or corruption violations.[41] The CPC is further constrained by underfunding and understaffing, particularly in antitrust: its budget was under US$500,000 in 2024.[42] Furthermore, the CPC does not publish the rationale behind its decisions, reducing transparency and weakening the development of a competition culture. Although basic information on decisions must be published, fully reasoned decisions are not publicly available; requiring publication of the CPC's reasoning would improve transparency and legal certainty.

Competition law is weakly enforced, with few cases, modest sanctions, and limited use of tools such as dawn raids and leniency. In the past three years, few investigations resulted in sanctions and fines were often light (e.g., €14,000 for refusal to deal in 2022).[43] Key evidence-gathering tools are underused: the CPC conducted only four dawn raids and received no leniency applications over the same period, while cartels may face criminal

[36] Relying on worldwide turnover thresholds or combined domestic turnover thresholds can trigger a review of mergers that leaves the structure of the market mostly unchanged. For example, such mergers may involve a firm that has a negligible presence in North Macedonia merging with a firm that exceeds the turnover threshold. Individual thresholds can help avoid unnecessary reviews, and international best practices include both combined and individual local turnover thresholds on a cumulative basis.

[37] These thresholds rely on the CPC's current definition of the market, which reduces legal certainty. Defining the relevant product and geographic market is difficult and time-consuming both for firms and the CPC. In addition to high administrative costs, this approach result in lack of predictability, since parties may not know how the relevant markets will be defined and thus whether merger notification is required or not.

[38] 2024 Annual Report of the CPC, pp. 16 - 113 decisions were cleared, six fell outside the scope of the Law, and two notifications were rejected for being incomplete. In 2022, the CPC adopted 101 merger decisions, all of which were cleared, 2023 Annual Report of the CPC.

[39] Regulation on the form and content of the notification and the necessary documentation to be submitted together with the notification, 2012.

[40] Article 15(7) of the Law which is reiterated in Article 19(1) of the Law here the CPC "shall examine the notification as of the day when it is received."

[41] Article 27(11) of the Law: only dismissals of the President and members on their request, or if they are absent for three consecutive sessions without justification

[42] Annual Report on the work of the Commission for Protection of Competition in 2023, page 55-57. This report stated that the CPC employed 26 staff in 2023, including the President and two members, as well as technical staff and all support staff (including IT, finance, administrative etc.) and had a budget of USD 448,370.

[43] A fine of approx. EUR 14,000 was imposed on Makedonski Telekom AD, a broadcaster of UEFA football matches in Macedonia. The CPC found that it abused its dominant position by refusing to transfer copyrights to another broadcaster unjustifiably.

prosecution and leniency covers only administrative penalties. Advocacy is also limited—only one market study was conducted in 2022, and no secondary legislation underpins the CPC's advocacy powers.

Box 3. Assessing the levels of merger notification thresholds

Merger notification thresholds should not be set so low that they unduly burden the competition authority and hamper the progress of (efficient) mergers. An international benchmarking model can be used to estimate a reasonable initial range for the merger notification thresholds, which can be then finetuned based on the list of companies qualifying over the proposed thresholds. The underlying benchmarking model employed posits a linear relationship between the level of the combined (or individual) threshold and the size of a country's economy, proxied by the logarithm of gross domestic product (GDP).

The model produces a benchmark estimate for both individual local and combined local thresholds—using merger threshold levels gathered by the Competition Policy team of the World Bank Group and GDP values from the World Development Indicators (WDI). Table 4 shows estimates of the benchmark thresholds for North Macedonia. Visual illustrations of each threshold as applied to North Macedonia are shown in Figure 54 and Figure 55. In each case, the logarithm of the threshold is on the vertical axis and the logarithm of GDP is on the horizontal axis. The model finds that a benchmark combined local turnover suitable for North Macedonia would be around 5 million EUR, double the current threshold. The model also produces a benchmark estimate for individual local thresholds of around 1.26 million EUR.

Table 4. Benchmark thresholds for North Macedonia in millions (USD, EUR and MKD)

Type	Threshold (USD)	Threshold (EUR)	Threshold (MKD)
Individual local	1.36	1.26	77.69
Combined local	5.27	4.86	299.86

Source: Benchmark estimates produced using North Macedonia's GDP in 2021 as reported by the WDI. The exchange rate used is the Macedonian Denar Per U.S. Dollar, Period Average. The latest available exchange rate value for MKD is for 2023, which is MKD 56.95 = USD 1, as reported by International Monetary Fund. The latest available exchange rate value for EUR is for 2024, which is EUR 0.9239 = USD 1, as reported by European Central Bank.

Figure 54. Individual local turnover threshold model

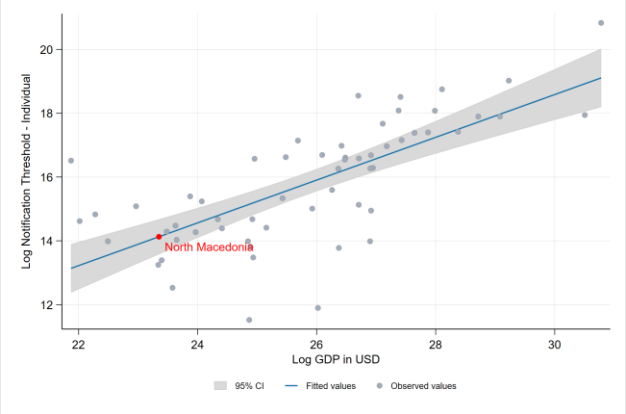
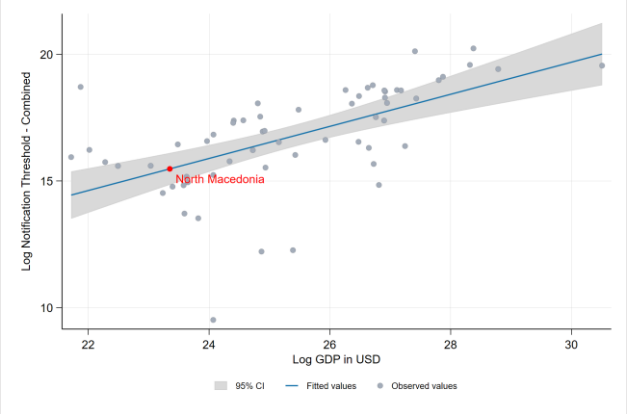


Figure 55. Combined local turnover threshold model



Source: Author's elaborations based on data from the World Bank Group Merger Notification Thresholds and Procedures Database, Version May 2023, and World Development Indicators (WDI).

3.2. State ownership across many sectors appears to constrain competition and labor reallocation

The state's economic footprint is substantial in competitive sectors. According to the World Bank's Global Business of the State[44] database, North Macedonia's government holds a direct or indirect equity stake of at least 10 percent in 191 domestic entities. Although these firms account for a relatively modest 9 percent of GDP (Figure 56a), they span across 13 percent of all industries (73 out of 563 at the 4-digit level). Notably, nearly half of these entities operate in competitive sectors, where private firms could function efficiently without state

[44] The database excludes sectors such as public administration, health, and education, where the state takes an active role in providing public goods. For more information, see Dall'Olio et al. (2022).

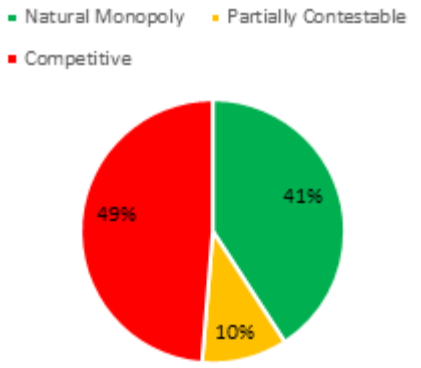
participation. State involvement risks crowding out private investment, weakening competition, distorting markets, and deterring private-sector growth.

Recent reforms have strengthened state-owned companies' (SOEs) governance, yet fragmentation, non-transparency, and exposure to political influence persist. The 2022 amendments to the Law on Public Enterprises strengthened qualification criteria for supervisory board members and require certain SOEs to appoint board members with at least five years of experience in finance and legal affairs. However, board members are not required to be independent, and the government can still appoint CEOs.[45] Furthermore, the government's SOE strategy lacks an articulated justification for state involvement, particularly outside strategic sectors or areas clearly characterized by market failures. SOEs are organized in a fragmented way, with responsibilities dispersed across multiple line ministries and municipalities, limiting effective oversight.[46] While the Ministry of Finance publishes quarterly financial information for some SOEs, it falls short of comprehensive disclosure. Many SOEs are not incorporated under the standard company law, which further skews the playing field between private and public firms.[47]

A coherent strategy and clear policy rationale for state ownership are essential to ensure that SOEs operate only where there is a well-defined public interest justification. Establishing a centralized oversight body would strengthen the state's capacity to act as an informed and effective owner, promote more consistent governance practices, and facilitate efforts to enhance transparency and accountability across the SOE portfolio. In addition, publishing a consolidated annual report that covers the entire SOE portfolio would increase accountability, enable informed stakeholder scrutiny, and ultimately enhance the financial discipline of SOEs. Finally, requiring SOE board members to be independent would help insulate strategic decision-making from undue government influence, support adherence to commercial principles, and strengthen oversight objectivity.

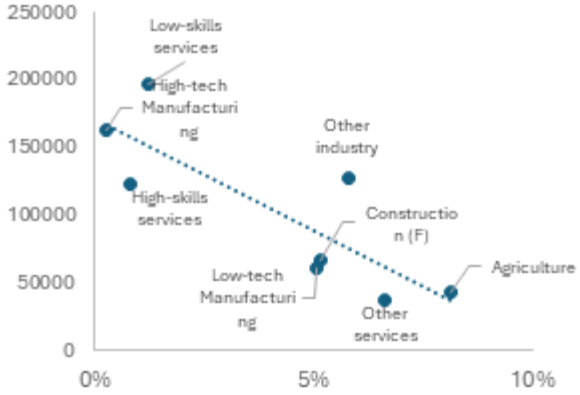
Figure 56. The state's economic footprint is extensive in competitive sectors

a. Revenue of firms in which the government has at least a 10% ownership stake as a share of GDP



Source: Based on a presentation in April 2025 that relied on the World Bank Global Business of the State database (2019).
Note: Data coverage on revenue for North Macedonia is 99%.

b. Correlation between labor productivity and SOE employment shares across subsectors



Source: World Bank staff calculations based on national business registry data. *Note:* Value added per worker is in US$ thousands in purchasing power parity terms. Data are for 2022 only.

Reallocating labor from SOEs to the private sector could boost productivity growth. Productivity levels are highest in sectors where SOE employment is minimal, such as high-tech manufacturing and high-skill services, and productivity levels tend to be lower in sectors with greater SOE involvement, including agriculture and other services (Figure 56b). Hence, SOE concentration may be associated with lower labor efficiency at the sector level and promoting private-sector competition or improving SOE performance could boost productivity. Between 2011 and 2022, sectors with shrinking SOE presence saw greater productivity gains. SOE employment fell from

[45] Companies with a single-tier board have a single board of directors with executive and non-executive members.
[46] WB BOS database (2019) and Western Balkans Competitiveness Outlook: North Macedonia (2024)
[47] Western Balkans Competitiveness Outlook: North Macedonia (2024)

21 to 8 percent of total employment in agriculture and approached zero percent in manufacturing and construction. Simultaneously, labor productivity rose, especially in high-tech manufacturing and agriculture.

3.3. Public support gets more firms to start innovating, but it reduces their own investments and does not improve productivity

North Macedonia's state-aid framework could be more efficient and aligned with the EU. In contrast to EU regulations, agriculture and fishery sectors are exempt from the law[48] and private operators may act as intermediaries for state aid, e.g. under the national innovation fund, potentially distorting competition.[49] Moreover, a broad definition of horizontal aid[50] allows the authorities significant discretion to exempt certain categories of state aid.[51] The distribution of state aid across providers is inconsistent, inefficient, and often ineffective. While there is a high number of state-aid providers, publicly available information is incomplete, the allocation criteria remain often unclear, and the monitoring lacks transparency.[52] The most prominent deficiencies are in outreach and promotion efforts, weaknesses in instrument design, high application and compliance burdens, limited predictability, excessive reliance on first-come-first-served selection criteria, and weak monitoring and evaluation systems. These factors diminish the efficacy of state aid and undermine confidence in the overall process.[53]

The framework for aid to distressed firms is underdeveloped and vulnerable to misuse. The competition law[54] does not precisely define a distressed firm (or "undertaking in difficulty"). The criteria for rescue and restructuring aid—including definitions of necessity, incentive effects, and proportionality—do not reflect the applicable 2014 EU guidelines.[55] The law also lacks specific provisions for granting aid to services of general economic interest. In addition, the CPC's power to interpret the EU's general interest regarding restructuring aid conflicts with EU law, which vests such authority in the European Commission.[56]

The improvement of the state aid regime hinges on a functioning inventory including de minimis aid, which would improve oversight. Also, publishing detailed information on aid providers, beneficiaries, amounts, intensities, and expected benefits would improve transparency and enable analyses of the sectoral or regional concentration of benefits. The CPC should be strengthened by addressing procedural limitations that prevent it from identifying distortive aid and restoring a level playing field. Interested parties should be more broadly

[48] See Article 3(2) of North Macedonia's law on state aid and Articles 42 and 107(1) of the Treaty on the Functioning of the European Union (TFEU). The extent to which the provisions of the Treaty on State Aid apply to the fisheries and aquaculture subsector is set forth in Regulation (EU) No 508/2014 of the European Parliament and the Council.

[49] Article 5(1) of the state aid law broadly reflects the notion of aid under EU law, but it omits key elements of Art. 107(1) TFEU, e.g, it does not refer to aid "granted by a [Member] State or through State resources". This could exclude from scrutiny state aid granted by a private operator as intermediary, because it is unclear if the "bodies or legal persons" mentioned in Art. 6 can also be private. Art. 6(1) of the state aid law reads "State aid provider within the meaning of this Law shall be the Government of the Republic of Macedonia, the bodies of the state administration, the units of the local self-government and other bodies or legal persons granting or planning to grant State aid."

[50] Article 8(2)(e) of the state aid law.

[51] Article 107(3)(c) TFEU, which has been the basis for the approval of so-called horizontal aid in the EU (e.g., aid for research and development) requires that such aid does not adversely affect trading conditions to an extent contrary to the common interest. Similar limits do not exist in North Macedonia.

[52] 2024 EU progress report for North Macedonia.

[53] Correa, 2014. Public Expenditure Reviews in Science, Technology and Innovation, A Guidance Note; Cirera et al., 2020. A Practitioner's Guide to Innovation Policy. Instruments to Build Firm Capabilities and Accelerate Technological, Catch-Up in Developing Countries. Washington, DC: World Bank.

[54] Article 4 and 9 of the competition law

[55] Republic of North Macedonia, 2014. Guidelines on State Aid (C 249/01).

[56] Section 8.1 of the Guidelines on State Aid states that "Restructuring aid will, as a rule, only be admissible in circumstances in which it can be demonstrated that it does not run counter to the interest of the Republic of Macedonia and the EU and that any distortions of competition will be offset by the benefits flowing from the firm's survival and, where appropriate there are adequate compensatory measures in favor of competitors." Under Article 17 of the TFEU, only EU institutions, especially the European Commission, are qualified to interpret the EU's interest.

defined (as in EU law) and legal action against CPC decisions should not suspend those decisions, as this can enable complaints to delay the recovery of unlawful or misused state aid, prolonging its distortive effects.[57]

Box 4. Introduction of State Aid Management Information System

To promote transparency and efficiency on state aid, the Ministry of Finance launched the procurement of comprehensive, custom-made State Aid Management Information System (SAMIS) in March 2025. This system, developed with the support of the World Bank and expected to be functional by the end of 2026, will be used by all state aid providers and the ex-ante control entity (CPC) to:

- Automate the whole process of state aid design, approval, implementation and reporting for institutions, Government and CPC;
- Simplify the application for state aid by creating a single portal where firms can apply electronically for all state aid instruments provided by various relevant institutions, eliminating paper-based applications and the need to provide supporting documents from other institutions;
- Ensure that *de minimis* criteria are met by automatically providing information on the prior state aid received by the applicants from different state aid providers;
- Improve transparency and accountability in use of public financing by creating a single registry of State Aid, providing public access to information dashboard, generating standardized reports and improving CPC reporting toward EU;
- Support evaluation of effectiveness of state aid by enabling analysis of key firm-level indicators (jobs, revenue, profit) from the annual financial statements of the beneficiaries;
- Improve the financial management of state-aid providers and increase efficiency of the disbursement to beneficiaries by enabling electronic submission of reports and requests for payments.

Source: Building Efficient, Transparent and Accountable Public Finance Management Institutions Project (P176366), World Bank

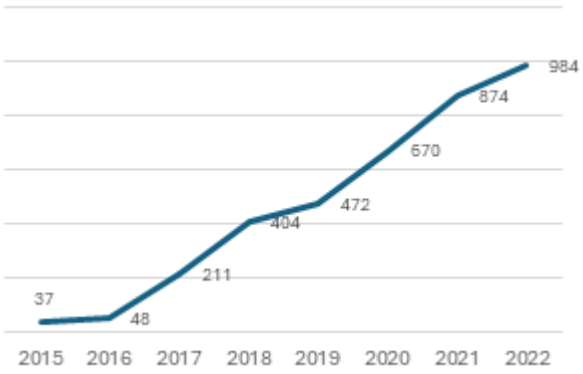
Innovation is widely recognized as a key driver of long-term economic growth and competitiveness. Public policy often seeks to stimulate innovation through direct funding, subsidies, and targeted programs designed to lower the financial risks associated with research and development (R&D). This is what had been observed in North Macedonia's Fund for Innovation and Technology Development (FITD), which has grown rapidly, supporting over 984 projects across 884 productive units between 2015 and 2023 (Figure 57a). In the short run, government spending appears to have increased firms' employment, investment, wages, and sales. However, beneficiary firms showed no significant gains in labor productivity or TFPR, with more than 30 percent exhibited declining productivity, while more than 60 percent of the firms remained in the exact same segment of productivity after receiving state support (Figure 57b). This mirrors the findings from Community Innovation Survey (CIS), which showed no significant gains in labor productivity.

Figure 57. Rapid growth in innovation funding boosts activity but fails to raise productivity

a. Cumulative number of projects receiving FITD support, 2015-2023

b. Productivity segments of beneficiary firms pre- and post-treatment (FITD innovation programs)

[57] Article 26 establishes that "The action for instituting an administrative dispute shall be taken within 30 days as of the day of receiving the decision, and it shall not defer the enforcement of the decision, except when the Commission for protection of competition orders the reimbursement of unlawful or misused state aid whose granting is not compatible." This provision is at odds with EU law, under which actions brought before the CJEU shall not have suspensory effect unless so ordered by the CJEU based on special circumstances (Article 278 TFEU).



Pre-intervention productivity	Post-intervention productivity: Low	Medium	High
Low	20.9%	3.7%	0.0%
Medium	7.9%	15.1%	1.0%
High	8.0%	15.8%	27.5%

Source: World Bank staff elaboration using data from beneficiaries provided by the FITD and the North Macedonia Central Business Registry. *Notes:* In panel (b), productivity is measured as the TFP using the Levinsohn-Petrin methodology. Each segment of low, medium and high productivity is based on the annual pre- or post-intervention distribution of TFP using three quantiles as cut-off points. For instance, low productivity indicates firms within the sector that appear among the lowest 33% of the distribution of TFP. The transition matrix shows the productivity segment of each beneficiary for years before and after the intervention. In red indicated cases represent a transition to lower productivity segments, whereas in green there are cases of productivity upgrading.

Chapter 4. Leveraging trade to support productivity and job creation

North Macedonia's trade sector, while historically a significant engine of growth, is showing signs of diminishing returns that call for a strategic reorientation. Exporting firms dramatically outperform non-exporters — with firms transitioning to exporter status generating up to three times more employment. Yet, the trade-promotion model in place since 2009 is losing momentum, with export contributions to growth declining since 2023. The country's export structure remains heavily concentrated, while domestic value-added content represents only 18 percent of total export value. Infrastructure weaknesses constrain the country's export potential, and services trade — at only 30 percent of total exports — remains underdeveloped due to regulatory barriers in professional and digital sectors. High tariffs on intermediate inputs further raise costs for domestic producers, undermining competitiveness. To revitalize trade-led growth, North Macedonia would need to shift from low-value assembly toward higher-value manufacturing and digital services, align tariffs with EU levels, invest in quality infrastructure and standards, and leverage its geographic position to become a regional logistics hub.

4.1. Exports has been a great generator of jobs, but is slowing down

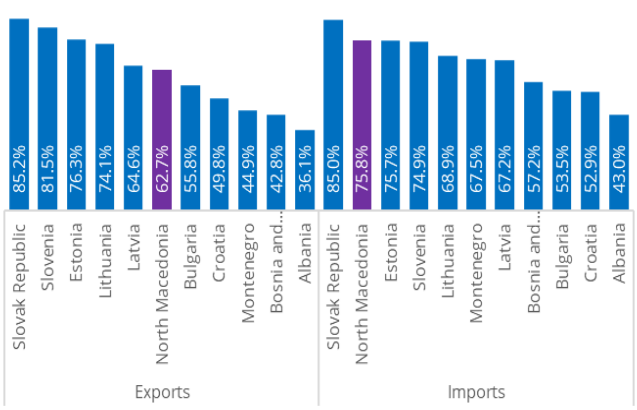
A recent World Bank analysis highlights trade as an important mechanism for fostering job creation. Worldwide, a 10 percent increase in exports is associated with a 3.1 percent increase in employment, a 3.9 percent increase in labor earnings, and a 0.95 percent increase in productivity.[58] Firms that export, and particularly those that shift from domestic production to exports, are among the most dynamic job creators. Export-oriented jobs often demand greater workforce skills and tend to offer better employment quality, with an emphasis on support, engineering, and managerial roles.[59] Empirical evidence also shows that increased trade leads to greater formal employment and narrower gender-wage gaps.[60]

North Macedonia has benefitted from a shift toward export-oriented manufacturing, but the gains from its current trade strategy are diminishing. Since 2009, the country has pursued a robust export-led growth strategy, resulting in accelerated trade expansion alongside overall economic development. The positive contribution of trade to GDP per capita shows a strong relationship between export growth and rising income levels. Compared to peer countries, North Macedonia demonstrates a relatively high level of trade openness for imports and an average level for exports (Figure 58a). The country achieved a sustained period of export-led growth from 2002 until 2008 and a resurgence following the 2008-09 global financial crisis, when the government adopted a strong strategy to attract export-oriented FDI. Despite a robust recovery from the COVID-19 pandemic, the contribution of exports to economic growth has diminished since 2023 (Figure 58b).

[58] Maliszewska et Winkler, 2024.
[59] Aguillar et al, 2023a.
[60] World Bank. Western Balkans Regular Economic Report Spring 2026.

Figure 58. Exports have a high degree of imported content and have slowed recently

a. Exports and imports of goods and services (% of GDP), 2024



Source: World Development Indicators.

b. Exports and imports of goods and services, 2000-2024 (% of GDP)

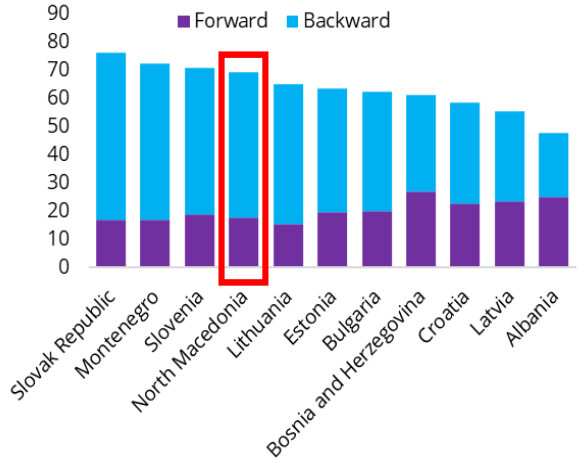


Source: EORA.

North Macedonia has made significant progress in integrating into global value chains (GVCs). Since adopting its export-oriented FDI strategy in 2009, the country has improved export performance and attracted advanced manufacturers to Technological and Industrial Development Zones (TIDZ), mainly in automotive-related industries centered on Germany and other EU member states. Integration into GVCs has led to greater domestic value addition. The country's exports are concentrated in intermediate stages of the value chain, with 51.6 percent of export value consisting of foreign inputs and embedded import value (Figure 59). By contrast, only 18 percent of export value is wholly created in the domestic economy.

Figure 59. North Macedonia is increasingly integrated into GVCs

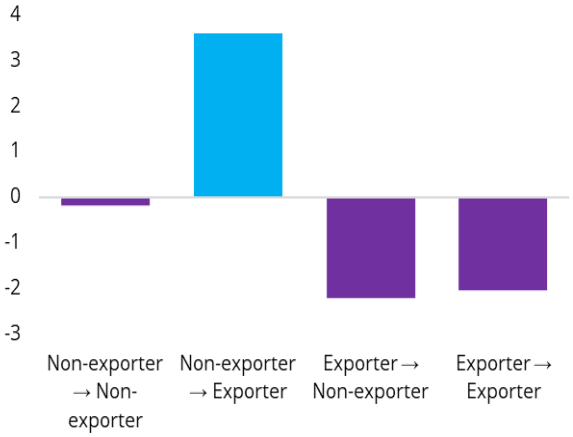
(GVC participation, 2024, percent share of total exports)



Source: EORA.

Figure 60. Transition to exporter status generates jobs

(Changes in employment by exporter status, 2019-2022)



Source: World Bank staff calculations based on firm and customs data.

Exporting firms consistently demonstrate higher productivity levels, greater sales, and stronger job creation compared to non-exporting firms. In addition to being among the largest firms in the economy, exporters are also the most significant contributors to job creation, generating up to seven times more jobs in certain years than their non-exporting counterparts. Moreover, firms that transition from non-exporter to exporter status generate three times more employment (Figure 60). Conversely, firms that cease exporting tend to experience job losses, and established exporters do not typically continue to create large numbers of additional jobs over

time. Exporters are also significantly more productive than non-exporters (Figure 61) and achieve higher average annual sales per worker (Figure 62).

Figure 61. Exporters are more productive…

(Total Factor Productivity Trends, 2011–2022)

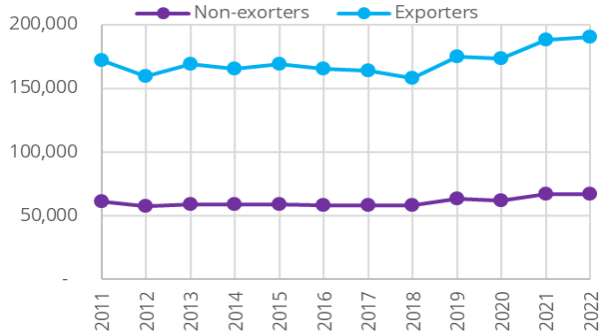
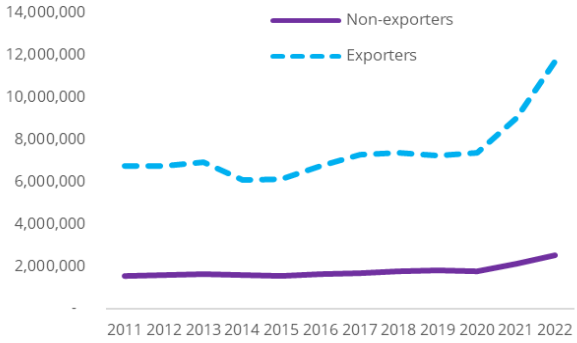


Figure 62….and have higher sales per worker.

(Sales per Worker Over Time, 2011–2022 in USD)



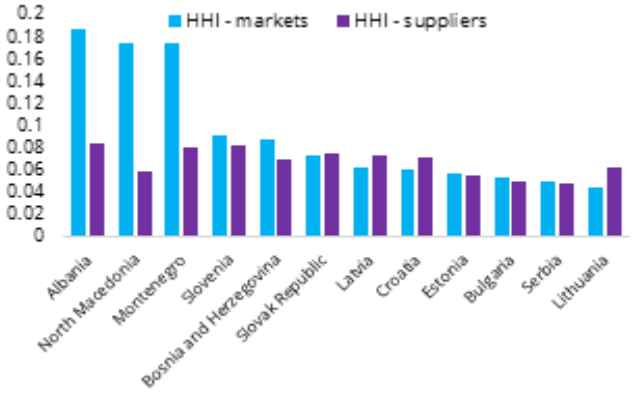
Source: World Bank staff calculations based on firm and customs data.

4.2. Diversifying exports and unlocking trade in services

North Macedonia's exports are concentrated in a limited number of markets and products, which increases vulnerability to external shocks. The Herfindahl–Hirschman Index (HHI) that measures market concentration indicates greater concentration in North Macedonia and dependence on the small number of export markets or import suppliers than with its aspirational peers (Figure 63). Exports are mostly to the EU (70 percent), and specifically Germany, following the country's export-FDI strategy that attracted investment in the automotive industry (Figure 64a). By contrast, regional markets represent less than 10 percent of North Macedonia's total exports and the number of firms exporting within the Western Balkan region has been declining, reflecting the constraints of fragmented markets. Finally, trade in services accounts for just 30 percent of total exports, lower than peers and remains a relatively small part of the economy (Figure 64b). Addressing regulatory frictions could unlock new opportunities, diversify the export portfolio, and reduce reliance on a narrow set of markets. Expanding into higher-value services and digital sectors would strengthen resilience and support more inclusive economic growth.

Figure 63. Highly concentrated export markets

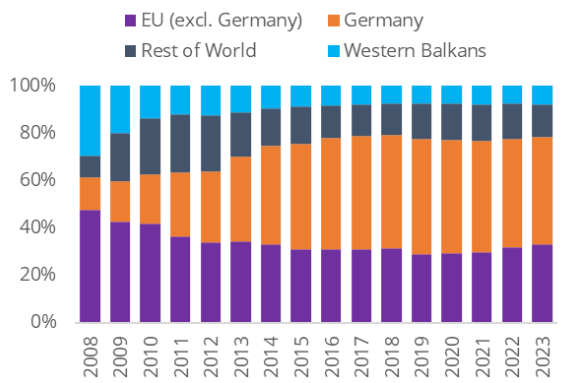
(Export-market and import-supplier concentration, Herfindahl–Hirschman Index)



Source: World Development Indicators.

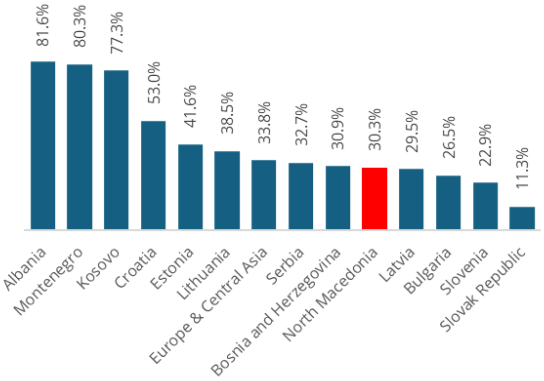
Figure 64. High focus on single export market and low service exports

a. Exports and imports of goods and services (% of GDP), 2000-2024



Source: EORA.

b. Services exports as a share of total goods and services exports (%), 2024



Source: WB staff illustration based on World Development Indicators.

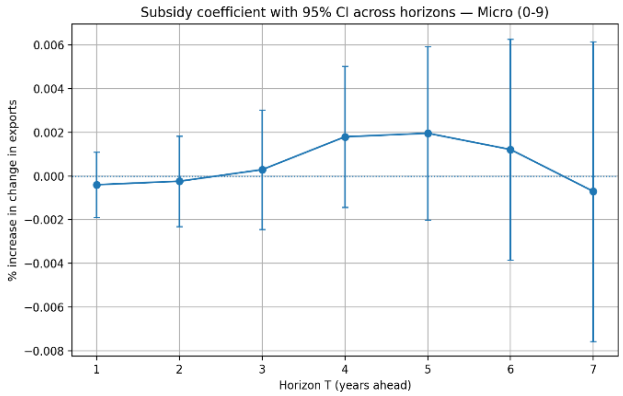
4.3. Removing policy distortions could unlock trade growth

Multiple constraints limit export growth and job creation. Government subsidies can do better to support exports, while high tariffs increase input costs for firms. Tariffs and non-tariff measures restrict access to markets; trade in services is underperforming due to highly restrictive regulations; and logistics remain a major bottleneck.

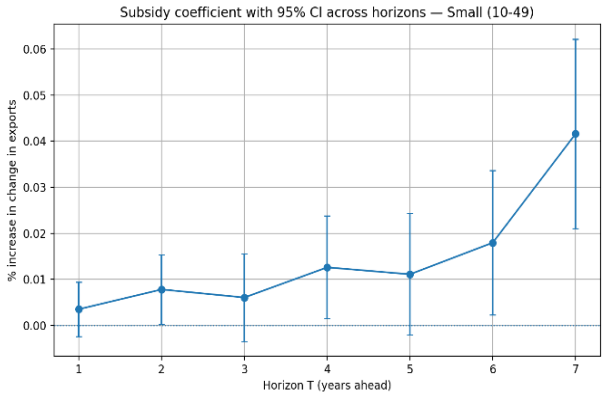
Recent analysis indicates that government subsidies were most effective in enhancing export performance among small and medium-sized enterprises (SMEs), while their impact was limited for microenterprises and largely negligible for large firms. The number of subsidized firms rose from fewer than 500 annually in the early 2010s to over 8,000 in 2021, driven by expanded support programs and the broader eligibility criteria introduced under the 2017 Economic Growth Plan and the 2018 Financial Support Investment Law. Microenterprises (defined as firms with fewer than 10 employees) received the largest share of subsidies. About 80 percent are located in Skopje, and over half operate in the trade and manufacturing sectors. However, microenterprises do not show significant improvements in export performance due to subsidies (Figure 65). By contrast, SMEs (those with 10–249 employees) benefit the most from subsidies, exhibiting consistent export growth over a seven-year period. Large firms (those with 250+ employees) demonstrate little improvement in export performance from subsidies, with long-term evidence suggesting even weakening effects.

Figure 65. The number of subsidized firms has risen sharply, but the gains from subsidies are concentrated among SMEs

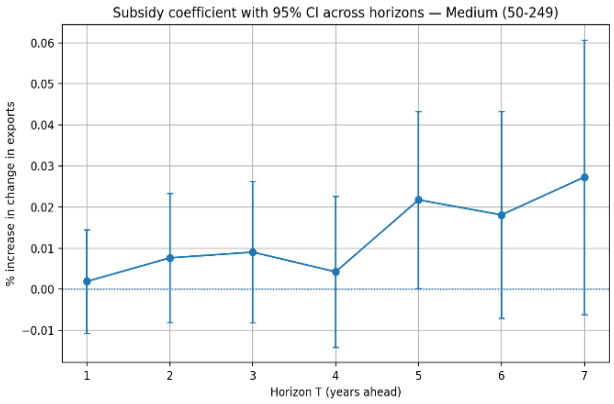
a. Change in exports among subsidized microenterprises



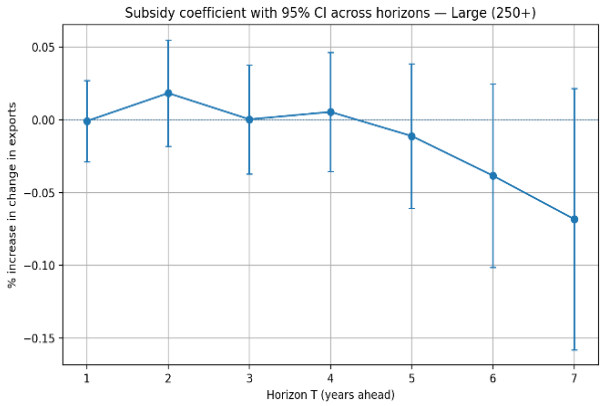
b. Change in exports among subsidized small firms



c. Change in exports among subsidized medium firms



d. Change in exports among subsidized large firms

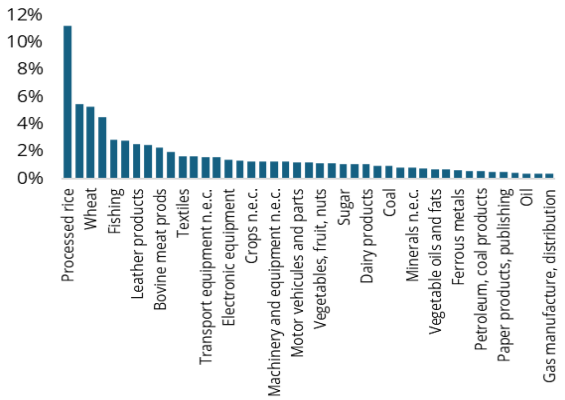


Source: firm-level data customs and business registry and subsidies data from Ministry of Finance. *Note:* the graph shows the 95 percent confidence interval from the time the firm is receiving the subsidy. Microenterprises are defined as having fewer than 10 workers, small firms as having 10-49, medium-sized firms 50-249, and large firms 250 or more.

Policy distortions, including high tariffs on intermediate and manufactured goods can undermine competitiveness. In North Macedonia, input tariffs range from 0.4 to 11 percent (Figure 66a), hitting labor-intensive sectors like food products and textiles the hardest, while applied tariffs on manufactured goods reached 6.4 percent—higher than that of most peer economies except Bosnia and Herzegovina (Figure 66b). This elevated tariff burden raises input costs for domestic producers and undermines the competitiveness of the country's manufacturing sector. Significant untapped potential exists to boost exports and employment, especially by aligning tariffs with EU levels.[61]

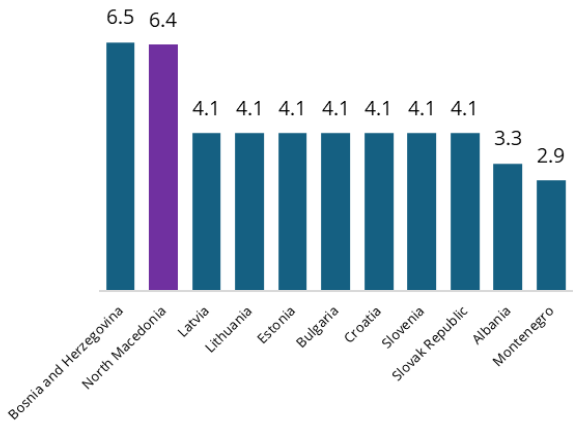
Figure 66. Reducing input tariffs can boost exports and employment

a. Upstream tariff costs as a share of total input costs by industry (%)



Source: UTAS, WB staff estimates using customs data for effectively applied tariffs. *Note:* Visualization has been truncated for better visualization.

b. Weighted average tariffs on manufactured products (%), 2021



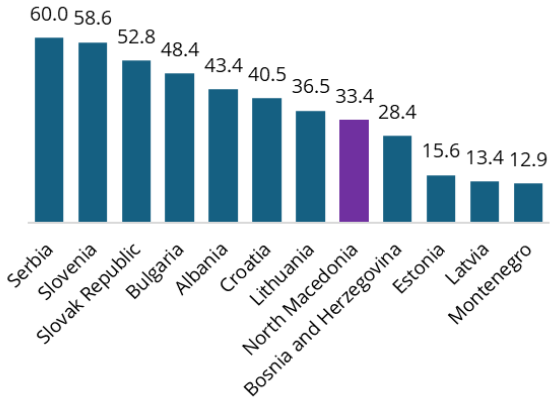
Source: UTAS, WB staff estimates using customs data for effectively applied tariffs.

Gaps in trade-related infrastructure further weaken export competitiveness. North Macedonia scores below most regional peers on the Quality Infrastructure for Sustainable Development Index (Figure 67a), which highlights deficiencies in the standards, accreditation, and metrology systems needed to support integration into GVCs. Equally, indicators of logistics performance underscore the need to improve transport, customs, and trade facilitation to enhance exports competitiveness (Figure 67b).

[61] Empirical evidence shows that lowering input tariffs improves firm-level productivity and output by expanding access to higher-quality intermediate good, as in the case of Indonesia (Amiti and Konings 2007), China (Yu 2015), Pakistan (Lovo and Varela 2020), and India (Khandelwal and Topalova 2011).

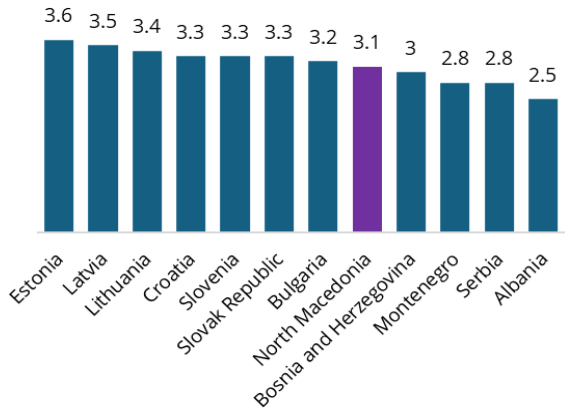
Figure 67. Weaknesses in infrastructure and logistics limit the ability of domestic firms to integrate into GVCs

a. Quality Infrastructure for Sustainable Development Index, 2021



Source: UNIDO.

b. Logistics Performance Index, 2023

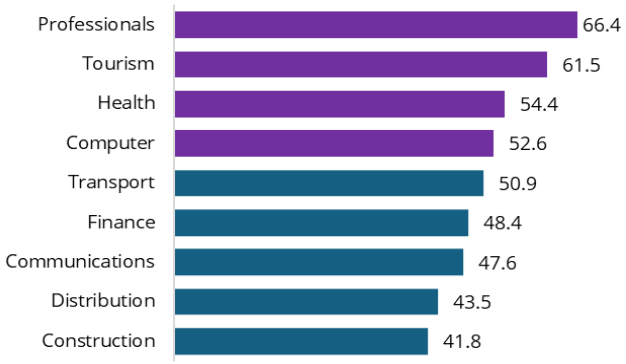


Source: World Bank.

Burdensome regulations hinder the services trade, particularly in professional, tourism, health, and computer services. Services have considerable potential to contribute to trade growth, but North Macedonia maintains restrictions that exceed those of comparator countries (Figure 68a). Many of these barriers focus on the cross-border supply of services, in which services are traded without the movement of people. Computer-related services alone account for more than 50 percent of total services exports, a significantly larger share than in most comparator countries but also faces highest restrictions among the peers (Figure 68b). Restrictions also inhibit the consumption of services abroad, the commercial presence of service providers through foreign investment, and the movement of service-supplying professionals, further constraining the sector's potential.

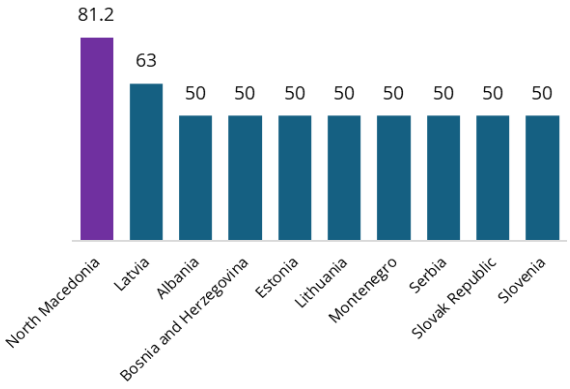
Figure 68. Restrictions on trade in services hinder the growth of exports

a. Services Trade Restrictiveness Index, 2023



Source: World Bank.

b. Services Trade Restrictiveness Index for Cross-Border Computers Service, 2023



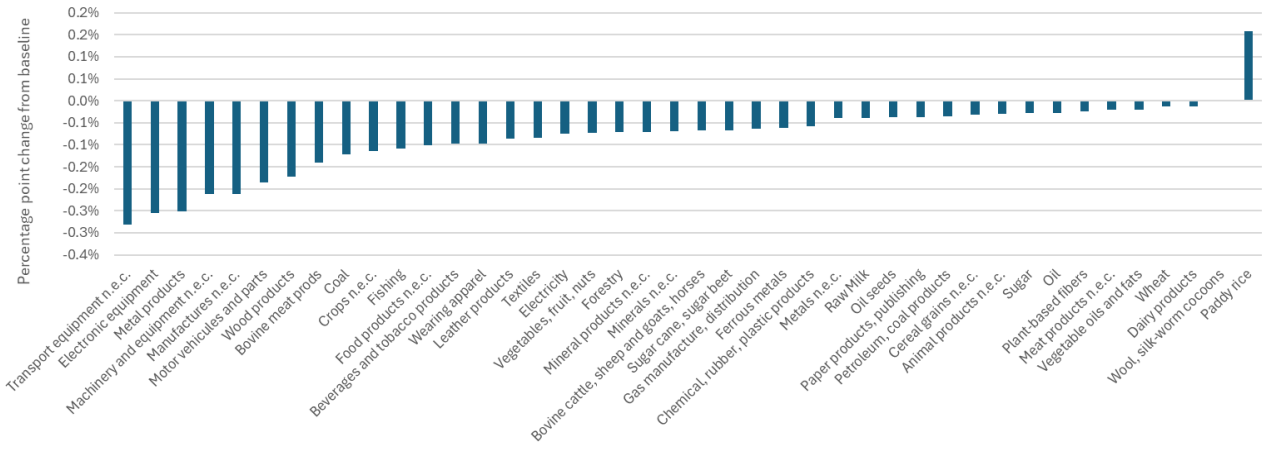
Source: World Bank.

Aligning North Macedonia's tariffs on intermediate and capital goods with EU levels would lower input costs—particularly in the manufacturing sector—and enhance firm competitiveness. Lower, harmonized tariffs would support the country's long-term objective of EU integration while yielding immediate productivity gains by allowing firms to access cheaper imported inputs and capital goods. Manufacturing industries such as transport equipment, electrical equipment, metal products, machinery and equipment, and motor vehicles would benefit the most from tariff reductions (Figure 69a), as they rely heavily on intermediate goods and capital equipment. Other key sectors, including chemicals and food processing, would also experience cost reductions, strengthening the overall industrial base.

Improving North Macedonia's infrastructure and logistics services could more than double the country's export potential (Figure 69b and Figure 69c). The gains from improving standards, accreditation, and metrology is to ensure that exporters can access high-value markets and are the largest among regional peers. Raising North Macedonia's QIS Index by 0.5 points would lead to an increase in potential exports of more than 100 percent, underscoring the substantial gains that could be achieved through targeted investments and institutional reforms in this area.

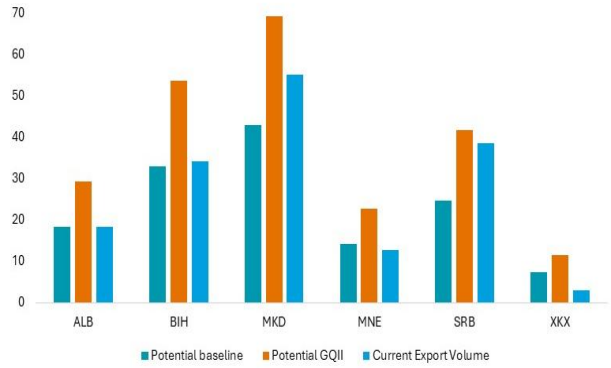
Figure 69. Impact of tariff and infrastructure reforms

a. Changes in upstream tariffs as a share of total input costs by aligning tariffs on intermediate and capital goods with EU Levels (percentage points)

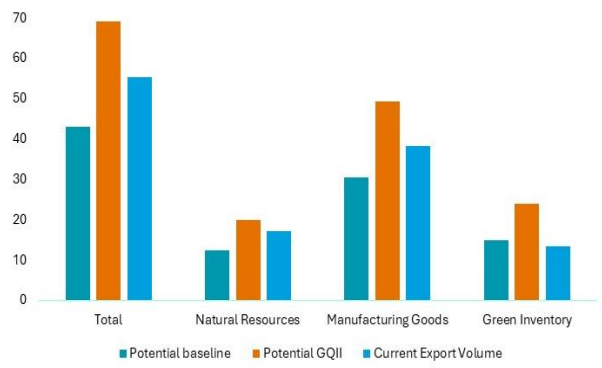


Source: National authorities and World Bank staff calculations. *Note:* The estimate for processed rice is 5.9 percent but is not shown in the figure due to scale issues.

b. Total Export Potential, 2023 (% GDP)



c. Total Export Potential by Sector, 2023 (% GDP)



Source: World Bank staff calculations based on an improvement in QI of 0.05 for all Western Balkans countries using the gravity model estimates.

North Macedonia trade strategy 2.0 requires a shift from low-value assembly to high-value, service-driven, inclusive trade. This will require: (i) reorienting the country's export-led growth strategy toward higher-value manufacturing, digital services, and green sectors; (ii) developing a national trade and investment strategy with a future-oriented and regional focus; (iii) enabling the growth of services and digital trade through regulatory reforms; (iv) investing in quality infrastructure to improve firm-level competitiveness and export readiness; and (v) leveraging transport services to establish North Macedonia as a regional logistics and trade hub.

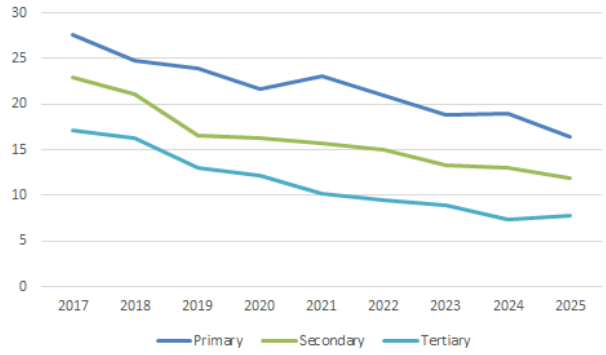
Chapter 5. Mobilizing untapped human capital

North Macedonia's labor market has made notable progress over the last two decades, with unemployment declining and employment rising. Yet job creation has slowed since the pandemic. As Chapter 2 showed, this deceleration is linked to weaker firm dynamism and a reduced contribution from reallocation and new firm entry, with gains increasingly coming from incremental improvements within existing firms. Structural challenges persist. While unemployment declines have been broad-based across education groups, job quality has improved only modestly, driven more by policy-led minimum wage and formalization dynamics than by productivity-led upgrading. Instability, informality, and skills mismatches remain widespread, with young and low-skilled workers most exposed. Tertiary education lowers the risk of poor job quality when skills are well matched, but North Macedonia struggles to absorb graduates into suitable roles: overqualification is high, wage premiums for education are weak, and emigration pressures persist. Looking ahead, demographic aging will intensify labor shortages, making it essential to mobilize underused labor—especially women—through care expansion, stronger work incentives, and more effective activation and training systems, complemented by strategic migration management. Finally, AI and the green transition will reshape labor demand and expose new skill gaps, raising the urgency of coordinated reforms to education, lifelong learning, and firm upgrading so workers can transition into more productive, resilient jobs.

5.1. Unemployment declined and job quality has improved but gains have stalled as informality and skills mismatches persist

Figure 70. Unemployment rates have fallen for all education groups

Unemployment rates by education level (15+), 2017-2025



Source: State Statistical Office.

Labor-market gains have been broadly shared across education groups. Between 2017 and 2025, unemployment rates fell from 27.6 to 16.4 percent among workers with only primary education, from 23.0 to 12.0 percent among those with secondary, and from 17.1 to 7.8 percent among tertiary graduates. In 2024, however, unemployment increased among workers with only primary education, while it continued to decline for tertiary graduates (Figure 70). This pattern suggests that low-skilled workers are the first to be affected when growth in low-productivity sectors slows. Beyond these stagnating employment outcomes, job quality remains a concern (Box 5). The multidimensional Job Quality Index combines informality, skills underutilization, low pay, and other dimensions.

Box 5. Job quality in North Macedonia

Wages remain the most commonly used measure of job quality, and for good reason: income determines living standards and shapes workers' ability to meet basic needs. Yet, wages alone do not capture the full picture. Whether a job comes with social protection, offers stability and a formal contract, imposes reasonable working hours, and matches a worker's skills and qualifications all matter for workers' wellbeing and long-term prospects.

Results from the construction of a Job Quality Index[62] (JQI) show that quality has improved, even if the gains reflect policy reforms rather than productivity growth or firm upgrading. The share of individuals working in poor-quality jobs

[62] The JQI is constructed from labor-force survey data using the Alkire-Foster multidimensional method. It measures job quality across five dimensions: (i) Income adequacy - wages exceed two-thirds of the national median; (ii) Access to benefits - the worker has access to at least one of six social insurance benefits (health, pension, unemployment, social assistance, paid annual, or sick leave); (iii) Employment stability - the job meets at least one of the following criteria: tenure, written contract, permanent status, or registered firm/employer;

declined from 8.1 percent in 2019 to 6.1 percent in 2023, a near 25-percent reduction in four years. Compared to its Western Balkan peers, North Macedonia outperforms Kosovo (26 percent) and lags only Bosnia and Herzegovina (3 percent). These improvements coincide with a gradual shift toward wage employment and formalization but are mostly explained by minimum-wage policies rather than by higher productivity or improvements at the firm level. Meanwhile, the deprivation intensity score, measured as the average share of unmet dimensions among workers in poor-quality jobs, remains high at around 45 percent, comparable to regional peers. These findings suggest that while the share of workers in poor-quality jobs has declined, those who remain in such jobs have seen little improvement in their working conditions. Roughly one third of wage employee still lack job stability and benefits, pointing to persistent vulnerabilities in employment protection.

Among workers in poor-quality jobs, employment instability and skills mismatches are the largest contributors, particularly among young and less-skilled workers. Employment stability is the single largest driver of poor job quality, especially in low-skill occupations and for younger workers, with the likelihood of holding a poor-quality job declining by 0.4 percent per additional year of age. Low-quality jobs are concentrated in traditionally vulnerable sectors such as agriculture, construction, trade, and hospitality, which are characterized by informality, seasonal work, and weak regulation. Educational attainment tends to lower the risk of poor job quality, but tertiary graduates face the greatest mismatch between employment and education. When overeducation is excluded from the JQI, tertiary education is positively associated with job quality, pointing to the economy's limited capacity to absorb high-skilled workers into suitable jobs. Among Western Balkan economies, the negative association between tertiary education and job quality is most pronounced in North Macedonia.

Simultaneously, skills mismatches are continuously increasing. According to Labor Force Survey data, about 22 percent of workers are formally overqualified for their jobs, up from only 14 percent in 2017 and 11 percent in 2006. This is in particular the case among tertiary graduates, where 32 percent are overqualified, i.e. working in occupations only requiring secondary education or less. Self-reported data from the Life in Transition Survey confirm these patterns, as 37 percent of workers feel overqualified for their roles. This trend in rising overqualification levels points to a structural demand shortfall, where labor demand has not kept up with the increasing supply of tertiary graduates – not solely a mismatch issue.

Informality remains elevated despite recent progress on formalization. More broadly, high labor taxes, social contribution burdens and benefit structures that discourage formal employment create persistent incentives for informality across the economy. Despite progress on formalization, many jobs remain insecure and poorly matched to workers' skills. About 13 percent of the workforce was informally employed in 2025, a rise compared to 2024, mainly in agriculture, construction, and household services, sectors with limited access to social protection and training.[63] Nearly two thirds of informal workers are in agriculture or construction, which is often seasonal, and where labor laws are weakly enforced. Informal employment undermines workers' rights and limits social insurance coverage while eroding fiscal revenues and distorting competition by allowing less-productive firms to operate at a lower cost. In addition, undeclared work in the formal sector remains pervasive and is estimated at around 21 percent of total hours worked. Undeclared work is particularly harmful in large sectors such as manufacturing and construction, where forgone gross value added is estimated at about MKD 37 billion per year.[64] Viewed against the wage-cost story in Chapter 1, informality is best understood as a symptom of underlying distortions rather than a barrier in itself. Rapidly rising minimum wages and public-sector pay, a tax and social-security wedge of around 33 percent of labor costs, and weak enforcement raise the cost of formal hiring, shift the margin between formal and informal work, and displace less-educated, lower-productivity workers. Tackling these cost and enforcement distortions is therefore more effective than treating informality as a standalone barrier.

(iv) Working conditions - the worker does not work more than 48 hours per week, does not hold a second paid job, and does not wish to work more hours; and (v) Skills match - the worker's education level is not higher than that required for the job.

[63] Informality is calculated using the LFS by identifying workers who report to work in registered firms but are not registered with relevant government institutions and workers who work in non-registered firms.

[64] ILO and FinanceThink. 2023. Granular Examination of the Informal Economy in North Macedonia: Insights into Informal Workers, Undeclared Work and Income Underreporting in Households.

5.2. Youth: turning education into opportunity

As North Macedonia's population ages and shrinks, sustaining growth will increasingly depend on converting education into productive employment. With fewer young labor-market entrants, each cohort's productivity and adaptability matter more. Despite falling unemployment and rising attainment, many young adults still struggle to secure stable, good-quality jobs aligned with their skills. The government's Human Capital Strategy and National Employment Strategy[65] aims to reduce the share of NEET young people to below 20 percent by 2027. The constraints on converting education into good jobs differ by group: for tertiary graduates the binding constraint is mainly on the demand side, while for women supply-side factors dominate, and for NEET youth and the long-term unemployed both sides bind (Table 5).

Table 5. Which constraint binds, by group

Group	Binding constraint	Evidence
Tertiary graduates	Mainly demand-side (too few firms creating high-skill jobs)	High over-qualification concentrated among tertiary workers; weak and falling wage premia for higher education
Women	Mainly supply-side (caregiving, social norms)	40 percent of inactive women cite care responsibilities; low participation despite attainment
NEET youth & long-term unemployed	Both sides bind	Stalled school-to-work transition and high NEET, alongside outdated skills, discouragement, and weak job-matching

The school-to-work transition has stalled, keeping NEET and youth unemployment high. The NEET rate fell from about 30 percent (2017) to 23 percent (2019) but then plateaued—24 percent in 2023—among the highest in the Western Balkans and more than double the EU average (Figure 71a). North Macedonia is the only country in the region without a NEET decline since 2019, and adolescent inactivity is rising: the NEET rate among 15–19-year-olds increased from 9.7 percent (2021) to 12 percent (2023), consistent with weak transitions and incomplete enforcement of compulsory secondary enrollment until age 16. Youth unemployment remained elevated at 29 percent in 2024 (down from 48 percent in 2017), and many unemployed young people—especially women—have exited the labor force.[66] Only 8 percent of students both study and work, far below the EU average (around 25 percent), highlighting limited early work experience.

Employment opportunities for graduates remain limited. Educational attainment is rising, but slow structural transformation and weak firm upgrading constrain the supply of productive, well-matched jobs. Since 2021/22, first-year university enrollment rose by 12 percent; by 2024, nearly 29 percent of the workforce held a university degree. Yet labor demand has not kept pace: an estimated 25,000 jobs requiring tertiary education were created between 2019 and 2024, mostly in mid-level technical or associate roles rather than high-level professional or managerial positions. Vacancy data similarly show demand concentrated in occupations requiring 1–3 years of tertiary education or extended experience, while demand for managerial roles is low (Figure 71b).[67] Only 8 percent of online vacancies are for managerial or professional roles, versus 54 percent for complex technical or practical jobs that may not require tertiary education.[68] Promoting short-cycle higher-education programs with firms could expand employment-oriented pathways that combine academic and practical training aligned with

[65] Human Capital Strategy, Government of Republic of North Macedonia, Skopje 2023; Ministry of Labour and Social Policy. 2021. National Employment Strategy 2021-2027.

[66] Revised LFS data reported by the Republic of North Macedonia State Statistical Office

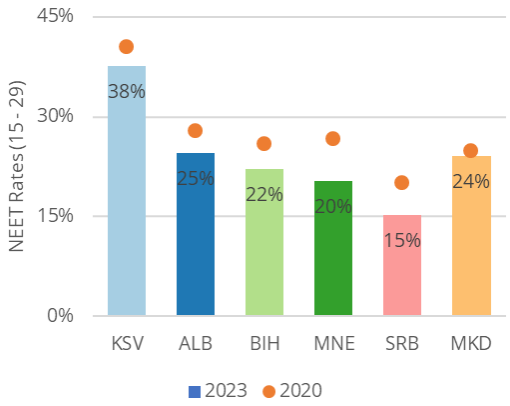
[67] World Bank Staff calculations based on more than 23,000 vacancies collected from North Macedonia's job portals between October 2024 and October 2025. Vacancies were classified into four-digit ISCO-08 occupations using machine learning algorithms and skill levels are classified according to ILO's four skill levels assigned to major ISCO groups.

[68] Managerial and professional roles refer to ISCO Major Groups 1 and 2, whereas Skill Levels 3 and 4 refer to Major Groups 2 - 3 and 4 – 8 respectively.

market needs. Among vacancies for managerial and professional roles, the largest share is in wholesale trade (25 percent), followed by manufacturing and machinery (15 percent) and computer programming (9 percent). Despite rising university attainment, too few firms create high-quality jobs or effectively use advanced skills, reinforcing a low-skill-use equilibrium that limits productivity growth. Promoting short-cycle higher-education programs with firms could offer a medium-term, employment-oriented pathway that combines academic and practical training aligned with market demand.

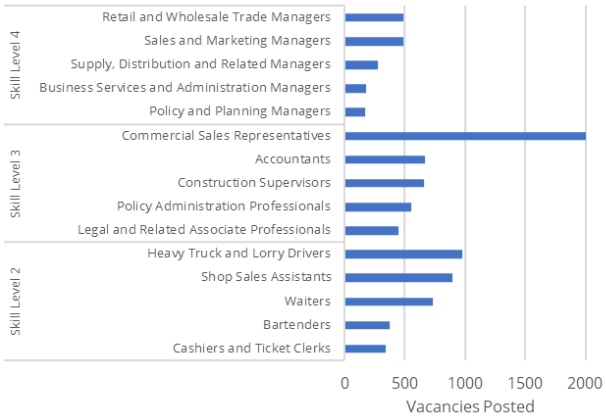
Figure 71. The NEET rate is relatively high and stagnant, and few advertised vacancies are for highly skilled positions

a. NEET Rates in the Western Balkans



Source: State Statistical Office based on LFS

b. Top 5 Occupations in demand by skill level required based on Online Job Vacancy Portals



Source: Online Job Vacancy Data collected by World Bank Staff between October 2024 and October 2025. Skill levels are based on ILO.

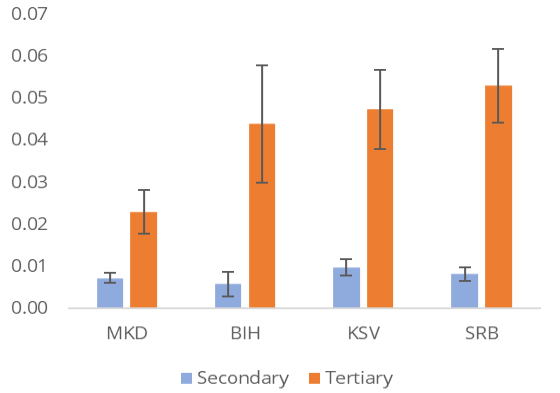
Weak demand for advanced skills is reflected in wage patterns. North Macedonia's premium for each year of tertiary education is 2.3 percent, well below the Western Balkans average (4.2 percent, Figure 72a).[69] Between 2016 and 2020, the conditional wage premium for tertiary graduates relative to secondary-educated workers even fell by 25 percent, while the premium for secondary over primary nearly disappeared.[70] The trend over time, measured on comparable LiTS regressions, points the same way: between 2016 and 2022 the wage return to completing secondary education (relative to less than complete secondary) fell sharply, to the point of becoming statistically insignificant, while the return to completing tertiary education also declined, though by less. As a result the tertiary premium over secondary-educated workers widened, but this reflects deteriorating wages for secondary-educated workers rather than strengthening demand for graduates, whose absolute returns still fell. Narrow and falling premia weaken incentives to invest in higher education and can encourage emigration. Regression evidence also suggests education and experience have become less predictive of earnings overall, reinforcing the need for more higher-productivity jobs and more relevant learning. On their own, these movements are consistent with weak demand for advanced skills but cannot separate a demand shortfall from an oversupply of graduates, qualification mismatch, or the wage compression caused by rapid minimum-wage increases.

Graduate outcomes point to skill underutilization and weak study-job matching. Employment rates rise with education (around 80 percent for tertiary-educated adults), yet many graduates work below their qualification level (Figure 72b). The share of young workers correctly matched to their jobs has not improved relative to older cohorts, indicating that education alone does not ensure productive, well-matched employment. Creating more jobs that use young workers' skills is essential and depends on faster structural transformation and firm upgrading.

[69] Unlike the figures indicating the downward trend these numbers are calculated based on Labor Force Surveys of the latest years.
[70] Based on regression analysis using the Survey of Income and Living Conditions (SILC).

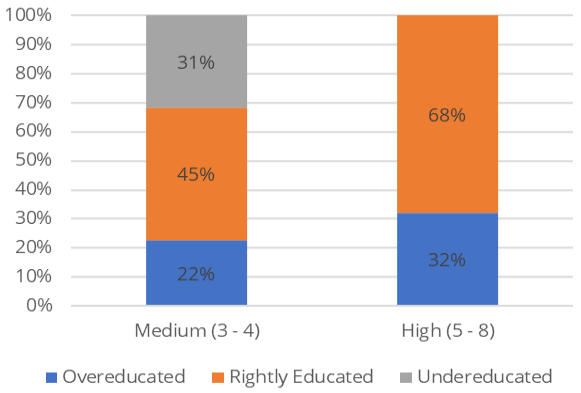
Figure 72. The returns to education are modest, and many workers are overqualified for their roles.

a. Premiums for each year of education by level



Source: World Bank staff calculations based on LFS. Note: Estimates are based on a Mincerian wage regression of log hourly wages and the gender of a worker, controlling for age, age squared and the sector.

b. Over/underqualification by level of education

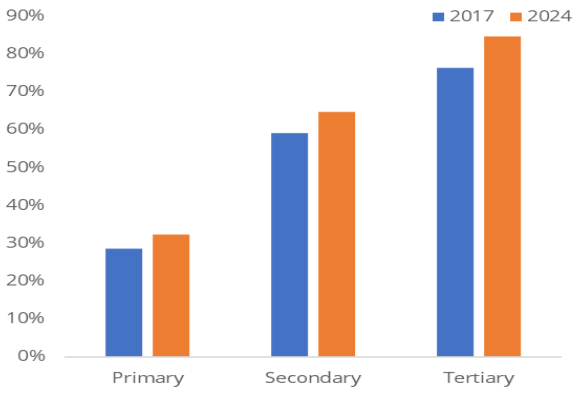


Source: 2023 LFS collected by the State Statistical Office, World Bank staff calculations. *Note:* Under- and overeducation are classified as per ILO skill levels for ISCO-08 major groups, though workers with professional experience may not be underqualified.

Persistent inefficiencies are evident in how skilled labor is absorbed. Eighteen months after graduation, 68 percent of bachelor graduates are employed (Figure 73a), 17 percent continue studying without working, and 15 percent are NEET.[71] Among those employed, over one third work below their qualification level and about 40 percent work outside their field; only 44 percent say their job is closely related to what they studied. Outcomes vary sharply by discipline: ICT graduates have employment rates of 89 percent, a similar level of formalization and strong matching (many employed before graduation), compared to 50 percent of arts, humanities, and social science graduates. Tourism graduates face weaker alignment, more low-skilled or informal work, and fewer formal contracts.

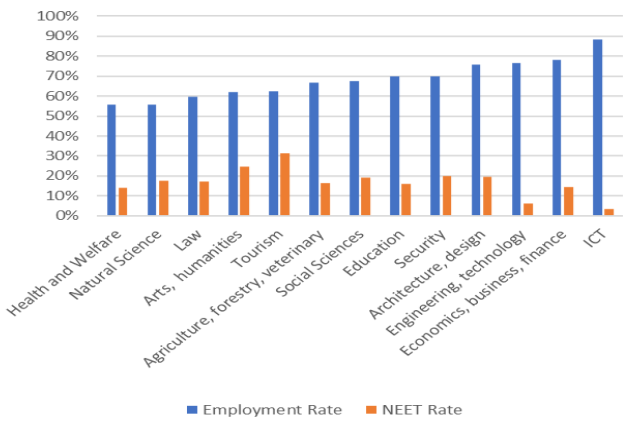
Figure 73. Employment rates rise with education level, but fields of study strongly influence labor-market outcomes

a. Employment rates by education level



Source: State Statistical Office. *Note*: The level of education required for a job is based on ILO's definition of skill required by ISCO's occupational groups at the 1-digit level.

b. NEET and Employment Rates by Field of Study



Source: Brodmann, S., Naceva, B., von Lenthe, C. and Vadle, Z. (2025). *Note*: The employment refers to employed students.

The field of study strongly shapes returns to higher education. Graduate Tracer Survey regressions show large earnings differences: ICT graduates earn about 64 percent more than education graduates; tourism graduates

[71] Brodmann, S., Naceva, B., von Lenthe, C. and Vadle, Z. 2025. Education-to-Work Transition in North Macedonia: Insights from the 2025 Graduate Tracer Survey. The Graduate Tracer Survey was implemented by the Ministry of Education and Science between April and May 2025 and targeted individuals who completed a bachelor's degree at a public university in North Macedonia during the 2022/23 academic year, approximately one and a half years following their graduation. The survey collected detailed information on graduates' labor market outcomes, education-to-work transitions, and perceptions of the relevance and quality of their studies. It also gathered data on job search strategies, job characteristics, skill use, study-job match, continued education, and barriers to employment.

25 percent more; engineering graduates 13 percent more. By contrast, law and business graduates earn 15–17 percent less than education graduates. Women earn 9 percent less overall, with large field-specific gaps (engineering: -22 percentage points; tourism: +33 percentage points). These patterns of underemployment and uneven returns underscore broader weaknesses in how the economy absorbs skills (Figure 73b).

Weak foundational skills deepen the education–work mismatch. International assessments show that by age 15, about two thirds of Macedonian students do not reach minimum proficiency in reading or mathematics—more than three school years behind EU peers—and learning losses between 2018 and 2022 were among the region's largest. Low learning quality affected all socioeconomic groups equally.[72] Early childhood education is expanding but remains uneven: preschool enrollment for ages 3–6 rose from 36 percent (2018) to 47 percent (2024). Over half of preschools had upgraded process-quality standards and over 90 percent of teachers received competency-based training, yet reading comprehension and basic mathematics bottlenecks remain.[73] Urban–rural and socioeconomic gaps persist, and at later stages short instructional hours, memorization teaching, and absenteeism weaken learning and reduce motivation for math/science, narrowing pathways into high-skill fields.[74] Employers report deficits in digital, analytical, and problem-solving skills and an inadequately educated workforce is cited as the second-most significant business constraint, especially among larger and exporting firms.[75]

Limited opportunities to use skills at home continue to push young people abroad. Emigration is disproportionately high among the most educated, eroding the domestic human-capital base. Wages in more advanced EU economies are up to three times higher than in the Western Balkans, even after cost-of-living differences.[76] In the 2024 OeNB Euro Survey, about 11 percent of people in North Macedonia intended to emigrate within 12 months (the highest in CESEE), and roughly one in five either planned to emigrate or was uncertain over a longer horizon.[77] Graduate-tracer evidence is consistent: almost 10 percent of young people plan to seek employment abroad,[78] and about 7 percent of NEET tertiary graduates intend to emigrate.[79] Beyond pay, limited career progression, few further-education options, and low professional prestige also drive outward mobility. Without higher-quality domestic jobs, education risks becoming an export rather than an investment.

5.3. Women: unlocking the largest underutilized source of human capital

With the workforce shrinking and youth cohorts declining, leveraging women's economic potential is the most effective way to sustain growth and counter labor shortages. The underutilization of female labor is a major constraint: in 2025, 56.7 percent of working-age women were economically inactive, compared with 38 percent of men (Figure 74a), despite women's stronger educational outcomes (31 percent of women in the labor force hold a university degree versus 27 percent of men). Even so, women are 15 percentage points more likely than men to be NEET by age 24 (Figure 74b).

Care responsibilities remain the most important barrier to women's labor-force participation. In recent surveys, 40 percent of inactive women cite unpaid caregiving as their reason for not working (Figure 75). The Life

[72] Shmis, Tigran; Cahu, Paul; Brajkovic, Lucia; Gresham, James. 2025. Unlocking Potential: Enhancing Education Quality in the Western Balkans Based on Insights from the PIRLS 2021 and PISA 2022 Studies. Education Working Paper; No. 7, January 2025. © World Bank. http://hdl.handle.net/10986/42680

[73] World Bank. Implementation Completion Report of Social Services Improvement Project. https://documents1.worldbank.org/curated/en/099122325174519076/pdf/BOSIB-d2fab876-6021-422b-9b10-abeef39ec4ba.pdf

[74] Shmis et al. 2025.

[75] World Bank. 2024. Enterprise Surveys: What Businesses Experience. North Macedonia 2023 Country Profile.

[76] Katharina Allinger, Elisabeth Beckmann, Söhnke Bergmann, Melanie Koch, Thomas Scheiber, Jette Leonie Weinel: OeNB Report 2025/13: OeNB Euro Survey in Central, Eastern and Southeastern Europe: first results for 2024. Available at: https://www.oenb.at/en/Publications/Economics/reports/2025/report-2025-13-euro-survey/html-version.html

[77] World Bank. 2025. Western Balkans Regular Economic Report No. 26: Retaining the Growth Momentum. Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/099101424043012347

[78] Gjeorgjievski, D., Stamenkovski, G. and Bozhanova, G. 2024. Technical Assistance to the Sector Reform Performance Contract 'EU for Youth': Tracer System Report. EuropeAid/140701/DH/SER/MK

[79] Brodmann, S., Naceva, B., von Lenthe, C. and Vadle, Z. 2025. Education-to-Work Transition in North Macedonia: Insights from the 2025 Graduate Tracer Survey.

in Transition Survey (2023) also finds persistent norms: 37 percent of women and 50 percent of men agree that a woman should handle most household chores even if her husband is not working; the 13 percentage-point gender gap in these attitudes is the second largest in the region, helping explain why caregiving continues to fall disproportionately on women and limits access to full-time, higher-quality jobs.

Figure 74. Despite rapid gains in educational attainment, women remain underutilized in the labor force.

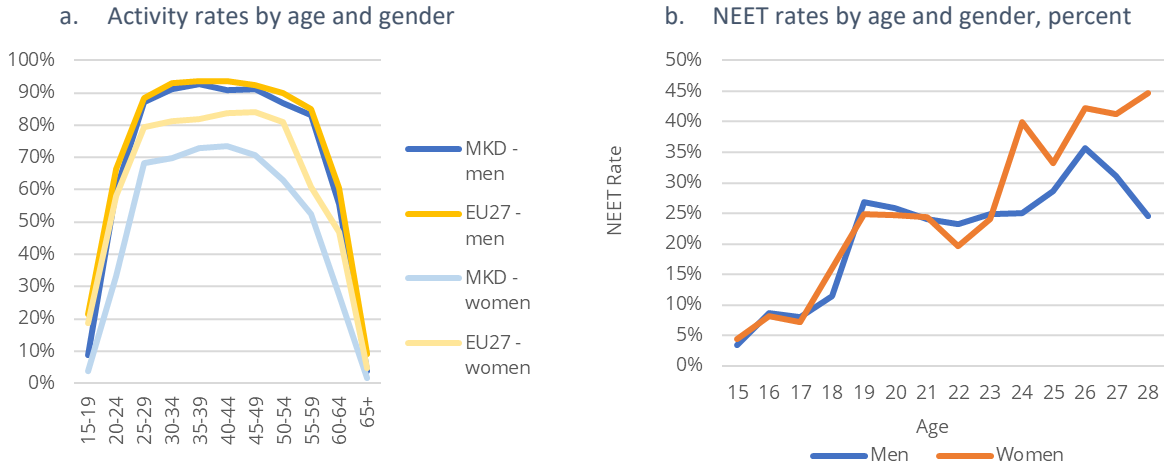
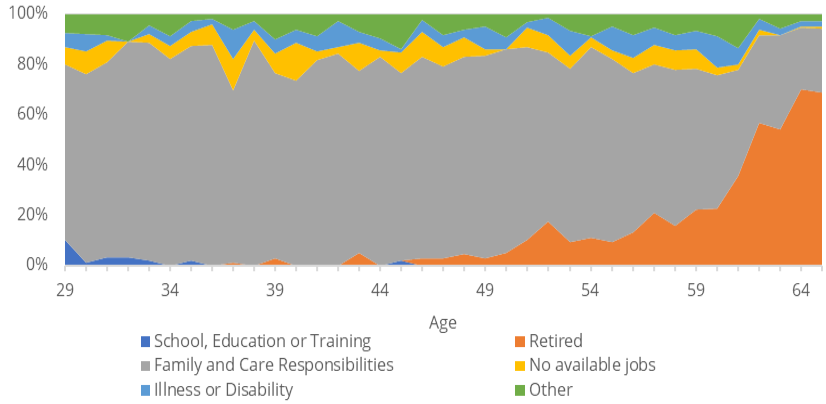


Figure 75. Reasons for inactivity among women by age



Source: 2023 LFS collected by the State Statistical Office, World Bank staff calculations

Expanding access to quality early childhood education and childcare is vital to increase women's labor-force participation and reduce intergenerational poverty and inequality. Provision has expanded and quality has improved: preschool enrollment among children aged 3–6 rose from 35 percent in 2015 to 46 percent in 2024, more than 90 percent of preschool teachers have received competency-based training, and new quality-assurance tools such as the Measuring Early Learning Quality and Outcomes (MELQO) instrument provide nationally comparable evidence. Yet access remains highly unequal—urban enrollment (47 percent) is more than double rural enrollment (20 percent), and 13 mainly rural municipalities still have no early childhood education and childcare facilities—leaving many mothers, especially in rural areas and poor households, without reliable childcare.

Demographic aging is increasing demand for long-term care, with major implications for women's employment and fiscal sustainability. By 2040, more than one fifth of North Macedonia's population will be over age 65, but formal eldercare remains limited and uneven. Under the 2019 social protection reform, a pilot system of licensed nonresidential services—largely in-home eldercare—has reached over 5,400 beneficiaries across more than half of all municipalities and created new formal jobs, many for previously unemployed women; still, only about one in ten elderly people currently receives formal long-term care. Public spending is just 0.3 percent of GDP (among the lowest in Europe) and could rise to 0.6 percent by 2040 from demographic pressures alone, while EU-

comparable coverage would require spending over 2 percent of GDP.[80] Expanding affordable home- and community-based care with a coherent financing framework is therefore essential.

Care constraints also translate into persistent gender gaps in job quality and pay, and policy can do more to support women's full participation. Women are more likely to work in lower-paying sectors and occupations; the adjusted gender wage gap is 9.9 percent[81] and the graduate wage gap is 13.5 percent even after controlling for field of study.[82] Women are overrepresented in public administration, education, and health (often lower pay but more flexibility), while men dominate higher-paying export-oriented industries such as ICT, engineering, and manufacturing. North Macedonia's labor framework provides employment protection and maternity leave but remains only partially aligned with EU directives on gender equality and work/life balance; paternity leave is rare and flexible work options are limited. Closing these gaps will require a coordinated package that expands childcare and eldercare in rural and underserved municipalities (including part-day and extended-hour models), supports reentry and career continuity through ALMPs (wage subsidies, reskilling grants, and digital upskilling), modernizes rules to enable flexible work and shared parental leave, enforces equal pay for work of equal value, and helps women enter high-productivity sectors by aligning education and training with demand in ICT, engineering, and green industries. North Macedonia's own experience shows these measures can deliver: recent care-economy reforms and a social mentoring pilot have moved previously unemployed and social-assistance-dependent women, many of them Roma, into formal jobs, with gains sustained a year on and returns exceeding program costs (Box 6).

Box 6. From social assistance to jobs in the care economy: results from recent reforms

North Macedonia's own experience shows that well-designed care and activation reforms can move caregivers into formal work. The 2019 social protection reform—operationalized through national service standards, licensing, pricing, and competitive grants for non-institutional services—has expanded community-based care nationwide. Licensed nonresidential (largely in-home) eldercare now reaches over 5,800 beneficiaries across more than half of all municipalities and has created several hundred new formal care jobs, in part filled by long-term unemployed women, establishing a credible jobs pipeline in a sector with high social returns and strong relevance for women.

Job creation alone, however, proved insufficient for the most vulnerable. A tracer study of Roma women in caregiver certification found that care responsibilities, social norms, income insecurity, and fear of losing benefits kept many from taking up formal jobs even after certifying. An EC-financed, World Bank–executed social mentoring pilot (May 2024–March 2025) tested a response across eight municipalities, enrolling 121 women (74 percent Roma) in government-accredited caregiver training, which requires only primary education, paired with peer mentoring by women from their own communities. Results were strong and durable. Of those enrolled in training, 97 percent completed certification; the number in employment rose from 14 at baseline to 60 at program exit, and 57 remained employed one year later. Among employed participants, 91 percent reported an improved financial situation; reported life satisfaction rose from 50 to 86 percent; and 90 percent of those still unemployed felt more employable.

The pilot was also a sound fiscal investment: against a public cost of about US$245,000, participants' earnings generated a net return of roughly US$108,000 in the first year, rising to about US$1.36 million over five years—an estimated four to seven dollars of earnings per dollar invested. Adding avoided social transfers (saved guaranteed minimum assistance, child, and educational allowances, counted from the second year onward) and new income-tax revenue, the combined net return could reach about US$2.1 million over five years, or US$9.41 per dollar invested.

The core lesson is clear and well-evidenced: skills alone are not enough for beneficiaries facing long-term unemployment and care duties, but integrated support, i.e., training combined with mentoring, family mediation, employer engagement, and post-placement follow-up, turns job creation into durable employment, and does so at a net fiscal and economic gain. North Macedonia should therefore scale these lessons by introducing income disregards to reduce disincentives to formal

[80] Coll-Black, Sarah; Hamandi, Ali; Shimbov, Bojan; Wiener, Mitch. 2025. Assessing current and future expenditure on Long-Term Care in North Macedonia.

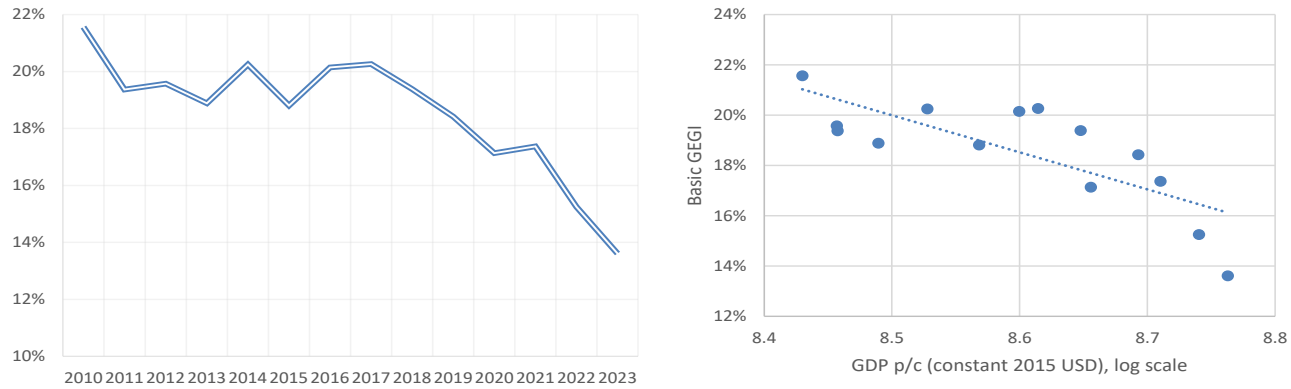
[81] Estimates are based on a Mincerian wage regression of log hourly wages and the gender of a worker, controlling for age, age squared, sector and occupation.

[82] Brodmann, S., Naceva, B., von Lenthe, C. and Vadle, Z. 2025. Education-to-Work Transition in North Macedonia: Insights from the 2025 Graduate Tracer Survey.

work, strengthening coordination between Centers for Social Work and the Employment Service Agency, and embedding social mentoring as an instrument in the active labor market toolkit.

Achieving gender parity in employment could raise long-run GDP per capita by roughly 13.6 percent. The Basic and Full Gender Employment Gap Index (GEGI, Annex 4)[83] results for North Macedonia show substantial improvement between 2010 and 2023. The Basic GEGI employment gap fell from 21.6 to 13.6 percent, and the Full GEGI declined from 20.1 to 11.9 percent, indicating markedly stronger labor-market inclusion of women over that period (Figure 76). Both indices are inversely related to GDP per capita, suggesting that economic development has coincided with (and may reinforce) narrower gender disparities; the close alignment between the Basic and Full measures also reflects a rising share of "better employment" over time. North Macedonia's employment gap is slightly better than the Western Balkans average on the Basic measure, though it trails regional averages on the Full measure. Even so, the remaining gap implies large unrealized economic gains. The analysis therefore points to targeted, gender-responsive labor-market policies—expanding access to formal and higher-quality jobs, reducing barriers to participation, and supporting skills and human-capital development—to sustain progress and translate inclusion into stronger, more equitable growth.

Figure 76. The Basic GEGI (left) and its relationship with GDP per capita (right)



Source: Author's calculations.

5.4. Global transitions will reshape labor demand and expose new skill gaps

The global technological and climate transitions are set to amplify North Macedonia's labor market challenges. Rapid advances in artificial intelligence (AI) and the shift toward carbon neutrality will reshape labor demand and required skills, creating both displacement risks and productivity/job-creation opportunities depending on how quickly workers, firms, and institutions adapt. Near-term pressure will rise with the EU Carbon Border Adjustment Mechanism (CBAM), while AI-enabled automation will increasingly transform even high-skilled work—raising the urgency of aligning education, training, and ALMPs with the changing nature of work. This is particularly relevant for the push toward higher-value services and digital trade: realizing productivity and jobs gains will depend on upgrading skills and ensuring technology adoption is AI-complementary.

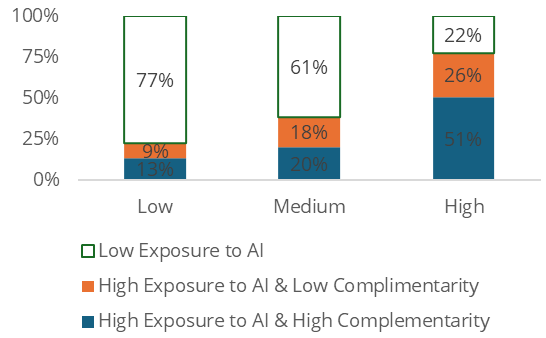
AI is expected to disproportionately affect workers with medium to high levels of education. Projections suggest that more than three quarters of tertiary-educated workers in North Macedonia are exposed to AI in their daily tasks (Figure 77a). For about two thirds, AI is expected to complement work and raise productivity, while for the remaining third it increases displacement risk. North Macedonia also has the Western Balkans' highest share of jobs with high AI exposure but low complementarity, implying a larger share of roles could be replaced rather than augmented.[84] Risks are especially acute in ICT (82 percent of jobs are highly exposed with low complementarity) and among prime-age workers, reinforcing the need for continuous learning and reskilling.

[83] The basic GEGI measures the gap between male and female employment as a share of total employment. The full GEGI is a weighted average of a "better employment gap" and "other employment gap," providing a more nuanced view of gender employment disparities.
[84] World Bank. 2025. Western Balkans Regular Economic Report No. 28: Towards Better Jobs. Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/099100625100539134

The green transition will mainly affect low-skilled workers and carbon-intensive industries. In North Macedonia, about 12 percent of jobs are expected to undergo substantial task changes, and 8 percent are directly tied to carbon-intensive industries likely to contract.[85] Unlike AI, the green transition hits low-skilled workers hardest: only 14 percent of tertiary-educated workers are estimated to hold a brown job or require substantial reskilling to access green jobs, compared with one in four poorly educated workers (Figure 77b). Mining, energy, and manufacturing are among the most exposed; without faster reskilling and regional diversification, displaced workers—especially outside major cities—may struggle to find alternatives, even as opportunities grow in renewables, building retrofits, and environmental services.

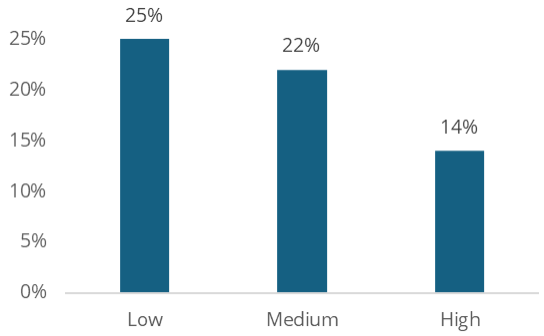
Figure 77. Workers in North Macedonia are highly exposed both to AI and to the green transition

a. Exposure to AI and complementarity of occupation by education levels



Source: World Bank Staff Calculations using 2023 LFS based on Cazzaniga et al. 2024 and Felten et al. 2021.

b. Share of workers in a "brown job" by education level



Source: World Bank. 2025. Regional Economic Report: Adapting for Sustainable Growth

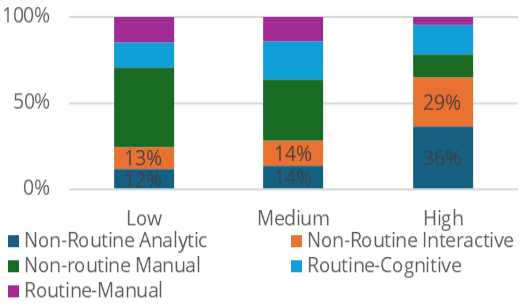
Labor demand remains concentrated in occupations most exposed to these twin transitions. Global employer sentiment suggests rising demand for analytical thinking, creativity, and problem-solving, alongside declining routine/manual tasks[86]. Yet vacancy data from Employment Service Agency (ESA) indicate North Macedonia's demand is still concentrated in occupations relying on physical abilities and foundational skills, with low demand for cognitive and analytical roles. This points to continued cost-based competition and leaves the economy vulnerable to disruption; without stronger incentives for technology adoption and skills upgrading, North Macedonia risks remaining in a low-skill, low-productivity equilibrium as global demand shifts toward knowledge- and innovation-intensive work.

The workforce is not prepared for the transformative impact of AI and carbon neutrality. Only 20 percent of all workers and 36 percent of tertiary graduates in North Macedonia are employed in jobs requiring non-routine analytical tasks, and vulnerable workers face skills gaps that limit access to green jobs (Figure 78)[87]. Women in brown jobs show slightly higher skill gaps than men; in AI-exposed jobs, women tend to have larger knowledge gaps but smaller ability gaps than men. These patterns point to the need for targeted upskilling/reskilling, especially for poorly educated, unemployed, and female workers whose current skills are less aligned with future labor demand. Given this will likely accelerate the dynamism of the labor market, there will be an increasing need for adaptive social protection systems that support workers through transitions.

[85] World Bank. 2025. Western Balkans Regular Economic Report No. 27: Adapting for Sustainable Growth. Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/099042325021511545.
[86] WEF. 2025. The Future of Jobs Report 2025. https://www.weforum.org/publications/the-future-of-jobs-report-2025/
[87] World Bank. 2025. Western Balkans Regular Economic Report No. 27: Adapting for Sustainable Growth. Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/099042325021511545.

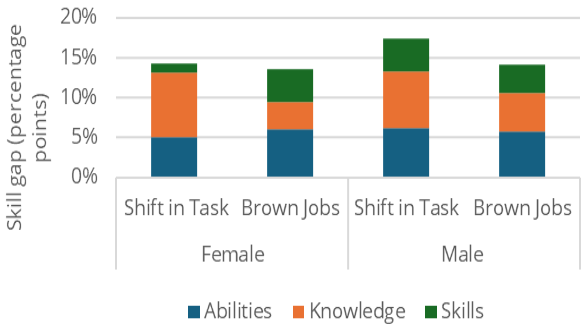
Figure 78. Workers in North Macedonia are highly exposed both to AI and the green transition.

a. Tasks content of North Macedonia's occupations by required level of education



Source: 2023 Labor Force Survey collected by the State Statistical Office, World Bank staff calculations based on Mihaylov and Tijdens (2019). Required level of occupation is determined using ILO skill levels assigned to ISCO-08 Major Groups.

b. Decomposition of general skill gaps for most vulnerable workers by risk type and gender



Source: World Bank Staff Calculations based on World Bank staff calculations based on the methodology of Gukovas, Garrote-Sanchez, Makovec (forthcoming, 2025) using 2023 LFS data.

To seize the opportunities and mitigate the risks arising from AI and the green transition, North Macedonia needs a coordinated strategy to future-proof its workforce and firms. Priorities include integrating AI and green skills across education and training (with early-to-lifelong emphasis on analytical thinking, problem-solving, and digital literacy), targeting at-risk workers with employer-linked reskilling for transferable skills through regional training centers, and accelerating firm upgrading through fiscal incentives, technology-adoption grants, and partnerships that connect training to productivity-enhancing investment.

Strengthening the link between education and employment is critical so graduates gain skills demanded by the labor market and can contribute productively from the start of their careers. While most tertiary graduates eventually find jobs, many enter late or accept positions below their qualifications, reflecting structural gaps: limited work-based learning, weak employer engagement, and insufficient coordination among universities, the ESA, and firms. Closing these gaps requires stronger collaboration and feedback loops to align study programs, internships, and career services with changing needs, and to complete implementation of the 2018 Higher Education Law—especially external quality assurance and graduate-outcomes data collection.

Continuous skill upgrading is increasingly important as technologies and job requirements evolve, but the skills ecosystem remains fragmented and insufficiently demand driven. North Macedonia has advanced lifelong learning through the Center for Adult Education (CAE)[88], short-cycle higher vocational programs, and ESA-led ALMPs, yet these operate largely in silos and labor-market intelligence weakly informs program design and funding. The result is duplication in some fields (e.g., personal care) and gaps in others (e.g., industrial and technical trades).[89] Training alignment relies heavily on ESA's Annual Skills Needs Survey, which is useful for ALMP planning but remains short-term and skewed toward lower-skill demand. The survey projects employer needs over the next 6–12 months; the latest results show over half of expected new hires in manufacturing, followed by wholesale and retail, administrative services, ICT, and construction, and about three quarters of projected jobs requiring secondary education or less. Without complementary forward-looking intelligence (e.g., graduate-tracer studies and sector analyses), education and training risk expanding in low-productivity areas while undersupplying technical, digital, and green skills needed for future growth.

The ESA remains the key provider of adult training, but its relative importance is declining as non-public providers expand in an uncoordinated system—and outcomes remain weak. Between 2023 and 2024, ESA-

[88] The CAE plays a central role in accrediting non-formal training providers, verifying programs, and managing recognition of prior learning. As of early 2025, 280 verified programs were registered with the CAE: these are short vocational courses typically ranging from 100 to 250 hours, leading to nationally recognized certificates delivered by various providers (private training companies, NGOs, and other institutions, often in partnership with the CAE).

[89] Mojsoska-Blazevski, Nikica. 2025. Mapping of formal and no formal post-primary education and training system in North Macedonia. Unpublished manuscript.

financed training participants for in-demand occupations fell 43 percent (2,062 to 1,184) while average costs per trainee rose by more than 25 percent; meanwhile overall adult-training participation increased (3,461 to 3,848) and ESA's share dropped from 85 to 54 percent. Within ESA training, provision shifted toward social services and personal care while industrial/technical trades (e.g., machine operators, welders, crane drivers) were scaled back despite employer demand.[90] Labor-market entry is limited: only about 1 in 5 beneficiaries is formally employed a year after training, with particularly low absorption in beauty and care services, suggesting risks of informality or underemployment. Despite a broadly sound framework, participation in verified non-formal programs is just 0.3 percent of working-age adults annually, far below EU averages (EU27: 21.1 percent participated in the last 4 weeks in 2024).[91] A smaller cohort with more intensive support could improve impact if funding prioritizes in-demand occupations, coordination among CAE/ESA/employers accelerates approvals, and financing is tied to verified employment outcomes.

Beyond annual ALMP planning, North Macedonia needs a coordinated, demand-driven lifelong-learning system and stronger labor-market intermediation to translate skills into productive jobs. VET remains small-scale and donor-dependent, even as the Law on Adult Education envisages recognition of prior learning to help scale reskilling. Programs should be modular, flexible, co-designed with employers, and informed by real-time intelligence.[92] Short-cycle higher education/vocational programs (less than 500 active students across eight private and one public institution) are promising but underutilized, mostly focused on ICT and business and attracting younger students; while expansion into technical/engineering fields relevant to the green transition and industrial upgrading is needed.[93] Institutionally, fragmented mandates and data systems across CAE, the Ministry of Education and Science, and the ESA limit accountability—a Labor Market and Skills Observatory could integrate administrative data, employer surveys, tracer studies, and sector analyses to guide funding and curricula. Finally, the ESA should modernize from administrator to strategic intermediary through improved staffing, digital tools, case management, employer outreach, coordination with social services and providers, and evidence-based monitoring—rebalancing ALMPs toward high-demand technical and green occupations and linking public resources to sustained placement and skill outcomes.

5.5. Mobilizing all available workers to sustain growth

Demographic pressures are already contributing to labor shortages that will deepen even if gender gaps narrow. Even with substantial gains in women's participation, North Macedonia's labor force is expected to begin shrinking within a decade; projections extrapolating 2019–23 trends suggest shortages could intensify starting in 2027 and widen further after 2033 as youth cohorts contract (Figure 79). Addressing future labor demand will therefore require a combined approach: supporting women's and youth entry/reentry, activating unemployed workers, extending working lives, and harnessing migration more strategically.

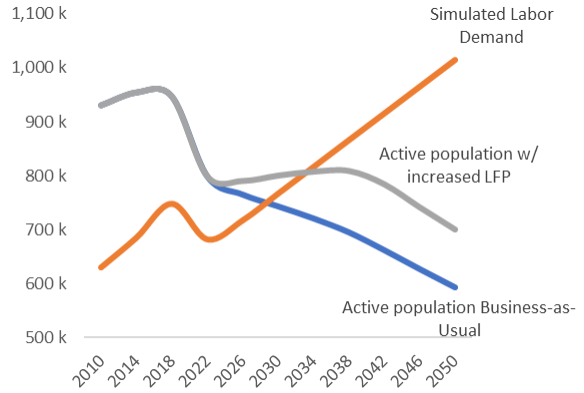
Reintegrating unemployed workers into the labor market requires intensive and personalized support. Although unemployment has fallen to 11.4 percent in 2025, more than 91,000 people remain unemployed, nearly three quarters of whom are long-term unemployed.[94] Many face multiple barriers—outdated skills, discouragement, and weak job-matching—and skills analysis shows much larger gaps among the unemployed, limiting access to occupations less exposed to the green transition. Standard low-intensity activation is unlikely to suffice: international experience underscores the impact of integrated services such as personalized case management, coaching, and post-placement support.[95] Promising pilots—including social mentoring in the care

[90] Mojsoska-Blazevski, Nikica. 2025.
[91] According to Eurostat, 21.1 percent of adults aged 25–64 in the EU-27 participated in formal or non-formal education or training in the four weeks preceding the survey in 2024. Source: Eurostat (2025). "Participation in Education and Training (last 4 weeks)," https://ec.europa.eu/eurostat/databrowser/view/trng_lfse_01.
[92] World Bank. 2025. Understanding Training Pathways into in-Demand Engineering Technician in North Macedonia. Forthcoming.
[93] Mojsoska-Blazevski, Nikica. 2025.
[94] Revised Labor Force Survey as reported by the State Statistical Office.
[95] European Commission. 2020. European Network of Public Employment Services: Integrated services for Longterm Unemployed.

economy[96] and targeted wage subsidies with strong employer engagement and job matching[97]—suggest that scaling will require stronger Employment Agency capacity and tighter coordination with training institutions and social service providers. Nevertheless, the employment effects of activation policies are likely to depend on the availability of suitable jobs and the intensity and targeting of support. Standalone activation measures may yield more modest aggregate employment gains if labor demand constraints or job quality concerns persist. North Macedonia's social mentoring pilot in the care economy illustrates the payoff from such integrated support: pairing accredited caregiver training with peer mentoring raised the number in employment among 121 participants from 14 at baseline to 57 still employed a year later, at an estimated five to eight dollars of earnings per dollar invested (Box 6).

Figure 79. Labor shortages will emerge even if participation increases, while unemployed or inactive will need support to enter the labor market

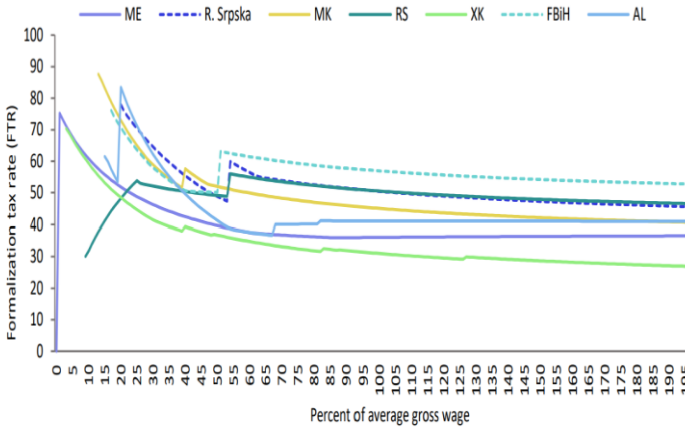
(Projected trends in the labor market)



Source: World Bank staff calculations based on LFS data. *Note*: The baseline scenario extrapolates supply and demand trends using data from 2019-23. Labor supply is modeled as the product of the working-age population and activity rates while labor demand is derived from the difference between GDP growth and labor productivity growth.

Figure 80. The cost of formalization is high across the Western Balkans

(Formalization rate)



Source: Western Balkans Labor Market Brief 2023, World Bank & wiiw (2025).

High reservation wages and structural disincentives discourage formal work. Extended joblessness, remittances, and informal income sources (including seasonal work in neighboring countries) keep reservation wages high and reduce incentives to accept available jobs.[98] These pressures are reinforced by labor taxation and social protection: taxes and social security contributions are about 33 percent of total labor costs, and minimum contribution requirements make formal work especially costly for part-time and low-wage workers (Figure 80).[99] This further underscores the importance of aligning wage developments with productivity gains, as higher wage floors alongside persistent formalization costs will continue to increase incentives for firms to rely on informal work. Many low-income workers also place limited value on uncertain future benefits relative to immediate earnings, while poverty-targeted benefits are withdrawn immediately upon formal employment, creating a steep "benefit cliff" (e.g., guaranteed minimum assistance can be lost entirely when one household member registers for formal work). To reduce disincentives, benefits should phase out gradually as income rises, minimum contribution thresholds should be lowered or eliminated, and the tax wedge for low earners reduced. The new

[96] World Bank. 2025. North Macedonia's New Care Economy: Changing Lives, Creating Jobs. Available at: https://www.worldbank.org/en/news/feature/2025/04/10/north-macedonia-s-new-care-economy-changing-lives-creating-jobs

[97] Alex Armand, Pedro Carneiro, Federico Tagliati, Yiming Xia. 2025. Can subsidized employment tackle long-term unemployment? Experimental evidence from North Macedonia, Journal of Development Economics, Volume 178, 2026.

[98] Petreski, Marjan. 2025. Analysis of Reservation Wages in North Macedonia. Unpublished.

[99] World Bank and wiiw. 2025. Western Balkans Labor Market Brief 2023. Available at: https://documents1.worldbank.org/curated/en/099061725062028120/pdf/P176230-dd9afa03-b9ea-45f0-8442-ef5eae1ba3f0.pdf

Law on Seasonal and Temporary Employment aims to simplify formalization for short-term and low-income work, though impacts on incentives and protection remain to be assessed. These reforms are compatible with overall fiscal sustainability. Guaranteed minimum assistance and related means-tested benefits absorb only about 0.5 percent of GDP and reach a narrow group of beneficiaries, so phasing benefits out gradually as income rises, lowering minimum contribution thresholds, and trimming the tax wedge for low earners can be designed at limited net fiscal cost, particularly as higher formal employment broadens the contribution base.

Extending working lives and managing migration will be crucial as demographic aging and emigration tighten labor supply. Men and women in North Macedonia exit the labor market earlier than EU peers: for men, participation falls from 83 percent in the mid-to-late 50s to 55 percent in the early-to-mid 60s, while for women it drops from 52 percent to 27 percent; meanwhile, the old-age dependency ratio is projected to double by 2050 (from one elderly person per four working-age adults to one per two).[100] Aligning working lives with life expectancy through gradual retirement-age increases, incentives for delayed retirement where possible, lifelong learning, and policies for healthy and productive aging can mitigate shortages and protect pension sustainability. Emigration continues to erode the workforce: North Macedonia's youth cohorts are expected to shrink by about one quarter by 2030.[101] The share of highly educated emigrants is about 12 percent (lower than some peers), but regional trends suggest rising tertiary-educated emigration; reintegration support, pre-departure counseling, and stronger diaspora engagement can help stabilize labor supply and foster knowledge transfer and investment. At the same time, immigration is increasingly used to address shortages: the Western Balkans hosted over 1.1 million foreign-born residents in 2020, often filling construction, agriculture, and services gaps,[102] but North Macedonia's integration framework remains weak, notably on education access and labor-market mobility.[103] A coherent migration-management strategy—built on bilateral agreements linked to forward-looking skills monitoring and streamlined visa/permit processes that reduce duplicative steps—can better align inflows with sectoral labor needs.[104]

To mitigate demographic decline and sustain growth, North Macedonia needs a coherent strategy to fully utilize its available labor force. Priorities include: (i) activating long-term unemployed workers through intensive case management (career guidance, tailored training, mentorship, post-placement support) and stronger partnerships among the Employment Agency, Centers for Social Work, and local training/service providers; (ii) reforming employment incentives by gradually phasing out social assistance under the Social Protection Law as income rises and reassessing labor taxes to reduce penalties for low-wage and part-time work; (iii) encouraging longer working lives by extending the statutory minimum retirement age for women, strengthening incentives to work longer, expanding lifelong learning, and supporting healthy and active aging; (iv) harnessing migration by facilitating returnee reintegration, engaging the diaspora in investment and skills partnerships, and aligning immigration policy with sectoral labor needs; and (v) strengthening cross-sector coordination so activation, tax-benefit design, migration, and care reforms work together to mobilize all available workers.

[100] UN DESA Population division, World Population Prospects: Revision 2024.
[101] World Bank. 2025. Western Balkans Regular Economic Report No. 28: Towards Better Jobs. Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/099100625100539134
[102] Ibid.
[103] North Macedonia scores 42 out of 100, with particularly low scores in access to education and labor-market mobility. See: Migrant Integration Policy Index 2019. Accessible at: https://mipex.eu/macedonia-fyrom
[104] World Bank. 2025. Western Balkans Regular Economic Report No. 26: Retaining the Growth Momentum. Washington, D.C.: World Bank Group. http://documents.worldbank.org/curated/en/099101424043012347

6. Policy reforms for growth and job creation in North Macedonia

Consistent with the national development priorities identified in the National Development Strategy 2024-2044 and with the EU Growth Plan for North Macedonia, the three reform clusters—strengthening economic governance and competition; strengthening technology infusion, energy efficiency and innovation; and developing skills and mobilizing workers—could lift long-term growth above 3.8 percent, increase real wages by 24.6 percent above the baseline scenario by 2050 and create the equivalent of 250,000[105] more and better paid jobs. This reform scenario envisages approximately 53,000 new jobs and 197,000 better-paid jobs – about one third of the current employment in the country. The policy simulations therefore underscore that a coherent set of reforms can fundamentally reshape North Macedonia's growth and employment trajectory over the long run.

6.1. Policy reform roadmap

The policy reform roadmap around three reform clusters represents the main channels through which reform policies generate economy-wide effects. The first cluster, *strengthening economic governance and competition*, targets the distortions and inefficiencies that prevent firms and markets from functioning at their potential, including barriers to competition, weaknesses in public spending, and constraints on trade and the business environment. The second cluster, *strengthening technology infusion, energy efficiency, and innovation*, pushes the productive frontier outward by fostering R&D, deepening FDI linkages, and accelerating the adoption of energy-efficient technologies. The third cluster, *developing skills and mobilizing workers*, shapes labor supply, participation rates, and the skills of the workforce. Taken together, governance and competition reforms raise the returns to factors of production, technology and innovation reforms push the productive frontier outward, and labor market reforms broaden participation and distribution of gains within the economy.

The prioritization of policy recommendations was conducted in three stages. Priorities aimed to target underlying constraints – the structural, institutional, or policy failures that drive observed challenges—have been assigned the high priority. These are foundational reforms. This was followed by filtering for sequencing as the order in which reforms are implemented matters. Finally, feasibility and ambition must be balanced. Reforms should be ambitious enough to drive real impact but realistic given the country's institutional capacity, political landscape, and available technical support. A phased approach — tackling high-impact, practically achievable reforms first — can outperform an overly ambitious program that stalls.

Healthy competition encourages private investment, boosts productivity, enhances market efficiency and thus competitiveness and job creation. To more effectively promote competition in North Macedonia, the authorities should consider: (1) reducing the potential distortions created by the state's presence in the economy by establishing a clear rationale to justify state presence, especially in competitive sectors, and by strengthening the corporate governance of SOEs through the adoption of best practices for transparency and accountability; (2) strengthening the regulatory and institutional frameworks for competition to foster greater contestability; (3) enhancing the state-aid framework to limit market distortions; and (4) addressing economy-wide and sectoral regulatory constraints, including administrative obstacles to entrepreneurship and the state's monopolies.

Recommendations for strengthening economic governance and competition	Priority	EU agenda
A. Reducing the potential distortions induced by the state presence		
1. Strengthen the legal framework for SOEs by formally defining the economic rationale for state ownership and reducing government stakes in SOEs that do not conform to this rationale.	High	√

[105] The more and better paid jobs estimate is based on the WBG scorecard methodology capturing both increases in employment (the extensive margin of "more jobs") and increases in earnings per worker (the intensive margin of "better-paid jobs").

2. Enhance corporate governance of SOEs, especially those in competitive sectors, by centralizing ownership, electing independent boards and CEOs, and publishing annual performance information.	Medium	√
B. Strengthening the regulatory frameworks for competition to foster greater contestability		
1. Consider implementing the "silence is consent" principle[106] as a standard for issuing business licenses and permits.	Medium	
2. Strengthen the enforcement of rules against anticompetitive conduct by: (a) clarifying public-interest exemptions; (b) developing a leniency program to fight cartels; (c) amending the law to apply an effects-based approach to all unilateral conduct; and (d) no longer using market share as the basis for the presumption of dominance.	High	√
3. Focus the CPC's resources on most problematic merger cases by: (a) eliminating legal notification thresholds to allow for regular review, adding individual thresholds to the aggregate ones, and eliminating market-share-based criteria; (b) including a simplified notification procedure for mergers that should have a limited effect on competition similar to the short form used by the European Commission; (c) specifying in secondary legislation the maximum number of 20-working-day extensions allowed after the original 90-day deadline, with a limited additional extension considered only during periods of remedy negotiations; and (d) splitting Phase I and II of the merger review procedure to ensure legal certainty and due process, with incentives for pre-notification meetings.	High	√
4. Foster greater transparency and develop the CPC's advocacy powers by publishing non-confidential versions of its decisions, as well as guidelines for market studies.	Medium	√
5. Increase the financial and human resources available to the CPC and implement staggered appointment terms, cooling-off periods, and objective criteria for dismissing senior officials.	High	√
C. Enhancing the regulatory framework for state aid to limit market distortions		
6. Limit exclusions in the state-aid law by adding agriculture and fisheries sectors to its remit; align legal definitions with EU standards, including those pertaining to small and medium-sized enterprises and de minimis aid; define aid granted by the state or through state resources in line with Article 107(1) of the TFEU; and ensure that "horizontal aid" does not distort competition and trade to an extent contrary to the national interest..	High	√
7. Align the state-aid categories with EU law by: (a) updating the regulation on the conditions and procedures for granting aid of minor importance to include provisions on de minimis aid and aid for services of general economic interest; and (b) including conditions and procedures for granting horizontal aid, compatible with the state-aid law, that reflect the latest EU aid categories, including assistance for low-carbon industries and sectors.	Medium	√
8. Strengthen the framework for aid to distressed firms by: (a) clearly defining "distressed firms" in Article 4 of the state-aid law; (b) ensuring that such aid is compatible with the EU definitions of necessity, incentive effects, and proportionality; (c) modifying Article 8.1 of the regulation on aid to distressed firms by removing the reference to the EU interest or at least establishing a benchmark for assessing that interest; and (d) adding specific provisions for granting aid to distressed firms involved in the providing services of general economic interest.	Medium	√
9. Enhance transparency and strengthen due process by: (a) updating the inventory of state aid, including de minimis aid; (b) modifying the sections dealing with interested parties and the suspensory effects of decisions; and (c) introducing a pre-notification phase in the regulation on the manner and procedure for notification of state aid.	Medium	√
10. Improve the design of state-aid instruments[107] by: (a) anchoring them in broader development strategies with clearly articulated objectives like energy efficiency, decarbonization and technology adoption and innovation; (b) targeting them to address specific market failures and avoid crowding out private capital; (c) providing sufficient	Medium	

[106] A "silence is consent" rule is a form of tacit approval under which a business license or permit application is considered automatically granted if the competent licensing authority fails to react or issue a decision within the prescribed time limit.

[107] Cirera et al., 2020. A Practitioner's Guide to Innovation Policy. Instruments to Build Firm Capabilities and Accelerate Technological, Catch-Up in Developing Countries. Washington, DC: World Bank.

funding per program and per beneficiary to generate the desired effect; and (d) establishing monitoring and evaluation processes with a clear results framework to assess the effectiveness of state-aid policies and adjust them as necessary.		
11. Ensure that the authorities have sufficient financial and technical capacity to implement the state-aid program and simplify compliance for beneficiaries by transparently announcing calls for applications, establishing clear eligibility criteria, allowing for electronic applications, using competitive selection by expert committees wherever feasible, manage conflicts of interest, providing a grievance mechanism, and publishing information on state-aid recipients.	Medium	
D. Strengthening the sectoral competition regulatory frameworks to foster greater contestability		
12. Open the rail transport to competition from domestic and foreign firms and/or limiting the state's participation in the sector.	Medium	

These policy recommendations are closely aligned with reform steps under the EU Growth Plan's policy areas on *business environment and private sector development* and *governance, public administration reform and public financial management*. In particular, measures to strengthen SOE governance through a clearer ownership rationale, independent boards, and enhanced transparency require the adoption of a State Ownership Strategy and Law on State Owned Companies, the establishment of a public register of SOEs, and the publication of financial reports and fiscal risk statements. Similarly, recommendations to reinforce the capacity and independence of CPC aim at ensuring financial sustainability, including the introduction of own-resource funding mechanisms, as well as strengthening enforcement through risk-based investigations. Finally, aligning the state-aid framework with the EU acquis, including improved targeting, transparency, and monitoring, are aligned with the reassessment and phase-out of non-compliant state-aid schemes and the operationalization of SAMIS. Taken together, these policy recommendations could help accelerate progress toward meeting specific EU Growth Plan reform steps and unlock additional financing.

Enabling North Macedonia to shift from low-value assembly to high-value, service-driven, inclusive trade will also require a clear strategy, bold reforms, and coordinated action. To more effectively promote trade and technology adoption in North Macedonia, the authorities should consider: (1) reorienting the exports strategy toward higher-value manufacturing, digital services, and green sectors; (2) developing a national trade and investment strategy with a future-oriented and regional focus; (3) enabling the growth of services and digital trade through regulatory reform; (4) investing in quality infrastructure to improve firm-level competitiveness and export readiness; and (5) leveraging transport services to establish North Macedonia as a regional logistics and trade hub.

Recommendations for strengthening technology infusion, energy efficiency, and innovation	Priority	EU agenda
A. Reorient the country's export-led growth strategy toward higher-value manufacturing, digital services, and green sectors		
1. Draft the new exports strategy that would go beyond traditional goods, support green transition and smart specialization as high-value-added sectors offer greater potential for job creation and growth	High	
2. Sign mutual recognition agreements for professional qualifications with regional partners that would create new opportunities for market diversification	Medium	√
3. Invest in export advisory services, market intelligence, and digital platforms for exporters that would reduce information asymmetry	Medium	
4. Enhance retention, expansion, and integration of existing FDI, introduce effective investor grievance mechanisms and stronger intellectual property rights, supplier development programs, joint innovation initiatives, and integration support (e.g., technology-matching and co-location services)	High	
B. Developing a national trade and investment strategy with a future-oriented and regional focus		
5. Launch a coordinated whole-of-government approach to export competitiveness that reflects evolving global trade dynamics and upgrade the existing export-led FDI strategy	Medium	

to strengthen regional integration to leverage new opportunities in high-value manufacturing and services		
6. Boost services trade and digital exports through innovation hubs, and align trade policies with the green transition to ensure access to valuable destination markets	High	√
7. Address labor shortages and skills mismatches to support export growth	High	
C. Enabling the growth of services and digital trade through regulatory reform		
8. Lower restrictiveness in key services sectors such as ICT, transportation, and finance, supporting firms in scaling up high-value-added services	High	√
9. Strengthen regional cross-border payment systems	High	√
10. Reduce regulatory frictions and modernizing the policy environment that would support the expansion of cross-border services and digital transactions	Medium	
D. Investing in quality infrastructure to improve firm-level competitiveness and export readiness		
11. Close infrastructure gaps to ease constraints on the ability of domestic firms to export and scale	Medium	
12. Expand supplier-development initiatives that would help integrate SMEs into global and regional value chains	Medium	
13. Review and streamline standards and certifications that would improve compliance with the requirements of export markets and multinationals	Medium	√
E. Leveraging transport services to establish North Macedonia as a regional logistics and trade hub		
14. Invest in trade facilitation and transport infrastructure and improve the efficiency of customs and other border services	Medium	√
15. Strengthen multimodal connectivity to reduce trade costs and improve reliability	Medium	√

These policy recommendations are closely aligned with reform steps under the EU Growth Plan's policy areas on *business environment and private sector development*, *energy and green transition*, and *digital transition*. In particular, measures to reduce regulatory burden, strengthen trade facilitation, and improve export readiness underpin reform steps on simplifying administrative procedures, implementing NCTS Phase 6 and the Automated Export System, advancing e-freight transport (eFTI), and expanding innovation ecosystems. Reorientation toward higher-value and green sectors aligns with reform steps to deploy at least 0.8 GW of renewable energy, reduce permitting times for installations, and implement electricity market integration. Finally, recommendations to boost digital trade and services, and reduce regulatory restrictiveness, support reform steps on electronic communications legislation, digital identity and trust services (eIDAS), e-services expansion, and cybersecurity and data infrastructure systems.

To sustain growth as the size of the youth cohort shrinks, North Macedonia must translate education into productive employment and mobilize all available workforce by: (1) addressing weak foundational learning outcomes, skill development and providing career support; (2) improving school-to-work transition; (3) increasing female labor force participation; (4) mobilizing all available workers to sustain growth, including diaspora; and (5) strengthening employment intermediation services.

Recommendations for developing skills and mobilizing workers	Priority	EU agenda
A. Addressing weak foundational learning outcomes, skill development, and career support		
1. Raise foundational learning outcomes by improving teaching quality, extending instructional time, and systematically tracking minimum proficiency (reading/math), with targeted support for low-performing schools	High	√
2. Strengthen career guidance and employer engagement across secondary and tertiary education by building structured pathways into work (career services, employer input to curricula, and placement support), responding to weak school-to-work transitions and limited early work experience	High	
3. Expand meaningful internships and practical placements in higher education through employer-linked programs that strengthen study-job matching and reduce overqualification/late labor-market entry	Medium	√

4. Align financing across formal and non-formal vocational education/training with labor-market demand and verified outcomes (placement/retention), to reduce duplication and reorient provision toward shortage occupations	Medium	
5. Rebalance reskilling/upskilling toward in-demand foundational, technical, digital, and green skills and deliver through modular, employer-linked programs, addressing fragmentation and weak employment outcomes from training. Link upskilling/reskilling efforts with adaptive social protection to support workers.	High	
6. Integrate AI and green skills more effectively into education/training curricula	High	
B. Improving school-to-work transition		
7. Implement targeted wage subsidies and apprenticeships tied to training and minimum retention, focusing on NEET youth and first-time jobseekers, to address stalled transitions and persistent youth disconnect from work	High	√
8. Introduce performance-based finance for training providers and programs that reward employment and skill-acquisition outcomes, building on the new financing model for secondary vocational education	Medium	√
9. Establish a Labor Market and Skills Observatory that integrates vacancy data, employer surveys, tracer studies, and administrative data to guide curricula, student intake, and funding decisions	Medium	
C. Increasing female labor force participation		
10. Expand childcare and long-term care coverage in rural and underserved municipalities, while improving quality and introducing flexible options	High	√
11. Offer targeted ALMP packages for women after caregiving breaks (skills refresh, placement support, wage subsidies where relevant)	Medium	
12. Modernize work-life balance provisions to enable flexible work and shared parental leave, complementing care expansion and addressing limited flexible work options (such as part-time)	Medium	
13. Create pathways for women into high-productivity sectors (ICT, engineering, green jobs) by aligning training/short cycle programs with demand and supporting transitions into these occupations	Medium	
D. Mobilizing all available workers to sustain growth		
14. Make work pay for social assistance beneficiaries by introducing earnings disregard and gradual benefit phase-out as income rises and review labor taxation rules penalizing part-time/low-wage formal employment for all other workers, taking into account fiscal sustainability and impact on pension entitlements	High	
15. Operationalize joint ESA-CSW case management for 'workable' social assistance beneficiaries	Medium	√
16. Provide intensive activation support for long-term unemployed and social assistance recipients (career guidance, soft-skills training, mentorship programs, reskilling, post-placement support, and/or psycho-social counselling)	High	√
17. Extend working lives through gradual retirement-age and incentive reforms, paired with lifelong learning and healthy aging measures	High	√
18. Adopt a coherent migration-management and returnee reintegration approach linked to sector shortages and skills monitoring	High	√
E. Strengthening employment intermediation services		
19. Shift ESA toward a segmented service offer—combining lighter case management (digital tools, group counselling, standard job-matching) for job-ready clients with intensive, personalized support for beneficiaries facing multiple barriers—paired with stronger employer outreach and targeted post-placement follow-up where retention risks are high	High	
20. Build integrated digital systems for data-sharing across education, social protection, and employer networks, enabling coordinated service delivery and better tracking of outcomes	Medium	√
21. Formalize partnerships between the ESA and local authorities, training providers, private employment agencies, and civil society to expand outreach and ensure that services are relevant to regional labor market demand	Medium	√

22. Strengthen labor-market intelligence and use it to rebalance training programs toward technical/green/digital shortages (not just short-term low-skill demand)	High	
23. Align ESA governance and management with evidence-based performance monitoring (placement, retention, earnings progression, and study-job match), reinforcing accountability and impact	Medium	√

These policy recommendations are closely aligned with reform steps under the EU Growth Plan's policy area on *human capital.* In particular, measures to strengthen foundational learning, teaching quality, and career guidance are based on reform steps requiring quality assurance systems in higher education, integration of digital competencies into curricula, and the provision of career guidance professionals in schools. Recommendations to improve school-to-work transition through apprenticeships, internships, and employer engagement align with targets to expand dual education, increase the number of firms involved, and scale up work-based learning in VET. Finally, rebalancing skills development toward market-relevant training and expanding validation of non-formal learning are aligned with developing VET programs, increasing adult certification, and operationalizing regional VET centers. Taken together, these policy recommendations support the implementation of EU Growth Plan reform steps while strengthening labor market matching and the effective use of available workforce.

6.2. Results from simulating policy impacts

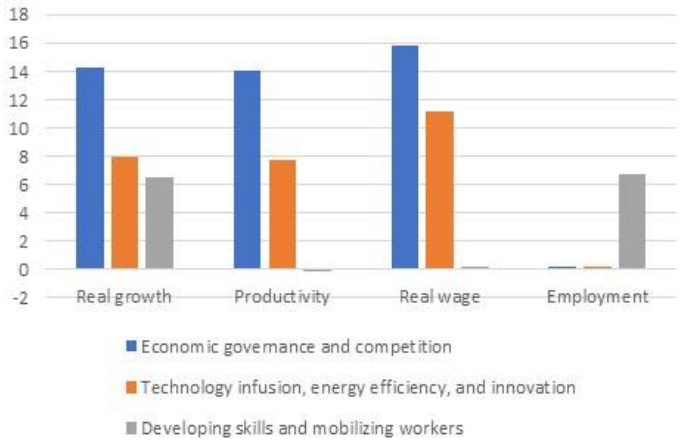
To estimate the economy-wide effects of each reform cluster, the policy simulations rely on MFMOD Stand-Alone (MFMOD-SA), a macrostructural model calibrated for North Macedonia. Economy-wide effects are estimated relative to a baseline scenario, with a projection horizon extending to 2050. The baseline scenario represents the evolution of the economy in the absence of the policy reform roadmap. Model results, including the impact on GDP, real wages, employment, and job quality, are reported as deviations from this baseline, allowing for a clear assessment of the incremental impact of each reform cluster and of the combined policy reform package. Annex 4 explains the clustering and simulation channels used to assess the reform impacts, while the below sections describe the theoretical background and mapping strategy used to cluster reforms and simulate their impacts.

Combined reform clusters

Combined, the policy reforms are expected to increase the average annual growth rate from an average business-as-usual (BAU) rate of 2.4 percent to 3.8 percent by 2050. Compounded over time, this higher growth path implies that GDP would be 18.8 percent higher by 2040 and 29.1 percent higher by 2050 than under the BAU scenario. Across reform clusters, the largest impact on growth during the forecast horizon stems from *strengthening economic governance and competition*, which raises output by 14.2 percent relative to the no-reform baseline by 2050, underscoring the strong growth dividends from pro-competition and -governance policies (Figure 81).

Figure 81. Simulated impact of reforms (percent deviation from baseline in 2050)

Reforms for higher growth and job creation (deviations from the baseline in 2050, percent)



Source: World Bank staff calculations using MFMOD-SA.

Real wages are projected to be 10.3 percent higher by 2040 and 24.6 percent higher by 2050 relative to the baseline. The policy simulations reveal that most of these wage gains stem from *strengthening economic governance and competition* reforms, which ease distortions that hinder the efficient allocation of resources across firms and sectors, raise economy-wide total factor productivity, and enable firms to offer higher wages in

a more competitive environment. On the other hand, while labor market reforms drive up employment supply, they result in modest wage increases by 2050, as the reforms primarily bring in previously inactive or underemployed workers into employment at initially lower productivity levels. In the absence of complementary productivity-enhancing reforms, the pass-through from productivity gains to wages remains limited, undermining the potential for aggregate growth to translate into more and better jobs.

The combined reforms are projected to raise the employment rate from a baseline of 49.3 percent to 52.8 percent in 2050. This would mean creating additional 53,373 jobs by 2050. Employment gains are predominantly driven by labor market reforms, which generate by far the largest number of new jobs across all three clusters through a reduction of structural barriers to participation and expansion of the labor supply, which create nearly 50,000 new jobs.

Table 6. Policy simulation results on growth, productivity, real wages and employment (percent deviation from baseline)

Policy Reform Clusters	GDP growth			Productivity			Real Wage			Employment		
	2030	2040	2050	2030	2040	2050	2030	2040	2050	2030	2040	2050
Combined reform clusters	3.6	18.8	29.1	0.9	10.9	20.4	0.9	10.3	24.6	2.7	7.1	7.2
Economic governance and competition	1.7	10.2	14.2	1.6	9.7	14.2	1.5	9.3	15.8	0.1	0.4	0.1
Technology infusion, energy efficiency, and innovation	0.2	1.8	7.9	0.1	1.7	7.7	0.0	3.1	11.1	0.0	0.2	0.2
Developing skills and mobilizing workers	1.1	4.8	6.5	-1.3	-1.1	-0.2	-0.6	-0.3	0.2	2.3	6.0	6.7

Source: World Bank staff calculations using MFMOD-SA.

Overall, the combined reforms are thus projected to generate both more jobs and better jobs. Using the World Bank's More and Better Jobs (MBJ) Indicator methodology, which quantifies the total contribution of reforms through increases in labor earnings stemming from both the extensive margin (more jobs) and the intensive margin (better-paying jobs), the combined reform clusters are estimated to create the equivalent of 250,086 more and better-paid jobs by 2050.

Strengthening economic governance and competition

Strengthening economic governance and competition encompasses reforms aimed at removing the barriers that constrain open and competitive markets in North Macedonia. This cluster includes reforms geared at: a) improving public spending efficiency; b) creating the level-playing field and improving governance of SOEs; c) improving the business environment and d) enhancing trade policy and infrastructure. The reforms which strengthen economic governance and competition produce the largest GDP gains among the three clusters, raising output by 14.2 percent relative to the no-reform baseline by 2050 (Table 7), while generating approximately 119,000 more and better jobs. The main transmission channel within this cluster is productivity enhancement, which provides a significant qualitative upgrading of employment. By removing distortions which hinder the efficient allocation of resources across firms and sectors, governance and competition reforms raise economy-wide TFP. As competition intensifies, firms invest in upgrading production processes and workforce skills, increasing output per worker and the wages they can offer, thereby creating better paid jobs in the economy.

- ***Improving public spending efficiency*** through eliminating inefficiencies in public investment management and improving public education spending efficiency, boosts output and supports the creation of 26,390 better-paid jobs by 2050. Public investment in infrastructure can reduce production costs for firms, shorten supply chains, and broaden their market access. This can raise economy-wide TFP, which in turn increases the marginal product of labor and puts upward pressure on wages, generating better jobs even as the net employment effect remains modest.
- ***Improving the business environment*** leads to the most significant gains for North Macedonia's economy. High administrative burdens, limited digitalization of G2B services, and corruption in areas such as public procurement and taxation raise the cost of doing business and constrain firm growth. Streamlining business permits, licensing, and customs procedures, especially for SMEs and growing firms, leads to the

largest creation of more and better jobs in the economy over the long run. Relative to a baseline employment level of 750,000, this reform can help generate more than 38,400 better-paid jobs.

- ***Creating the level-playing field and improving governance of SOEs*** reduces market rigidities which can enhance the country's competitiveness. As product market regulations are improved, entry barriers fall and incumbent firms face greater contestability, compelling them to lower costs and improve product quality. Moreover, stronger competition and better SOE governance reduce the scope for rent extraction, reduce input production costs, allowing workers to earn higher wages in more productive firms, while the economy produces more output with fewer distortions. Reducing restrictions captured by the Product Market Regulation (PMR) index by one standard deviation by 2050 raises productivity growth and leads to the creation of more than 11,400 better-paid jobs. At the same time, the total number of jobs declines marginally, as less productive firms exist, and resources are reallocated toward new or more productive firms in the economy. This reallocation effect, along with efficiency gains from stronger competition and better SOE governance, still yields a net positive impact on the economy: output is around 1.4 percent higher than in the no-reform baseline by 2050.
- ***Enhancing trade policy and infrastructure*** through reduced services trade restrictiveness, eased technical barriers to trade, improved trade facilitation and lower import tariffs also yields meaningful gains. Lower import tariffs provide access for domestic firms to an increased variety of higher-quality and/or lower-cost inputs. At the same time, exposure to international competition incentivizes continuous upgrading and adoption of best practices, reinforcing the feedback loop between trade openness and productivity growth. Reduced trade costs also expand the range of goods and services that can be competitively traded, deepening the country's integration into global value chains. The cumulative impact on exports is estimated to be 8.8 percent by 2050. Productivity gains associated with better access to inputs and expanded export market opportunities drive the creation of close to 41,600 better-paid jobs by 2050, accompanied by significant wage increases. Overall, these reforms raise output by 5 percent relative to a no-reform scenario by 2050, underscoring the role of trade policy and infrastructure for long-term growth, structural transformation, and higher-quality employment.

Table 7. Impact of proposed economic governance and competition reforms (% deviation from baseline)

Time period	GDP (% deviation)	Productivity (% deviation)	Real Wage (% deviation)	Employment (% deviation)
2030	1.7	1.6	1.5	0.1
2040	10.2	9.7	9.3	0.4
2050	14.2	14.1	15.8	0.2

Source: World Bank staff calculations using MFMOD-SA.

Fostering technology infusion, energy efficiency, and innovation

Fostering technology infusion, energy efficiency and innovation include reforms that can expand the productive frontier of the economy of North Macedonia. This cluster delineates reforms focusing on: a) boosting firm innovation capacity; b) maximizing the impact of FDI; and c) reducing persistently high energy intensity of firms and decarbonizing the economy. While the first cluster primarily operates by removing barriers which prevent firms from reaching their potential, the second cluster pushes the production frontier outward by introducing new technologies, processes, and energy-saving measures. By adding productive capacity rather than reallocating it, reforms under this cluster tend to produce positive employment effects from the outset and generate a particularly strong wage premium as workers in technology-intensive firms capture a share of the innovation rents (Table 8).

- ***Boosting firm innovation capacity*** generates approximately 48,400 better-paid jobs by 2050. The effects of R&D investment build over time, reflecting the cumulative generation of knowledge embodied in improved products, processes, and organizational methods and then the diffusion across firms in the economy. As the economy's R&D intensity rises, the demand for workers with specific skills needed to develop, adapt, and implement new technologies outstrips supply in the short run, bidding up wages for

these workers. Over time, this wage premium creates incentives for workers to acquire relevant skills, which gradually eases the talent constraint and supports further innovation from firms.

- ***Maximizing the impact of FDI*** yields the strongest output gains, but this requires sustained efforts to strengthen linkages between foreign investors and the domestic economy. In North Macedonia, weak integration between foreign and domestic firms has historically limited these benefits, underscoring the importance of targeted reforms to unlock the gains from these linkages.[108] Policies to enhance the retention, expansion, and integration of existing FDI, such as effective investor grievance mechanisms and stronger intellectual property rights, can help embed foreign investors more firmly in North Macedonia's economy. At the same time, supplier development programs, joint innovation initiatives, and integration support (e.g., technology-matching and co-location services) can facilitate stronger connections between foreign and domestic firms, ensuring that FDI translates into broader productivity gains.
- ***Reducing the persistently high energy intensity of firms and decarbonizing the economy*** raises output and generates close to 19,000 more and better-paid jobs relative to the no-reform baseline scenario, primarily by boosting economy-wide productivity, and in turn, generating more and better employment. When firms invest in energy-efficient technologies, they reduce the amount of energy required per unit of output, which directly lowers production costs and raises TFP. Furthermore, through the implementation of a domestic carbon tax, North Macedonia reduces its exposure to EU's CBAM, which would otherwise impose costs on the country's carbon-intensive exports. The domestic carbon tax directly incentivizes firms to accelerate their adoption of cleaner production methods, while simultaneously ensuring that carbon-related revenues remain within the country rather than being collected at the EU border. Moreover, the price signal from the carbon tax strengthens incentives for energy efficiency investments, while the availability of energy-saving technologies lowers the cost of tax compliance.

Table 8. Impact of proposed reforms to foster technology infusion, energy efficiency and innovation (percent deviation from baseline)

Time period	GDP (% deviation)	Productivity (% deviation)	Real Wage (% deviation)	Employment (% deviation)
2030	1.1	-1.3	-0.6	2.3
2040	4.8	-1.1	-0.3	6.0
2050	6.5	-0.2	0.2	6.7

Source: World Bank staff calculations using MFMOD-SA.

Developing skills and mobilizing workers

Developing skills and mobilizing workers includes reforms that remove structural barriers to full and productive participation. North Macedonia's labor force remains underutilized, with low female participation, high youth unemployment, persistent skills mismatches, and long-term unemployment that can trap workers in inactivity. This reform cluster focuses on: a) enabling women's equal participation and opportunities in the labor market; b) strengthening school-to-work transitions for youth; c) supporting lifelong learning and longer working lives; and d) reducing labor market frictions to support participation and formalization. Importantly, these reforms improve both job quantity and quality: by year 2025, the reform package generates more than 51,200 more and better-paid jobs, alongside higher real wages relative to baseline (Table 9). Impacts by reform area should not be interpreted as additive, as reform packages interact and share beneficiaries.

- ***Enabling women's equal participation and opportunities in the labor market*** creates nearly 9,000 additional jobs, with childcare expansion reducing the opportunity cost of work for mothers and supporting labor-force entry. The model shows an initial productivity dip as employment expands into

[108] World Bank. 2022. North Macedonia Trade and State Aid Report.

lower-experience workers or lower-productivity sectors, but this reflects a transition/composition labor effect rather than lower firm productivity.

- ***Supporting an effective school-to-work transition for youth*** generates around 2,000 new jobs, by speeding entry into employment and reducing early-career scarring that can depress earnings and productivity over the life cycle, raising the long-run productive capacity of young workers in the economy.
- ***Encouraging lifelong learning and longer working lives*** adds more than 32,300 more and better-paid jobs. Training temporarily diverts time and resources from production, so productivity gains materialize gradually, raising output by 4.2 percent relative to a no-reform baseline. Adult participation in upskilling and reskilling programs leads to the acquisition of new competencies that can raise productivity and wages. Despite the initial negative impact on productivity (as resources are diverted from production to training and the expanded workforce has not yet fully realized the returns to new skills), productivity gains begin to materialize and accumulate toward the end of the projection horizon. The extension of the working life prevents the premature withdrawal of experienced workers whose specific knowledge and institutional memory are valuable but difficult to replace. Extending working lives would gradually depress average productivity as an aging workforce becomes less adaptable, but along with lifelong learning programs, these reforms substantially mitigate the adverse effects on productivity and support sustained gains in output and job quality.
- ***Reducing labor market frictions to support participation and formalization*** creates more than 7,100 more and better-paid jobs, but may initially reduce average productivity as long-term unemployed workers or social assistance beneficiaries re-enter with depreciated skills and weaker networks. Here, the policy intent is to make work pay by avoiding abrupt benefit withdrawal of social assistance when earnings begin alongside measures that encourage formal employment at low wages rather than discouraging work altogether. Phasing out unemployment benefits gradually preserves the financial incentive to work while maintaining a safety net during the transition. Reducing hiring frictions can also lower the rate at which it becomes profitable for firms to hire workers who have been out of the labor force for an extended period. The impact on productivity can be negative as the long-term unemployed returning to work have lower initial productivity due to skills depreciation, loss of work habits, and detachment from professional networks.

Table 9. Impact of proposed labor market reforms (% deviation from baseline)

Time period	GDP (% deviation)	Productivity (% deviation)	Real Wage (% deviation)	Employment (% deviation)
2030	1.1	-1.3	-0.6	2.3
2040	4.8	-1.1	-0.3	6.0
2050	6.5	-0.2	0.2	6.7

Source: World Bank staff calculations using MFMOD-SA.

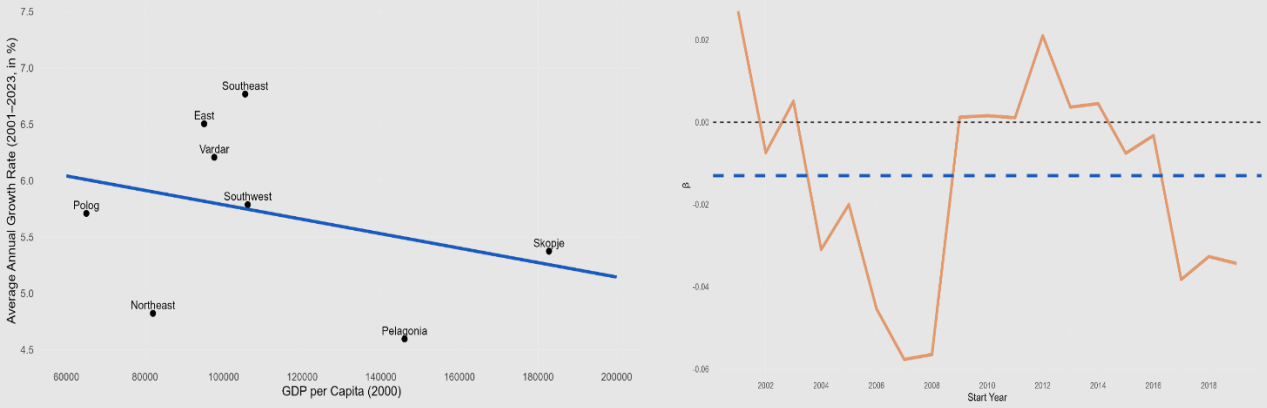
Annex 1. Spatial convergence has been slow, limiting sharing growth dividends

While within-country regional income disparities in North Macedonia have narrowed, convergence remains slow. GDP growth rates differ substantially across North Macedonia's eight statistical regions. Between 2000 and 2023, the economy grew by an average of 5.7 percent; however, regional growth performance varied: Pelagonia recorded the lowest average growth rate of 4.6 percent, while the Southeast recorded the highest at 6.8 percent. Regional disparities are also reflected in income levels (Figure 82a). For example, in 2000, Skopje's real GDP per capita was almost three times higher than Polog's.

Figure 82. Regional income disparities have narrowed

a. Average real GDP growth (2001-2023) and real GDP per capita in US$ PPP terms (2000)

b. β-estimate for five-year rolling-windows



Source: World Bank staff calculations based on MakStat. *Notes: T*he start year of each five-year window is denoted on the x-axis i.e. the estimate for 2002 refers to the period 2002-2006. The dashed blue line shows the β-estimate for the full time period.

Convergence is episodic, accelerating during periods of economic disruption such as the global financial crisis and the COVID19 pandemic, rather than being driven by consistent structural improvements. In the early 2000s, regional differences remained stable. However, the financial crisis resulted in disruptions that caused a surge in convergence, peaking between 2008 and 2012 with a β of -0.056 (Figure 82b).[109] Regional differences stagnated again in the early 2010s, before convergence regained momentum with the onset of the COVID19 pandemic. If North Macedonia was to sustain the most recent momentum, with a β of approximately -0.034, the convergence period could be reduced to just 20 years.

Regions with lower employment rates and fewer firms tend to grow more slowly, making it harder for them to catch up with richer regions. But when these differences are accounted for, a clearer pattern of convergence emerges: poorer regions can actually close the gap with wealthier ones relatively quickly, in around nine years, if they are able to address these underlying disadvantages.

[109] All β-estimates were obtained from panel models with time fixed effects and cluster-robust standard errors. β-convergence examines whether countries or regions with lower levels of GDP per capita experience higher growth rates on average. In theory, this should be the case due to diminishing returns on capital, meaning regions that are further from their steady-state level of capital tend to grow faster as they 'catch up'. This can be empirically analyzed using univariate panel regressions, where GDP growth is modelled as a function of lagged and logged GDP per capita. If β, the GDP per capita coefficient, is negative, regions converge in relative terms. Conditional β-convergence refines this approach by controlling for heterogeneous characteristics in a multivariate regression framework. Unlike unconditional β-convergence, this approach allows for different steady-state growth rates, reflecting differences in structural and institutional factors.

Annex 2. North Macedonia's firms are highly vulnerable to natural and climate-related hazards, and potential costs of inaction are substantial

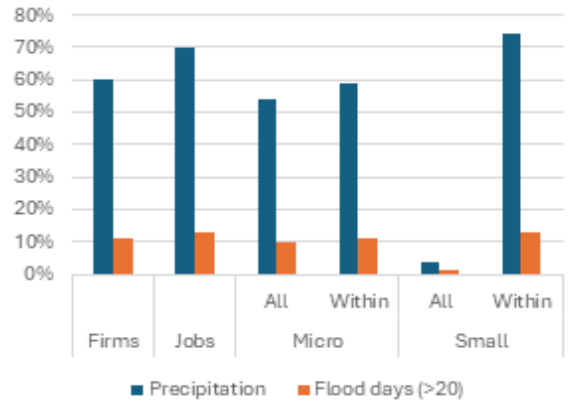
North Macedonia's escalating climate and natural hazard risks pose a serious and growing drag on productivity. These would manifest by disrupting labor supply, damaging capital, and undermining sectoral performance. The country has a long history of devastating floods, including the 2015 floods that affected more than 100,000 citizens and resulted in economic losses of US$107.4 million. Without adaptation, climate change could cost close to 4 percent of GDP by 2050, with risks compounding existing socioeconomic vulnerabilities and potentially producing far more dramatic localized impacts than averages suggest.[110]

Exposure to climate hazards can affect already vulnerable firms. Around 60 percent of firms and 70 percent of jobs are located in areas facing higher-category precipitation hazards (Figure 83). While exposure to frequent flooding affects a smaller share of firms (about 11 percent) and jobs (about 13 percent), it remains concentrated among micro firms that have less capabilities to weather shocks. Flood exposure is also not pronounced in Skopje; however, firms in the capital are likely to be exposed to flood events in other areas via industrial linkages, supply chains, and labor markets, amplifying productivity and employment impacts well beyond directly exposed firms.[111]

The results indicate that climate shocks are already weighing on firm performance and jobs. Higher flood risk is associated with lower productivity and weaker wage growth, while increased precipitation is linked to declines in annual earnings, reflecting persistent operational disruptions even outside extreme events (Figure 84). Although the estimated impact of a one-standard-deviation-increase in precipitation (i.e., around US$100 per worker per year) appears modest in isolation, it becomes economically meaningful given that around 70 percent of jobs are exposed to recurring higher-category precipitation risks. This pattern indicates that climate impacts in North Macedonia operate less through direct physical damage and more through frequent disruptions to operations, logistics, and worker availability, which affect earnings before firms adjust employment. By contrast, rising temperatures have the strongest effects on jobs, with firms reducing employment as heat stress raises costs and constrains normal production, underscoring the growing challenge climate risks pose to both competitiveness and labor markets.

Figure 83. Exposure to climate hazards is widespread, and more so for already vulnerable firms

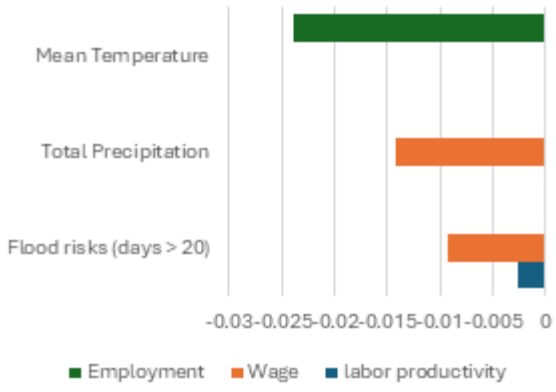
(Share of firms exposed to higher category climate hazards, by group, 2012-21)



Source: Firms and Climate Tool based on matched firm-level and weather data. *Notes:* Visualization has been truncated for better visualization. For precipitation, the share is calculated for firms experiencing 2.1 to 3.2mm, while flood days (days with above

Figure 84. Exposure is already translating into firm productivity, wage, and employment losses

(Effect of a one-standard deviation in climate hazard on firm-level outcomes, 2011-21)



Source: Firms and Climate Tool based on matched firm-level and weather data. *Notes*: The figure illustrates the result from estimating the relationship between climate hazards and firm performance using firm-level panel regressions with standardized

[110] World Bank (2025) Country Climate and Development Report: North Macedonia.
[111] Rentschler et al, 2021

20mm of precipitation) is calculated based on the highest quartile of the distribution.

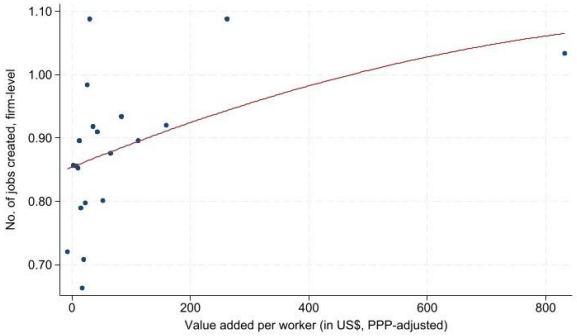
variables, controlling for firm- and year- fixed effects. All bars are statistically significant at 5% level. A coefficient of −0.02 would imply that a one-standard deviation increase in the climate variable is associated with a 0.02 standard deviation decline in the firm outcome.

Annex 3. How does firm-level productivity impact employment and wages?

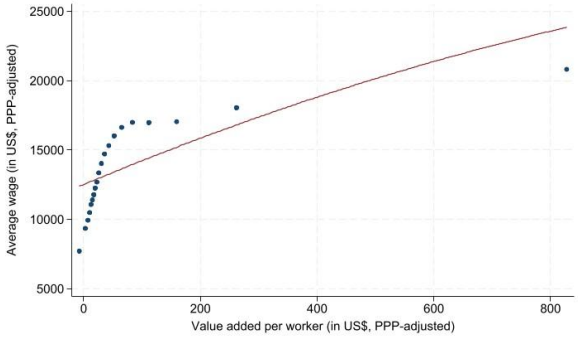
More-productive firms tend to create more jobs and pay higher wages. A binscatter analysis (Stepner, 2014)[112] shows a strong positive relationship between firm-level productivity, job creation, and wages (Figure 85). This relationship remains robust after controlling for year and year-industry fixed effects, indicating that even within the same industry and year, higher productivity is consistently linked with stronger employment and wage outcomes.

Figure 85. Firms with higher productivity levels tend to create more jobs and pay higher wages

a. Binned scatterplot showing value added per worker and job creation



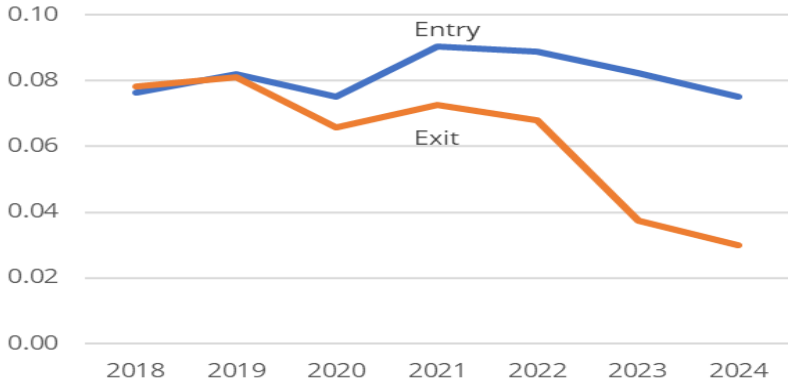
b. Binned scatterplot showing value added per worker and the average wage



Source: ICA 2.0 Toolkit based on national business registry data. *Notes:* Controls for year-fixed effects. Results are comparable to an analysis with year-2-digit-NACE fixed effects.

Figure 86. Firm entry and exit rates have been declining since 2021

(Firm entry and exit rates, 2018-2024)

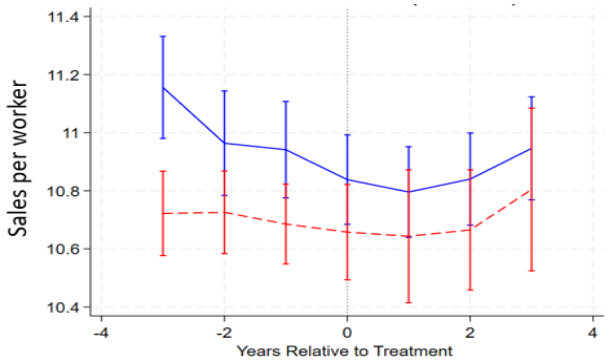


Source: ICA 2.0 Toolkit based on WB Entrepreneurship Database. *Notes*: Entry rate is calculated as percentage share new firms to total number of firms, while exit rate is based on percentage of closed firms to total number of firms in previous yeat. The units of measurement are private companies with limited liability.

[112] Michael Stepner, 2014. "Binned Scatterplots: introducing -binscatter- and exploring its applications," 2014 Stata Conference 4, Stata Users Group.

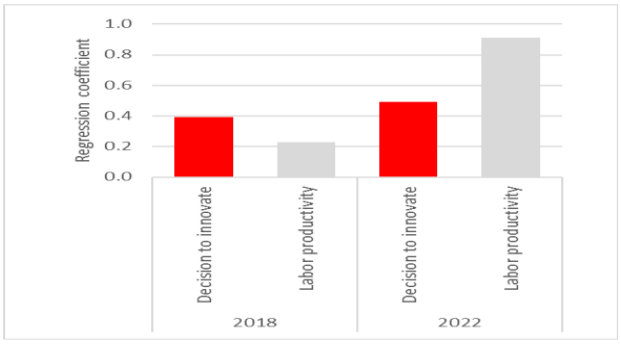
Figure 87. Innovation support and firm performance

a.



Source: WB staff calculation using data from Central Registry and beneficiaries provided by FITD. *Notes*: The figure illustrates the average sales per worker and 95% confidence interval on the matched sample of beneficiary (blue lines) and non-beneficiary firms (red lines).

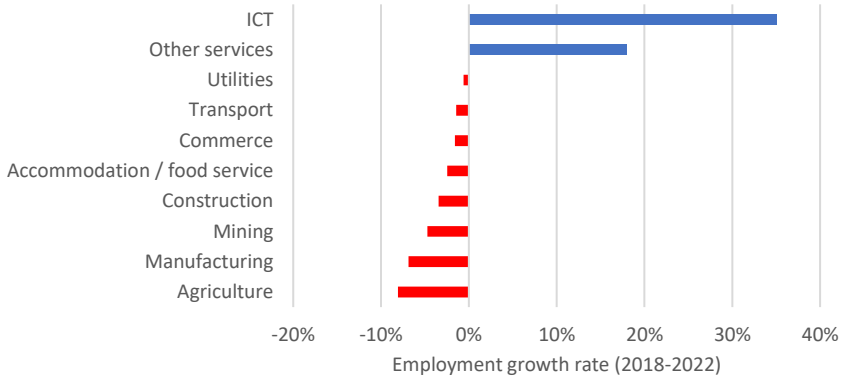
b.



Source: World Bank staff calculation using Community Innovation Survey (CIS) 2017-2018 and 2021-2022.*Note:* Red bars denote statistical significance at 10% level.

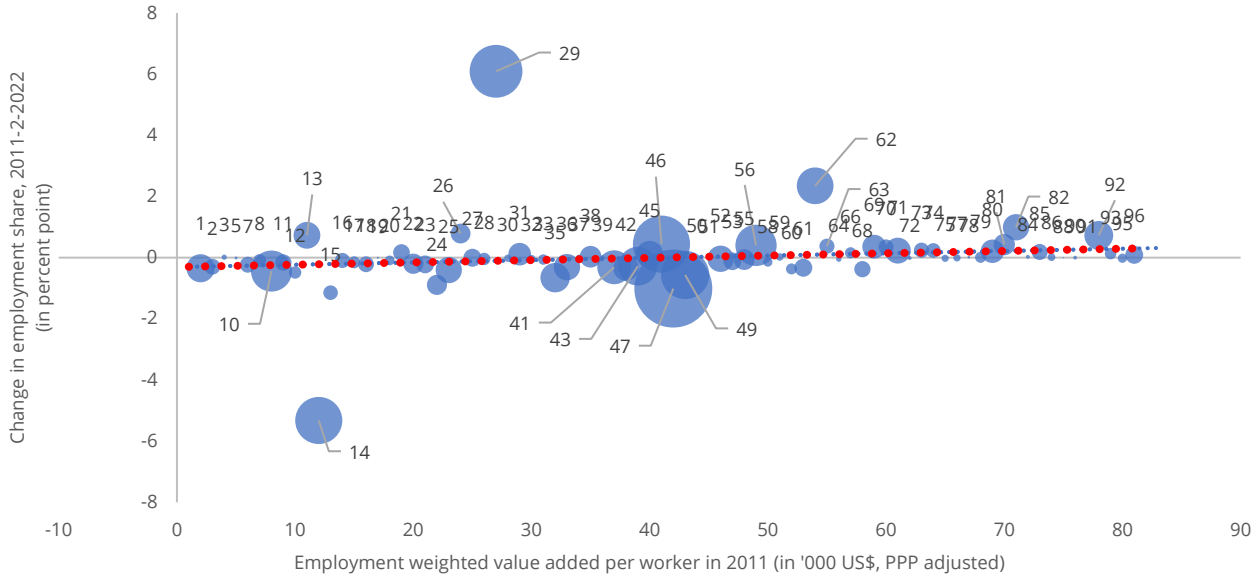
Figure 88. From 2018-2022, growth in employment is driven by ICT and services

Employment growth rate 2018-2020



Source: ICA 2.0 toolkit based on Central Registry

Figure 89. Over the past decade, the shift of workers across sectors has contributed little to overall labor productivity growth in North Macedonia



Source: ICA 2.0 toolkit based on Central Registry

Annex 4. Gender Employment Gap Index (GEGI)

The Gender Employment Gap Index (GEGI) is a simple and transparent macroeconomic tool designed to estimate the long-run GDP per capita gains from closing gender employment gaps. Developed by Steven Pennings (2022) under the World Bank's Pacific Gender Equality Program, the GEGI responds to the policy need for a broadly applicable and easily replicable measure of the economic cost of female underemployment, particularly in data-scarce environments.

The core idea is straightforward: by quantifying the employment rate gap between men and women and translating it into an equivalent gain in GDP per capita, the GEGI offers a policy-relevant metric to inform gender-inclusive labor market strategies.

The GEGI has two main variants:

The **Basic GEGI** is calculated as:

$$\text{Basic GEGI} = \frac{\text{(Men's Employment Rate - Women's Employment Rate)}}{\text{Total Employment Rate}} \times 100\%$$

This formulation expresses the gap in employment between men and women as a percentage of total employment, thereby approximating the **percentage increase in labor supply** (and hence GDP per capita in the long run) that would result from equalizing male and female employment rates.

The **Full GEGI** refines the basic measure by disaggregating employment into:

- **Better employment**: formal, higher-quality jobs, and
- **Other employment**: informal, lower-quality jobs.

It is calculated as a **weighted average** of the gender gaps in both categories:

$$\text{Full GEGI} = \theta \cdot \text{Gap in Better Employment} + (1-\theta) \cdot \text{Gap in Other Employment}$$

Here, the weights (θ and 1 − θ) reflect the relative importance of better and other employment types in each country's labor market, allowing for a more nuanced estimate of gender disparities in employment quality. Better employment is defined through non-agricultural employees and employers, while other employment through the reminder.

The weights assigned to better employment (θ) and other employment (1− θ) in the Full GEGI formulation are derived from their respective contributions to GDP, as specified in the following equation:

$$\theta = \frac{hL_{better}}{hL_{better} + L_{other}}$$

These contributions depend not only on the number of workers in each employment category but also on their relative productivity. For analytical clarity, we assume that individuals in “other employment” contribute only as much as raw labor, with a normalized productivity level of 1. Their human capital is assumed to have no bearing on productivity—meaning, for example, that a doctor working as an agricultural laborer would be just as productive as an uneducated person performing the same task. In contrast, workers in better employment are assumed to be as productive as their human capital allows; they are h times as productive as raw labor, where h represents average human capital per worker. We further assume gender parity in productivity, such that men and women in each employment category possess equal human capital and therefore equal productivity (i.e., $h_F = h_M = h$).

The calculation of the weight θ requires data on human capital per worker h, which requires taking a stand on timing. Because we are usually interested in evaluating the effect of today’s policy changes in the future (forward looking) we want a measure of the human capital of the next generation: which is what the World Bank Human

Capital Index seeks to measure. However, the HCI is expressed relative to a theoretical maximum of complete education and health (which are assigned a score of one), rather than relative to the productivity of raw labor, as we need here. Hence, we need to calculate $h=HCI/HCI_{min}$, where HCI_{min} is the theoretical minimum HCI (for someone with no education and poor health); a value of 0.2 is used.

We express the employment variables and their input in the calculation of θ through their rates, i.e. as shares of the respective working-age population, and not in absolute levels, i.e. as number of employed.

The methodology behind the GEGI rests on a set of simplifying but transparent assumptions. First, it assumes that the closing of the gender employment gap is exogenous—that is, female employment rises to match that of males without changes in productivity. It also presumes a linear relationship between labor and output, meaning that increases in labor supply translate proportionally into GDP gains. Importantly, the model does not factor in any crowding-out effects, such as the potential displacement of male workers or the diminishing returns that could result from a sudden increase in labor supply. Finally, it assumes homogeneous productivity across workers, treating the average productivity of newly employed women as equal to that of the existing workforce. These assumptions aim to keep the GEGI transparent and easy to interpret, even if they abstract from some real-world complexities.

The GEGI relies on **aggregate national statistics**—employment rates by gender and, where available, disaggregated by employment type. However, to better capture the full form of the GEGI, we rely on micro datasets of the Labor Force Survey of North Macedonia (2010-2023) to calculate such rates ourselves. We do the calculations on the working-age population 15-64.

Annex 5. Policy recommendations and policy assumptions used for simulations

Policy Area	Policy recommendations	Macro hooks	Policy assumptions	References
Strengthen economic governance and competition				
A. Improving public spending efficiency	1. Eliminate inefficiencies in public investment management	Public capital expenditure increases from 2027, reaching 1 pp of GDP by 2030.	Reduced under-execution of public investments adding 1% GDP to public capital spending to the baseline.	
	2. Improve public education efficiency to improve education quality	The increase in human capital as a result of the rise of learning-adjusted years of schooling (LAYS) enters the production function and affects the economy-wide productivity. The increase in LAYS is equally affecting both secondary and tertiary education, which means that different cohorts will gradually enter the labor market. The first cohort enters the labor market in 2036 and then the other ones gradually follow. Education spending to improve learning outcomes increases by 0.05 pp from 2027 per year.	Spending on education (such as targeting instructions to a child's learning level) can yield cost effective increases in LAYS or around 3 additional years of high-quality student per $100 per child as in Angrist et al., (2020). Expenditure assumptions used for the policy simulation are slightly more modest reflecting the lower average effectiveness of system-wide spending compared to best-practice interventions. The estimated rise in the equivalent years in schooling is 0.3 (by 2036), and then 0.6 years (by 2050).	https://documents1.worldbank.org/curated/en/801901603314530125/pdf/How-to-Improve-Education-Outcomes-Most-Efficiently-A-Comparison-of-150-Interventions-Using-the-New-Learning-Adjusted-Years-of-Schooling-Metric.pdf
B. Creating the level-playing field and improving governance of SOEs	1. Enhance the competition regulatory framework to limit potential anticompetitive actions and ensure more efficient merger control	1.1 pp higher TFP level by 2040, compared to the baseline, improving from 2030.	Bourles et al. (2010) find that one standard deviation reduction in Product Market Regulation for OECD countries can raise TFP growth by 0.1-0.15 pp. We assume that PMR will improve by one standard deviation starting from 2030 until 2040, which would lead to an alignment with the new EU member states. TFP will gradually increase to correspond to an increase in TFP growth of 0.1 p.p., accumulating to about a 1.1% higher level of TFP relative to the baseline by 2040.	https://www.nber.org/system/files/working_papers/w16520/w16520.pdf
	2. Improve the State aid regulatory framework to limit distortions and foster a level playing field			
	3. Adopt a clear state ownership policy to articulate the economic rationale for retaining state ownership and adopt good corporate governance principles for SOEs to improve their performance			
C. Improving the business environment	1. Align minimum wage policy with productivity growth to retain competitiveness	Not modeled.		
	2. Streamline business permits, licensing, and customs procedures to reduce regulatory burdens, especially	0.215% compounded annual TFP growth which will be 3.27% higher	Using WB Enterprise Surveys for North Macedonia, each firm that consider licensing a constraint is found to have 1.122 log points	https://www.enterprisesurveys.org/en/enterprisesurveys

	for SMEs and growing firms. In the short-run, consider a comprehensive review of business regulations and procedures.	TFP level introduced from 2030 until 2045.	lower TFPR. From Chapter 3, North Macedonia is 2.92 pp higher in terms of share of firms that consider licensing a constraint compared to the aspirational peer average. Aggregate TFP gain is = (.0285 – .0577) x (-1.122) = (-0.0292) x (-1.122) = 0.0327 log points or 3.27 percent.	
	3. Digitize G2B services, establishing business portals and online one-stop-shops, and support firms in digitalization. Enact the law on security of networks and information systems strengthen the capacity of the MDT to drive the digital transformation agenda.	TFP level is 0.38% higher by 2050 (20% of firms have 1.9% higher productivity i.e. the average of 2.2% and 1.6%) . This level increase is gradually introduced from 2027 onwards (around 0.016 pp higher growth every year) to eventually reach 0.38% higher TFP level in 2050.	Using Enterprise Survey data, each firm adopting email experiences 1.6% higher TFP relative to those that do not adopt, and each firm launching a website experiences 2.2% higher TFP compared to those that do not launch a website. The assumption is that 20 percent of firms adopt these changes by 2050. These partial elasticities are taken from Cusolito et al., (2020).	https://openknowledge.worldbank.org/server/api/core/bitstreams/c030fe17-dcdc-5b15-be94-e113b2c14d27/content
	4. Fight corruption in public procurement, taxation, and access to government services	Not modeled to avoid double counting.		
D. Enhancing trade policy and infrastructure	1. Reduce service trade restrictiveness by: i) reviewing and adopting transparent and predictable rules for cross-border e-commerce and digital trade; ii) aligning domestic laws with international best practices on digital trade and services regulation.	Higher export services of 10% by 2050 with a gradual increase per year due to a reduction in the STRI by 0.05. The impact on services are estimated via changes in relative export prices (i.e., price competitiveness).	Following Benz et al, (2023),service reforms cutting 0.05 from a country's STRI are associated with a short-run growth in services trade of less than 10% in all sectors, while in the medium to long term trade values can increase by 20% to 50%, depending on the sector. We assumed a modest reduction in STRI of 0.05 and services exports' increase by 10%.	https://www.oecd.org/content/dam/oecd/en/publications/reports/2023/04/right-here-right-now-new-evidence-on-the-economic-effects-of-services-trade-reform_da6f0ada/1159657f-en.pdf
	2. Negotiate mutual recognition of professional qualifications with trading partners and Mutual Recognition Agreements to reduce technical barriers to trade.	Not modeled.		
	3. Enhance the National Quality Infrastructure through targeted sector-level improvements and improve trade facilitation.	An increase in exports by 9.5% from 2030 to 2040, and 5.8% 2041-2050 relative to the baseline. The shock is introduced through higher export competitiveness (ie. relative prices).	Djankov, Freund, and Pham (2010) find that each additional day in export delays reduces trade by at least 1 percent. We assume an increase in Logistics Performance Index LPI to 3.4 (aspirational peer average) from 3.1 for North Macedonia by 2040 and best performing aspirational peer 3.60 by 2050; that is (3.4-3.1)/3.1 = 9.7 % assuming an elasticity of 1%. And by 2050 (3.6-3.4)/3.4 = 5.8%.	https://documents1.worldbank.org/curated/en/761201468175464382/pdf/wps3909.pdf

	4. Reduce import tariffs for input and intermediate goods in selected sectors for FDI, but also extend those to other domestic firms, to ensure that firms produce goods and exporting services have access to high quality and cost competitiveness inputs and capital goods.	2% higher TFP level by 2050.	According to Ahn et al. (2016), a 1 percentage point decline in input tariffs is estimated to increase total factor productivity by about 2 percent in the sector considered. We assume 2 percent higher TFP economy -wide in North Macedonia by 2050.	https://www.imf.org/en/publications/wp/issues/2016/12/31/reassessing-the-productivity-gains-from-trade-liberalization-43828
Foster technology infusion, energy efficiency, and innovation				
A. Boosting firm innovation capacity	1. Support continuous upgrading within firms, especially in manufacturing and services, through targeted programs for technology adoption and management practices. Interventions may include innovation vouchers and supply chain platforms for financing, among others. 2. Enhance the impact of innovation funding on firm performance by refining selection, monitoring, and support, potentially through fewer, larger programs that build capacity in viable firms.	TFP increases gradually and reaches about 5% above baseline level by 2050.	Using Wieser (2005), which reports R&D elasticities of productivity in the range of 0.03-0.38, we apply a conservative elasticity of around 0.04. This implies a long-run TFP level increase of approximately 5 percent from raising R&D intensity from 0.4 to 1.5 percent of GDP (aspirational peer) from 2025 to 2050. We assume 100 percent publicly financed increase in R&D.	https://onlinelibrary.wiley.com/doi/epdf/10.1111/j.0950-0804.2005.00260.x
B. Maximizing the impact of FDI	1. Develop and implement a targeted FDI promotion strategy that has high potential to achieve better job creation, technology transfer to local firms via industrial and spatial linkages, and sustainability/green economy, shifting focus from low-cost assembly to high-value sectors. 2. Design and implement policies to enhance retention, expansion and linkages from existing FDI (e.g., grievance mechanism; intellectual property rights, etc.) 3.Facilitate stronger linkages between foreign and domestic firms through supplier development programs and joint innovation initiatives, and integration support (e.g., technology matching services, co-location services) 4.Liberalize reciprocity requirements and other discriminatory restrictions to access real estate	Increase in net FDI stock (cumulative FDI inflows – depreciation) by 2.56% from the baseline (conservative) which enters private investment.	The analysis draws on Mistura and Roulet's (2019) elasticity estimate of approximately 0.213, which implies that every 1 percent reduction in the FDI Regulatory Restrictiveness Index (FDIRRI) is associated with a 0.21% increase in FDI stock. MKD FDIRRI is reduced to match the level of its structural peers, implying about 12% reduction in the index. Thus, projected change in FDI stock is 12 × 0.213 = +2.56% above baseline.	Mistura, F. and C. Roulet (2019), "The determinants of Foreign Direct Investment: Do statutory restrictions matter?" https://doi.org/10.1787/641507ce-en.

	5.Allow foreign participation to railway freight transportation			
C. Reducing persistently high energy intensity of firms and decarbonizing the economy	1. Shift focus from monitoring to deep decarbonization investments by incentivizing adoption of energy-efficient technologies and capital upgrades while prioritizing support for sectors with high technical potential for energy savings 2. Develop and expand energy service companies (ESCOs) and green credit lines that repay based on verified energy savings. 3. Reallocate funding program to support energy management skills among technical staff and managers. 4. Implement carbon taxation to minimize the CBAM impact on export competitiveness	Carbon tax is introduced and additional fiscal revenues increase investments in energy efficiency int eh amount of 50 percent of carbon tax revenues. The other half of carbon tax revenues are transferred to households as social assistance.	The assumption is a full implementation of carbon tax relative to a baseline with CBAM (starting in 2030 at 4 EUR/tCO2, converging to 60 EUR/tCO2 by 2050.), where 50 percent of revenues are recycled into energy efficiency, and other half to household transfers. The carbon tax shifts the relative price of fossil fuels and renewables. The tax is directly modeled in MFMod. Romero-Jordan et al., (2025) find that a 1 percent improvement in energy efficiency is associated with a 0.06 percent increase in TFP for 26 European economies). 50 percent of the carbon tax revenues invested into EE leads to a productivity gain of 0.05 percent	https://documentsinternal.worldbank.org/Search/40016840
Developing skills and mobilizing workers				
A. Enabling women's equal participation and opportunities in the labor market	1. Expand access to childcare and eldercare services to address a key barrier preventing women employment 2. Include measures to reduce the gender wage gap and introduce labor code reforms that support flexible and inclusive work arrangements. 3. Design and implement employment counselling and ALMPs in accessing quality jobs, such as skills gaps.	An increase in the number of women participating on the labor market using a fertility rate of 1.4 (9944/1.4=7103) leads to an increase in the overall structural participation rate by 0.88 pp. The impact on participation is introduced gradually from 2030 and reaches the full impact by 2050. An increase in fiscal expenditure of MKD3.4 bn over the forecast horizon.	The assumption is that pre-school enrollment rate (ages 3-6) will increase from baseline 47 percent to 60 percent in 2050. The increase of 13 p.p. is moderate relative to the recent improvement in an aspirational peer (18.6 p.p. between 2015 and 2024 for Croatia). This would imply enrolling addition 9,944 children through the government current investment boom into kindergartens. Halim et al., 2019 find that an additional public preschool per 1,000 children increases the work participation of mothers of preschool age eligible children by 11–16 percent from the baseline mean. We assume a cost of 120 million MKD of constructing a kindergarten for 350 kids based on a report from the State Audit Office for North Macedonia.	https://documents1.worldbank.org/curated/en/393431561484915075/pdf/Preschool-Availability-and-Female-Labor-Force-Participation-Evidence-from-Indonesia.pdf https://openknowledge.worldbank.org/entities/publication/c11a6885-bc74-5e12-bcbd-fa91ae627300

B. Supporting effective School-To-Work Transition for youth	1. Scale up workplace-based learning, align vocational and tertiary education with labor market needs, and explore feasibility of establishing tertiary vocational education, especially in relation to ICT.	Assuming that all youth workers (15-24) participate, structural youth employment increases by 1.40% by 2030 modeled through an equivalent reduction in the NAWRU (youth unemployment makes up roughly 18 percent of total unemployed in 2024). This leads to an overall increase in structural employment of 0.25% (=1.4%*0.18).	According to Card et al., (2018), stronger job search assistance (for those that participate) leads to an average increase in employment of 1.4% by 2030.	Card, D., Kluve, J., & Weber, A. (2018). What Works? A Meta Analysis of Recent Active Labor Market Program Evaluations. Journal of the European Economic Association, 16(3), 894–931.
	2. Facilitate youth employment by strengthening career guidance systems, expanding school-to-work transition programs			
	3. Stimulate demand for young workers by offering targeted hiring incentives, subsidizing work-based training and apprenticeship programs, and supporting employer participation in youth employment initiative.			
C. Encouraging lifelong learning of workers and longer stay at the labor market	1. Expand access to tailored upskilling programs for unemployed individuals focusing on foundational, digital, and job-relevant skills to improve employability.	Increased productivity by 5% and related earnings for 7.5% of total working-age adults by 2050 leading to a TFP boost of 0.375% (0.075 x 5%) with respect to the baseline.	OECD (2019) study shows that non-formal training (e.g. seminars, courses) is associated with 11% higher wages of training participants. Informal learning (e.g. learning on the job through practice, colleagues) is associated with a 3.5% wage premium. Participation in adult non-formal learning (7.5% of working-age adults) increases from 0.3 percent baseline to 5 percent in 2050 leading to a more modest 5 p.p. increase in earnings.	OECD (2019), *Returns to Different Forms of Job-Related Training*, OECD Social, Employment and Migration Working Papers No. 231. https://doi.org/10.1787/b21807e9-en
	2. Introduce financial and regulatory incentives for employers—particularly SMEs—to co-invest in the upskilling and reskilling of their workforce.			
	3. Develop and regularly disseminate labor market intelligence to support evidence-based decision-making by employers, education providers, and policymakers.			
	4. Increase retirement age for both gender and introduce pension bonus for work after the statutory retirement age	The projected rise in the labor force, relative to the baseline, by 8.7% by 2050 (starting from 1.4% in 2035).	Raising the retirement age to 65 from 2035 and align for both genders. This would yield initially 1.4 percent of new labor force growing to 8.7 percent of projected LF by 2050.	https://www.worldbank.org/en/country/northmacedonia/publication/north-macedonia-public-finance-review
D. Reducing labor market frictions to better support participation and formalization	1. Reform labor code and labor tax system to allow more flexible use of fixed-term contracts, particularly for first-time hires, gig economy or seasonal needs.	Decrease in structural unemployment (NAWRU) corresponding to the assumed employment increase while accounting for the share of long-term unemployment in total unemployment of 66.1 percent. . Each year, 0.5*2% of long-term	Job search assistance program for the long-term unemployed increases employment in the first year after assignment by about 4 percentage points (Cottier et al 2024). The job search assistance program is assumed to increase employment among treated long-term unemployed individuals by 4 p.p. in the	https://www.aeaweb.org/articles?id=10.1257/pandp.20241093
	2. Make work pay: Phase out social benefits for social assistance recipients gradually (rather than immediately			

	upon employment) to avoid benefit cliffs.	unemployed persons are employed through this program according to the policy assumption. The shock starts in 2030 and lasts for 10 years	first year after treatment. We assume that 50% of those who are in long-term unemployment will participate in the program and will get a job.	
	3. Strengthen activation measures (e.g., job search support, tailored training) for recipients of social assistance.			

Standardized tables

	2000-2013	2014-2024	Aspirational peers (2014-2024)	Structural peers (2014-2024)	Source
Real/Income					
Real GDP growth, market prices	2.8	2.3	2.1	3.2	MPO
Investment, % change	6.4	2.8	5.6	5.4	MPO
Public	5.2	0.8	4.1	3.0	Calculated from MPO
Private	3.7	3.8	7.2	4.0	Calculated from MPO
Consumption, % change	2.3	2.7	2.9	2.9	Calculated from MPO
Public	1.6	1.6	1.8	2.8	MPO
Private	2.7	3.0	3.3	3.0	MPO
Net exports, % change	11.3	6.6	69.2	6.5	Calculated from MPO
Exports	6.4	6.7	6.1	8.7	MPO
Imports	7.0	6.6	6.4	6.4	MPO
Investment, % of GDP	22.4	24	23.7	24	Calculated from MPO
Public	5.4	5.5	4.6	6.1	Calculated from MPO
Private	17.1	18.5	16.1	20.9	Calculated from MPO
Consumption, % of GDP	93.1	88.7	75.3	92.6	Calculated from MPO
Public	17.3	14.2	20.2	16.5	Calculated from MPO
Private	75.8	74.5	57.9	76.2	Calculated from MPO
Net Exports, % of GDP	-16.5	-22.5	-2.4	-18.0	Calculated from MPO
Exports	35.4	63.6	59.8	31.5	Calculated from MPO
Imports	51.9	86.1	62.3	49.5	Calculated from MPO
Potential GDP growth	3.1	2.8	2.7	3.9	MPO
Distance to frontier, GDPpc as % of USA's GDPpc	24.6	30.7	46.4	23.2	Convergence Tool
Real GDP growth (LCU)	2.8	2.3	1.6	3.2	MPO
TFP, contribution (ppt)	0.8	0.5	1.5	1.2	Calculated from MPO
Capital accumulation, contribution (ppt)	1.4	1.7	1.7	2.1	Calculated from MPO
Labor, contribution (ppt)	0.6	0.2	-0.2	0.4	Calculated from MPO
Inclusion					
Poverty rate, $2.15 poverty line, percent	10.5	4.8	0.6	3.0	MPO
Gini Inequality Index	39.5	34.3	30.9	33.9	WDI
Demographics and Jobs					
Age dependency ratio (% of working-age population)	46.5	48.9	49.5	53.6	WDI
Employment to working-age population ratio	41.1	52.1	64.2	52.2	ILO
Male	49.8	61.5	68.3	59.9	ILO
Female	32.7	42.8	60.1	44.9	ILO
Labor force participation rate	61.8	64.6	71.3	63.4	ILO
Male	74.4	76.8	76.1	71.8	ILO
Female	49.5	52.5	66.6	55.3	ILO
Unemployment rate	33.5	19.4	10.1	17.9	ILO
Male	33.1	20.0	10.3	16.9	ILO
Female	34.0	18.7	9.9	19.2	ILO
Structural Transformation					

Real GDP growth, factor prices, % change	2.7	2.4	2.6	3.3	MPO
Agriculture	1.3	-1.6	0.5	2.0	MPO
Manufacturing	4.3	-0.3	2.0	3.0	MPO
Services	2.6	3.8	2.9	3.6	MPO
Agriculture, % of GDP	10.1	8.3	3.3	11.0	WDI
Manufacturing, % of GDP	20.8	23.5	25.8	24.1	WDI
Services, % of GDP	54.9	55.1	58.5	49.6	WDI
Sectoral composition of employment					
Agriculture	19.6	13.6	10.1	27.0	WDI
Manufacturing	32.0	30.8	30.7	20.8	WDI
Services	48.5	55.5	59.3	52.2	WDI
Employment Structure					
Share of waged employment in total working age population	29.8	40.9	54.7	31.7	ILO
Average years of schooling, share of HIC average			107.1	91.5	PWT
External					
Current account balance, % of GDP	-4.5	-2.0	-3.2	-10.6	MPO
FDI, % of GDP	5.0	3.7	4.1	6.9	WDI
Exchange rate, LCU/USD	51.1	54.4			MPO
REER	97.9	100.4			MPO
Reserves, months of imports	4.7	4.3	2.8	4.6	WDI
Memo items					
Credit to private sector, % of GDP	47.9	52.1	55.7	40.1	WDI
Interest rate, annual average	6.5	3.5	1.2	4.1	MPO
Inflation, CPI	2.8	3.1	121.9	9.8	MPO
Fiscal deficit, % of GDP	-2.1	-3.9	-3.5	-3.9	MPO
Public debt, % of GDP	32.3	53.2	53.4	46.4	MPO
-of which external, % of GDP	22.1	35.1	22.8	28.7	MPO

Employment Table

	Annualized growth, %				Share of total employment, %, 2022		
	2000-2013	2014-2022	Structural Peers 2014-2022	Aspirational Peers 2014-2022	North Macedonia	Structural Peers	Aspirational Peers
Total Employment	1.3	2.2	1.3	0.6			
Agriculture	-0.8	-5.9	0.8	-4.5	9.5	20.3	16.9
Mining	0.2	1	-0.8	-3.7	1	0.8	1.6
Manufacturing	-0.6	2.5	2.7	0.6	19.7	12	17.7
Utilities	2.6	3.1	0.2	0.6	3.3	2.6	3.4
Construction	2.6	1.2	2.7	0.9	6.5	6.4	8.2
Wholesale & Retail Trade	2.9	4.1	1.3	0.3	15.6	14.6	14.1
Transport & ICT	3.1	3.1	2.8	1.4	8.2	7.2	7.6
Hospitality	5.7	2.1	3.3	-0.8	3.6	5.9	4.7
Finance	0.9	5.1	0.7	3	1.5	1.5	1.7
Business Services	7.4	6.3	4.1	2	5.6	6.5	3.1
Public Administration	2.7	2.8	0.5	0.2	7.4	6.7	6.8

Education	1.8	4.4	0.2	1.4	7	5.7	5.2
Health	3.2	5.3	2.6	2.3	6.8	4.9	5
Other Services	1.4	7.6	-1.4	1.9	4.5	4.7	4

Source: World Bank Structural Change and Employment Database (based on ILO modelled estimates).

Productivity Table

Country	Year range	Between - Manufacturing	Within - Manufacturing	Total - Manufacturing	Between - Services	Within - Services	Total - Services
North Macedonia	2011-2022	0.20	0.20	0.40	0.10	0.16	0.25
Bosnia and Herzegovina	2011-2022	-0.22	0.16	-0.06	-0.12	0.27	0.15
Croatia	2010-2022	0.07	0.18	0.25	0.01	0.21	0.21
Kosovo	2018-2024	0.04	0.04	0.08	-0.01	0.00	-0.01

Source: Preliminary results from the World Business Dynamics Database.

Notes: OP decomposition computed as the cumulative sum of the difference over time of log labor productivity.

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